

Opinion

Capturing the facets of evolvability in a mechanistic framework

Jana M. Riederer ^{1,2,*}, Stefano Tiso ^{1,2,*}, Timo J.B. van Eldijk ^{1,2,*}, and Franz J. Weissing ^{1,*}

‘Evolvability’ – the ability to undergo adaptive evolution – is a key concept for understanding and predicting the response of biological systems to environmental change. Evolvability has various facets and is applied in many ways, easily leading to misunderstandings among researchers. To clarify matters, we first categorize the mechanisms and organismal features underlying evolvability into determinants providing variation, determinants shaping the effect of variation on fitness, and determinants shaping the selection process. Second, we stress the importance of timescale when studying evolvability. Third, we distinguish between evolvability determinants with a broad and a narrow scope. Finally, we highlight two contrasting perspectives on evolvability: general evolvability and specific evolvability. We hope that this framework facilitates communication and guides future research.

Evolvability is an important yet elusive concept

Understanding adaptation to changing environments is more important than ever. Climate change, antibiotic resistance, and viral vaccine evasion represent major societal challenges. Is an endangered species able to adapt to environmental change? Will a bacterial pathogen evolve antibiotic resistance? Can a virus evade vaccine-based immunization? At the core of these issues lies a common element: the capability of organisms to adapt – evolvability [1]. Evolvability research sheds new light on genomic architecture [2], the structure of regulatory networks [3,4], and many other features of **biological systems** (see [Glossary](#)). It has yielded surprising new insights, such as: adaptive evolution can proceed at a pace similar to ecological change, resulting in intricate and unexpected ecoevolutionary dynamics [5,6]; evolvability and **robustness** do not conflict but mutually reinforce each other [3,7,8]; and organisms with high evolvability can ‘generalize’ over environments [9,10]. Furthermore, evolvability research may add new perspectives to the formulation of a predictive theory of evolution ([Box 1](#) and see [Outstanding questions](#)).

Evolvability is studied by diverse approaches; for instance: Johansson *et al.* [11] inspect the genetic variance–covariance matrix; Woods *et al.* [12] compare the speed of adaptation of bacterial strains; and Martín-Serra *et al.* [13] focus on morphological integration and **modularity**. These approaches, although all valid, are widely disparate: all aim to understand evolvability, yet each focuses on a different facet. This plurality is also reflected in the fact that evolvability has been defined in many different ways ([Box 1](#)). We aim to highlight the different facets of evolvability and how they relate to each other, to facilitate a more nuanced and cohesive discourse on the topic. Throughout, we define evolvability as the capability of a biological system to undergo adaptive evolution (see [Box 1](#) for a justification).

Toward a mechanistic approach to evolvability

Evolvability is often viewed in terms of outcomes (e.g., speed of adaptation). As the same outcome can be achieved in many ways, it is useful to study evolvability by a mechanistic approach

Highlights

Evolvability, the capability to undergo adaptive evolution, is determined by a staggering diversity of mechanisms and organismal features. When discussing evolvability, it is useful to distinguish three categories of determinants: those providing variation, those shaping the effect of variation on fitness, and those shaping the selection process.

Some determinants of evolvability have a broad scope in that they affect adaptive evolution across many different environments; others have a narrower scope in that they impact evolvability only with respect to particular challenges. Being explicit about the scope of evolvability determinants would largely facilitate communication across disciplines.

On different timescales, the comparison of organisms regarding their evolvability and the comparison of mechanisms regarding their effects on evolvability can lead to very different conclusions.

¹Groningen Institute for Evolutionary Life Sciences, University of Groningen, Groningen, The Netherlands

²These authors are co-first authors.

*Correspondence: j.m.riederer@rug.nl (J.M. Riederer), s.tiso@rug.nl (S. Tiso), t.j.b.van.eldijk@rug.nl (T.J.B. van Eldijk), and f.j.weissing@rug.nl (F.J. Weissing).
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Box 1. Definitions of evolvability

Here we briefly discuss some definitions of evolvability, as they provide a good overview of the diversity of approaches in the field of evolvability research [63,72,73]. One important early definition revolves around the additive genetic coefficient of variation. It defines evolvability as the ability to respond to selection as governed by the presence or absence of standing genetic variation (often assessed in the G matrix) [38,58,74]. Another important perspective was given by Wagner and Altenberg [59], who made a distinction between variation and variability (i.e., the propensity of characters to vary). Evolvability is then considered not as the currently present variation but instead as the ability to generate new variation. A third, different perspective on evolvability considers the ability to generate major innovations [75,76]. For a comprehensive review of these developments, the reader is referred to [1]. Note that these definitions of evolvability are reflected in our first category of determinants, as they all view evolvability as determined by the presence or provisioning of variation.

One additional aspect of the definitions of evolvability (also called 'evolutionary potential' or 'adaptive capacity') is the relationship between variation and adaptation. While earlier treatments (as discussed in [1]) define evolvability as the ability of a biological system to evolve, irrespective of whether evolution is adaptive or not, recent definitions tend to restrict the concept to adaptive processes. For example, Payne and Wagner [14] combine the aspects of variation and adaptation when they define evolvability as '...the ability of a biological system to produce phenotypic variation that is both heritable and adaptive'. Other definitions in this vein have been provided in [7,38,60,77–81].

Following the general trend in the field, we here focus on adaptive evolution as well. When defining evolvability as the 'capability of a biological system to undergo adaptive evolution' we do not mean the presence or absence of this capability, but rather we consider its degree in a continuous fashion. Adopting an adaptive perspective does by no means imply that nonadaptive processes (e.g., genetic drift) are irrelevant; many of the determinants of evolvability reflect such processes (e.g., mutation). However, there are at least two reasons for focussing on adaptive evolution. First, many applications of evolvability (e.g., evolution of antibiotic resistance, adaptation to anthropogenic change) consider the adaptation to environmental challenges. Second, relating the rate and outcome of evolution to underlying selection pressures provides a yardstick, making it possible to differentiate between organismal and environmental features [38] and allowing comparisons across organisms. Both features are important first steps toward a predictive theory of evolution [82].

[14]: viewing evolvability not as a phenomenon *per se* but as a product of the mechanisms and organismal features that underlie it. A mechanistic perspective also clarifies discussions on the evolution of evolvability [1,14]: while questions regarding the evolution of 'the capability to undergo adaptive evolution' easily turn abstract, they become more obvious and transparent when translated into questions regarding the evolution of concrete mechanisms (e.g., the mutation rate).

Categorizing the determinants of evolvability

We refer to the mechanisms and organismal features that govern evolvability as determinants of evolvability. These affect different aspects of adaptive evolution and consequently shape evolvability in different ways. We here identify three ways in which determinants can shape evolvability, based on what aspect of adaptive evolution they affect, and categorize them accordingly (Figure 1). First, determinants may affect evolvability by providing variation. The mutation rate is the most obvious example of such a determinant [2,15–17]. Second, determinants may affect evolvability by influencing the effect of variation on fitness. For example, **developmental biases** may predispose mutations toward being beneficial [18–21]. Third, determinants may affect evolvability by shaping the selection process; shorter generation times, for example, may accelerate adaptation [22,23].

Category 1: providing variation

Heritable variation serves as the raw material for evolution. Hence, our first category refers to those mechanisms that generate and maintain variation. For example, mutations generate variation in many ways, ranging from point mutations to genome rearrangements. Interestingly, mutation rates vary widely between organisms as well as within genomes [24–26] and they can be regulated based on the environment (e.g., stress-induced mutagenesis, [27]), indicating that evolvability can evolve through the evolution of the mutation rate. Examples of determinants maintaining variation include **evolutionary capacitors** such as heat shock proteins. HSP-90 in *Saccharomyces*

Glossary

Biological system: we here define a biological system to be any biological entity that can be subject to evolution by natural selection.

Cryptic genetic variation: standing genetic variation that has little effect on phenotypic variation under normal conditions but generates variation under changed conditions. The release of this variation can facilitate (or hamper) adaptation and thus impact evolvability.

Developmental bias: the developmental mechanisms underlying a trait can introduce biases in the variation in the phenotype, even if the underlying mutations are unbiased. These biases can be (but need not be) aligned with the direction of selection, in which case they facilitate adaptive evolution.

Developmental canalization: robustness to genetic or environmental perturbations frequently exhibited by developmental systems, leading to a stable phenotypic outcome.

Evolutionary capacitor: mechanism that prevents the expression of genetic variation under some conditions, thus allowing the accumulation of cryptic genetic variation, and 'releases' this variation under other conditions, thus exposing it to selection.

Gene regulatory network (GRN)

model: model that explicitly represents the genotype-to-phenotype map as a complex network of regulatory interactions between genes. GRN models have been studied extensively in the context of evolutionary developmental biology and evolvability.

Modularity: the ability of subsets of a system ('modules') to function independently of other parts of the system (see [51]). Modularity can impact evolvability in various ways – for example, independent modules can be easily combined in different ways and furthermore do not interfere with each other's functioning.

Phenotypic plasticity: the expression of different phenotypes by the same genotype in response to environmental conditions. The impact of phenotypic plasticity on evolvability is subject to much debate (e.g., [10,47–50]); their relationship is complex and not yet well understood.

Robustness: the capability of the state of a biological system to persist under (environmental or genetic) perturbation. For example, robustness may refer to

cerevisiae, for instance, acts as a chaperone protein aiding correct protein folding. Chaperoning can shield sequence mutations from selection, thus maintaining variation. This can later be released under stressful conditions [28,29]. **Developmental canalization** can affect evolvability in a similar manner, in that it allows the accumulation of **cryptic genetic variation** [30]. Furthermore, capacitors may also be behavioural in nature, as parental care and thermoregulatory behaviour also allow cryptic genetic variation to accumulate [31,32]. Horizontal gene transfer may also be viewed as a category 1 determinant, as it allows variants to be maintained that would otherwise be lost from the population; for instance, by establishing an ‘accessory genome’ [33,34] or through the so-called rescuable gene hypothesis [35]. Not all heritable variation is genetic: epigenetic inheritance, inheritance of environmental features, and cultural inheritance can also affect adaptive evolution [36,37]. Hence, category 1 also includes mechanisms providing non-genetic heritable variation.

the ability to maintain a certain phenotype in the face of environmental fluctuations or genetic mutations. Intuitively, one might consider evolvability (the ability to change) and robustness (the ability to withstand change) to be opposed; however, it has been shown that they can be two sides of the same coin [3,7].

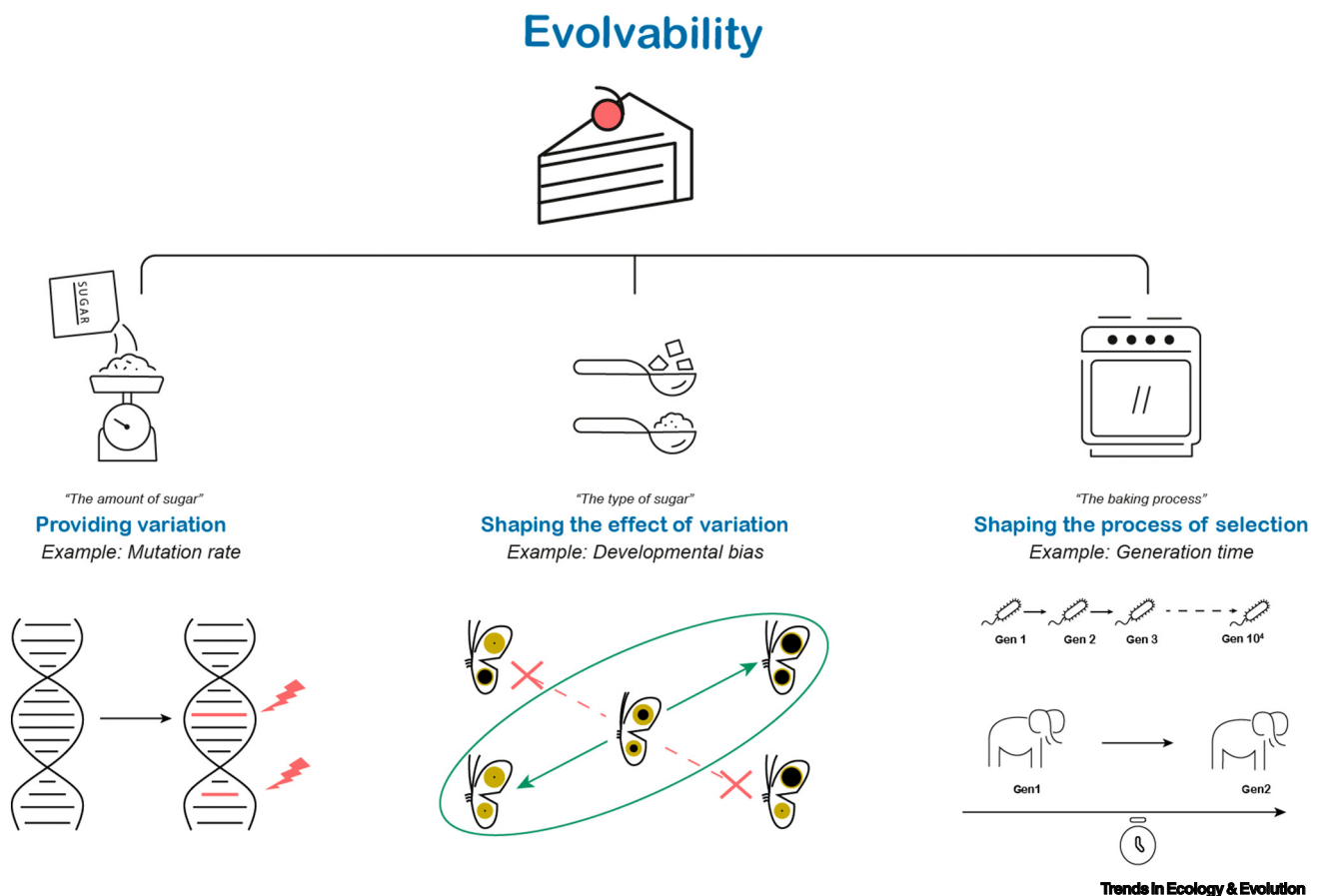


Figure 1. The way that mechanisms and organismal features affect evolvability can be classified into three categories. Each one contributes to evolvability in a different way. This can be compared with the process of baking a cake: the end result depends on several fundamentally different aspects – the amount of ingredients, the quality of ingredients, and the baking process. We suggest that evolvability is analogously affected by three different classes of determinants: those providing variation (‘the amount of sugar’), those shaping the effect of variation on fitness (‘the type of sugar’), and those shaping the selection process (‘the baking process’). An example of a determinant providing variation is the mutation rate, where mutation encapsulates a wide variety of phenomena ranging from point mutations to genome rearrangements. Developmental biases are examples of determinants that shape the effect of variation on fitness. Consider, for instance, the developmental system underlying the eyespot pattern on the wings of the butterfly *Bicyclus anynana*. This system is organized in such a manner that mutations can easily change the colour composition of the two wing eyespots in the same direction, while mutations changing the colour composition in opposite directions are extremely rare (depiction based on [40]). Depending on whether the selective pressure favours eyespots with the same colour composition or not, this bias may facilitate or impede evolvability. Finally, an example of a determinant that shapes the selection process is generation time: a shorter generation time allows faster adaptation – in absolute time, bacteria evolve faster than elephants.

Category 2: shaping the effect of variation on fitness

The mapping from mutation to fitness is affected by a variety of mechanisms: mutations may be random with respect to the genotype, but their effects on the phenotype and consequently fitness are often not [8]. Through features such as genomic, developmental, and regulatory architecture, the genotype-to-phenotype-to-fitness map can bias the fitness effects of mutations [19,38]. Category 2 thus contains determinants that influence the effect of variation on fitness. Examples can be found in the evo-devo literature [19,21,39,40], which describes various biases introduced through the developmental process (developmental bias; for an example see the butterfly *Bicyclus anynana* in Figure 1). The effect of variation on fitness can also be biased by the genomic and regulatory architecture [4,41]. In yeast, for example, genes for which upregulation is selectively favoured in a higher-temperature environment are grouped on the same chromosome. Therefore, a duplication of this chromosome suffices to achieve upregulation of all relevant genes; without such genome organisation, many independent mutations would be required to obtain an equivalent high-temperature adaptation [42].

Category 3: shaping the selection process

Starting from the same variation, evolution can still proceed at a very different pace and/or can lead to very different outcomes. Thus, category 3 contains determinants that impact evolvability not by shaping variation, but rather by shaping how the selective process acts on this variation. Examples are organismal features influencing population structure (e.g., dispersal tendency, mating patterns), as population structure may strongly affect adaptive evolution [43]. For instance, limited dispersal is hypothesized to have aided the rapid evolution of eusociality in diverse clades of insects [44]. Two other examples of category 3 determinants are generation time and the mode of reproduction. In coevolutionary host–pathogen arms races, the shorter generation time of pathogens provides them with an evolvability advantage, as they can evolve faster per time unit than their host [23]. Considering the Red Queen hypothesis, it becomes evident that hosts need other adaptations (e.g., sexual reproduction, a variation-generating immune system) to cope with pathogens on a longer-term perspective [45]. In the coevolution of hosts and their symbionts, the rapid evolution and/or diversification of the symbiont is often not in the interest of its host. Accordingly, the hosts of various symbiotic systems reduce the symbiont's evolvability by actively interfering with the symbiont's sexual reproduction [46].

How a mechanistic categorisation aids our understanding of evolvability

Some determinants of evolvability can be classified into more than one category. This is a deliberate feature of the proposed categorisation, as it highlights that a determinant can affect evolvability in different ways. The categorisation prompts the researcher to critically consider how mechanisms and processes shape adaptive evolution. An illustrative example can be found in the literature on evolvability and **plasticity**. Some have argued that plasticity impedes evolvability: plastic responses shield organisms from selection, preventing genetic adaptation (category 3) [47]. Others have argued that plasticity allows the accumulation of cryptic genetic variation (category 1) [10,48], thus potentially enhancing evolvability, because plastic traits are expressed only under particular environmental conditions. Finally, arguments derived from **gene regulatory network (GRN) models** conclude that the evolution of plasticity can restructure the genotype–phenotype map in such a way that random mutations are more likely to produce adaptive phenotypes (category 2) [49,50]. Our categorisation of determinants thus showcases these often subtle but nevertheless crucial distinctions.

The effect of modularity on evolvability provides another example. Inspection of the underlying mechanisms reveals that modularity has not one but two impacts on evolvability. First, it facilitates innovation by allowing pre-existing modules to be combined in different configurations, thus

providing variation (category 1) [51]. Second, it also allows individual modules to vary independently without affecting the functionality of the entire system (reducing antagonistic pleiotropy); this makes deleterious mutations less impactful, thus creating an adaptive bias that shapes the fitness effects of variation (category 2) [52]. This reduced impact of deleterious mutations (category 2) may also allow organisms to tolerate higher mutation rates (category 1), showing that determinants in different categories can interact in a reciprocal manner: the processes that provide variation (category 1), bias the fitness effects of variation (category 2), and shape the selection process (category 3) are not independent of each other.

Our view on the determinants of evolvability is suited to different approaches to evolution and evolvability. The first category (providing variation) contains not only mechanisms that provide new mutations, but also mechanisms that facilitate major innovations (however, the relationship between evolvability and major innovations is not yet well understood). Similarly, the second category not only considers instances of genotypic and developmental biases, but also includes broader ideas such as phenotypic accommodation and the theory of facilitated variation [20,53–55]. Finally, the third category considers not only genetic mechanisms (e.g., horizontal gene transfer, which also allows beneficial variants to spread more quickly [56]) but also – among others – niche construction, where organisms shape their own selective environment [57].

Explicitly considering timescale resolves apparent incongruencies

Determinants differ in the timescale on which they act; thus, when comparing evolvability across biological systems, the outcome is crucially dependent on timescale (Figure 2). Considering the timescale can help to resolve several apparent discrepancies.

This is exemplified by comparing determinants that provide variation (category 1): consider the impact of standing genetic variation [58] and the impact of mechanisms generating variation [59] on evolvability ([1]; Box 1). In the short term, adaptation is more strongly influenced by standing genetic variation, whereas mechanisms generating and maintaining variation are of greater significance when considering longer-term evolutionary trajectories [60].

Some approach evolvability in terms of speed of evolution [61], while others approach it in terms of the attained level of adaptation [38,62]. Both aspects are relevant [63], but they are often two

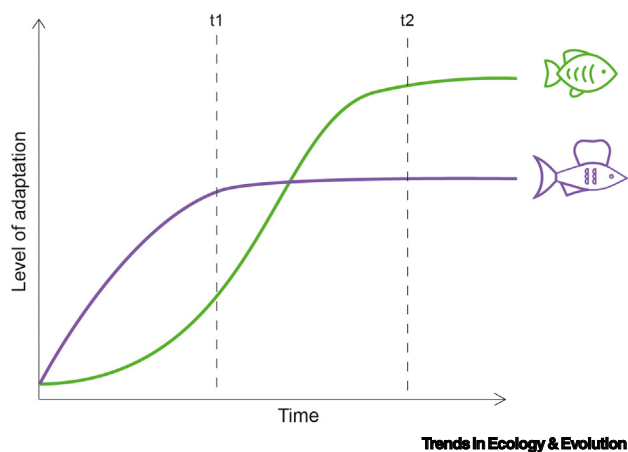


Figure 2. When studying evolvability, it is important to explicitly consider the timescale. Observing at different times can lead to different conclusions. Suppose that we observe the ability of two fish species to adapt to a new food source. Our conclusions on which of the two is more evolvable (is better able to adapt to the new selective challenge) will depend on the time at which we observe their level of adaptation. At time t_1 , the purple fish species is more adapted (and hence seems more evolvable), but at time t_2 the green fish species is more adapted (and hence seems more evolvable). This occurs because rates of adaptation are not constant across time; thus, being

explicit about the timescale of observation can resolve apparent discrepancies when comparing the evolvability of different organisms.

sides of the same coin, which becomes apparent when explicitly considering timescale. Consider, for example, different modes of inheritance. Epigenetically inherited traits can provide fast adaptation, yet this adaptation is often relatively inaccurate, given that the relative instability of epigenetic marks impedes the reliable maintenance of a certain optimal phenotype. By contrast, genetic adaptation proceeds more slowly, but in view of the high fidelity of genetic inheritance it may, in the long term, result in a higher level of adaptation. Therefore, epigenetic inheritance confers higher evolvability in the short term and genetic inheritance confers higher evolvability in the long term [36].

Another discrepancy that can be resolved by considering the timescale is the debate over the evolvability benefits of sexual reproduction, with both sexual and asexual reproduction being linked to increased evolvability [1,64]. All other things being equal, the response to selection (and hence the rate of adaptive evolution) is higher under asexual reproduction, as in the case of sexual reproduction, selection can act only on the additive component of genetic variation [65] (category 3). Therefore, asexual reproduction facilitates evolvability in the short term. In the longer term, sexual reproduction confers a higher evolvability, as the slower speed of evolution is outweighed by the ability to better explore the fitness landscape and reach global rather than local peaks. The claims that sexual reproduction increases evolvability and the claims that asexual reproduction increases evolvability can thus both be true, just at different timescales (Figure 2). Overall, the earlier examples show that the effects and relative importance of determinants vary over time. Therefore, explicit consideration of the timescale is crucial when studying evolvability.

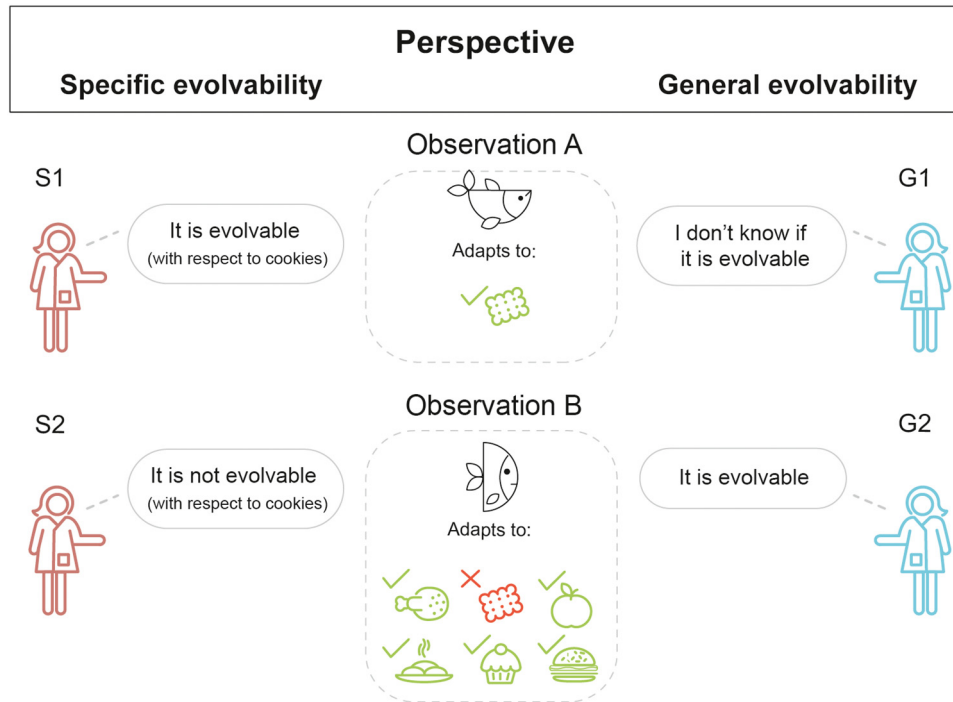
Accounting for environmental context shows that determinants differ in scope

Evolvability is the capability to undergo adaptive evolution; it is therefore necessary to consider in relation to which environmental challenge such adaptation arises. This reveals an additional property of determinants: their scope. Some determinants affect evolvability across many different environmental challenges; we consider these to have a broad scope. For example, mutation rates impact evolvability in virtually all environments. Other determinants have a narrow scope as they shape evolvability in only a restricted set of environments. For example, the grouping of temperature-relevant genes on one chromosome in yeast [42] enhances evolvability only to a change in temperature; it does not impact adaptation to other environmental challenges.

The scope of determinants pushes the researcher to consider the range of environments in which a determinant is relevant. Determinants relevant for adaptation to one environment may not be as relevant when considering adaptation to another. For example, in the radiation of Darwin's finches, developmental biases in beak development have been implicated in their adaptation to different seed sizes [66,67]. However, evolvability regarding beak shape will not be relevant with regard to other environmental challenges, such as temperature regulation or predator escape. By contrast, a higher mutation rate will affect evolutionary adaptation with regard to many different environmental challenges.

Two perspectives on evolvability

A very different distinction does not refer to the determinants of evolvability, but to the scholars studying evolvability. Depending on their scientific discipline, research question, or model system, scholars differ in whether they view evolvability as 'general' or 'specific' with regard to environmental challenges (Figure 3). Scientists adopting a specific evolvability perspective refer to the capability of a biological system to undergo adaptive evolution to a specific environment or a specific challenge; the main focus therefore is on how well a biological system can meet a specific selective target. This perspective is useful when studying adaptation to a particular challenge; for example, when exploring the ability of bacteria to evolve resistance to a particular antibiotic,



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Figure 3. Specific and general evolvability represent two different perspectives on evolvability. This influences questions and the interpretation of results in evolvability research. Different scientists can reach different conclusions from the same observations. This figure illustrates how scientists with different perspectives (on the left: specific evolvability; on the right: general evolvability) interpret the same observations regarding adaptation very differently. Observation A: a fish species can easily adapt to using cookies as a food source. From the specific evolvability perspective (S1), this is interpreted as high evolvability with respect to cookies. From the general evolvability perspective (G1), it is not possible to draw conclusions since no information is available regarding the ability to adapt to other food sources (environments). Observation B: a fish species cannot adapt to using cookies as food source, but (e.g., due to modular mouth parts) can undergo adaptation to a wide range of other food sources (environments). From a specific evolvability perspective (S2), this is interpreted as a lack of evolvability with regard to cookies. From a general evolvability perspective (G2), the ability to adapt to a wide range of different environments (ability to deal with dietary shifts) indicates high evolvability. Notice that the two perspectives characterise scholars of evolvability rather than the determinants of evolvability. The two perspectives should therefore not be confused with determinants acting at short versus long timescales or with determinants having narrow versus broad scope.

the ability of a virus to evolve resistance to a vaccine, or the ability of an endangered species to evolve adaptations to a specific anthropogenic threat [68,69]. By contrast, scientists adopting a general evolvability perspective view evolvability as the capability of a biological system to adapt to a wide spectrum of environments or of challenges, thus effectively considering evolvability irrespective of the environmental context. This perspective is useful when considering adaptation to unpredictable environments and is frequently used in studies exploring the link between evolvability and diversification [62,70,71].

While specific and general evolvability are both useful conceptualizations of evolvability, insights gained from one do not necessarily translate into insights about the other. Depending on the chosen perspective, the same observation can lead to different conclusions (Figure 3); it informs what questions are asked and affects how results are interpreted. Consider a population that is able to adapt rapidly to a specific challenge, such as a bacterial strain quickly evolving resistance to a particular antibiotic. From the perspective of specific evolvability, this strain has a high evolvability, while viewed from the perspective of general evolvability this single instance of

rapid adaptation says nothing about the ability of the strain to adapt to other challenges (heat stress, pH stress, etc.). The distinction between general and specific evolvability should not be confused with the scope of a determinant: the latter is a property of a determinant, whereas the former concerns two different ways of viewing evolvability.

Concluding remarks

Evolvability is an intricate concept with many facets. Different facets are at centre stage in different lines of research. Furthermore, evolvability is conceptualized in two different ways: specific and general evolvability. Being aware of these differences is crucial to foster an integrated and structured view on evolvability research.

Throughout we argue that evolvability should not be studied as a phenomenon *per se* but as a product of the mechanisms underlying it. Moreover, it is useful to clearly distinguish between determinants that provide variation, shape the effect of variation on fitness, and shape the selection process. A structured mechanistic approach clarifies debates in the literature and provides a sound basis for studying the evolution of evolvability.

Evolvability cannot be quantified by a single number. Both speed of evolution and level of adaptation are relevant, but they are not independent. Scholars should explicitly consider this when conducting evolvability research.

We hope that the proposed mechanistic approach facilitates communication across disciplines, helps to address major questions regarding evolvability (see Outstanding questions), and provides guidelines for the design of future studies on evolvability.

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Declaration of interests

No interests are declared.

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Outstanding questions

What mechanisms are most important in determining evolvability?

While we are slowly gaining a better understanding of the mechanisms that underlie evolvability, the relative importance of these mechanisms is not always apparent. Which mechanisms are the key drivers of evolvability and its evolution? How does this depend on the ecoevolutionary context?

How do plasticity and evolvability interact?

The relationship between evolvability and plasticity is currently hotly debated and should remain a topic of future study. This discussion includes various questions and concepts, such as the ‘plasticity first hypothesis’, the ‘flexible stem hypothesis’, and the question of whether genes are ‘leaders or followers’ in evolution.

Does evolvability evolve and if so how?

A mechanistic perspective may help in understanding the evolution of evolvability, as the mechanisms underlying evolvability can clearly evolve. Still, the question remains about whether and how organismal features that are mainly relevant on a long-term perspective (like the ability to adapt to novel environmental challenges) can be shaped by the myopic process of natural selection. Can evolvability be the target of selection or is it the by-product of other processes?

Can evolvability help in the formulation of a predictive theory of evolution?

The question of whether or to what extent evolution is predictable has fascinated evolutionary biologists for decades. Evolvability research may contribute new perspectives to this question: Understanding evolvability may be a first step toward a probabilistic prediction of evolutionary trajectories.

How can practical applications benefit from evolvability research?

While it is clear that evolvability is relevant for major societal problems, such as the evolution of antibiotic

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resistance and the implications of climate change, the full benefits of an explicit consideration of evolvability – as well as the methodology for doing so – remain to be explored.

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Pleiotropy, epistasis and the genetic architecture of quantitative traits

Trudy F. C. Mackay^{1,2}✉ & Robert R. H. Anholt^{1,2}✉

Abstract

Pleiotropy (whereby one genetic polymorphism affects multiple traits) and epistasis (whereby non-linear interactions between genetic polymorphisms affect the same trait) are fundamental aspects of the genetic architecture of quantitative traits. Recent advances in the ability to characterize the effects of polymorphic variants on molecular and organismal phenotypes in human and model organism populations have revealed the prevalence of pleiotropy and unexpected shared molecular genetic bases among quantitative traits, including diseases. By contrast, epistasis is common between polymorphic loci associated with quantitative traits in model organisms, such that alleles at one locus have different effects in different genetic backgrounds, but is rarely observed for human quantitative traits and common diseases. Here, we review the concepts and recent inferences about pleiotropy and epistasis, and discuss factors that contribute to similarities and differences between the genetic architecture of quantitative traits in model organisms and humans.

Sections

Introduction

Pleiotropy

Epistasis

Pleiotropy and epistasis

Conclusions and perspectives

¹Center for Human Genetics, Clemson University, Greenwood, SC, USA. ²Department of Genetics and Biochemistry, Clemson University, Clemson, SC, USA. ✉e-mail: tmackay@clemson.edu; ranholt@clemson.edu

Introduction

Quantitative traits display continuous phenotypic variation in populations, which is attributable to the simultaneous segregation of the many polymorphic loci that affect each trait and their sensitivity to environmental effects^{1–3}. As quantitative traits (such as morphology, behaviour, physiology, disease susceptibility, survival and fertility) account for most of the phenotypic variation observed in populations, understanding the genetic basis of quantitative variation is important for human health, agriculture and the study of evolution. Understanding the genotype–phenotype relationship for quantitative traits can help to identify genes associated with human diseases that can be targeted for treatment with approved drugs⁴, and understanding the genetic basis of differential drug responses can identify patients for which the therapy will be effective, while minimizing deleterious side effects⁵. Polygenic risk scores can predict the genetic predisposition to disease, enabling appropriate lifestyle changes before the age of disease onset⁶. Genomic prediction of breeding values in agricultural species can markedly increase selection responses⁷. The genetic basis of phenotypic variation for ecologically relevant quantitative traits determines how rapidly populations can respond to natural selection⁸; conservation efforts to maintain this variation can reduce inbreeding depression and limit loss of biodiversity.

Our understanding of the genetic architecture of quantitative traits is shaped by the results of studies that map quantitative trait loci (QTLs) by linkage to or association with polymorphic molecular markers in model organisms and agriculturally important animal and crop species³ and, more recently, by large genome-wide association studies (GWAS) for human quantitative traits and complex diseases⁹. It is now clear that the ‘many’ loci that affect quantitative traits equate to hundreds or thousands of associated molecular polymorphisms. Most highly powered genome-wide association analyses have been performed in human populations, but there are limits to mapping genes and variants associated with quantitative traits, including molecular phenotypes, in humans. The limit to the precision of association mapping in any species is linkage disequilibrium, that is, the extent to which molecular markers in close physical proximity are not independent. The human genome is organized in blocks within which linkage disequilibrium is high; therefore, determining which of the multiple polymorphisms in linkage disequilibrium are truly causal for the trait is difficult. Furthermore, only human tissues that can be obtained through blood draws, from biopsies or from post-mortem samples can be studied and these are not always relevant to the trait of interest. In humans, not all pleiotropic traits are measured on the same individuals, and neither is it possible to control the environment to which each person is exposed. These limitations can be overcome, to some extent, in studies using model organisms, in which the genetic background and environment can be controlled, mutations can be created to systematically evaluate the gene–quantitative trait phenotype relationship and gene editing can be used to functionally validate candidate genes and polymorphisms statistically associated with quantitative traits *in vivo*. Model organism studies also enable genetic variation to be quantified and multiple quantitative trait phenotypes measured on the same genotypes in trait-relevant tissues to assess the contribution of molecular phenotypes to organismal-level quantitative genetic variation. General features of the genetic architecture of quantitative traits inferred from model organisms are likely to be applicable to humans, and vice versa. As fundamental biological pathways are evolutionarily conserved across taxa, orthologous genes may affect variation in the same quantitative traits between humans

and genetic model organisms. However, the extent to which specific QTLs are conserved between model organisms and humans is limited by differences in genome size and organization, demographic history and biology.

Most QTL mapping studies utilize simple statistical models that evaluate the effects of variation at each polymorphic molecular marker for one or a few traits individually. Greater complexity arises when considering that genes affect multiple quantitative traits (pleiotropy) and act in non-linear genetic interaction networks (epistasis). Pleiotropy implicates unexpected mechanistic associations between traits¹⁰; can improve the precision of mapping genes and variants associated with quantitative traits¹¹; and affects responses to artificial or natural selection¹². Molecular networks associated with quantitative traits can be deduced from pleiotropic effects of high-dimensional molecular phenotypes such as gene expression traits and metabolites¹³. With epistasis, alleles have different effects in different genetic backgrounds, which causes variable allelic effects among individuals and populations¹⁴ and affects the magnitude of expressed quantitative genetic variation¹⁵. Genetic networks inferred from epistatic interactions can be used to improve the accuracy of predicting quantitative trait phenotypes from genomic data¹⁶.

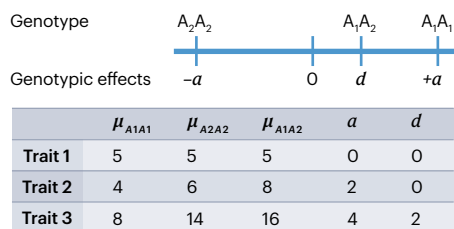
Here, we discuss the concepts of pleiotropy and epistasis for quantitative traits, methods for detecting and quantifying pleiotropy and epistasis, and the evolutionary consequences of pleiotropy and epistasis. We compare inferences about pleiotropy and epistasis between genetic model organisms and human populations, and discuss reasons for some apparent discrepancies in these features of genetic architecture between the model systems and humans.

Pleiotropy

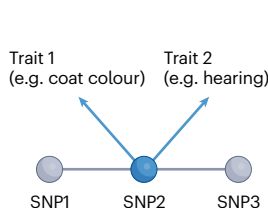
Pleiotropy was initially defined as the phenomenon whereby a single gene independently affects two or more phenotypes (that is, traits)^{17,18}. A more rigorous definition of pleiotropy is when the additive effects and/or dominance effects of a polymorphic variant are non-zero for two or more traits (Fig. 1a). True pleiotropy indicates shared genetic architecture affecting the traits and can occur when a polymorphism independently affects more than one trait (‘horizontal’ pleiotropy) (Fig. 1b) or when a polymorphism affects one trait which, in turn, affects another¹⁹ (‘mediating’ pleiotropy) (Fig. 1c). Mutations affecting hair and coat colour in mammals often have pleiotropic effects on vision and hearing – examples of horizontal pleiotropy²⁰. An example of mediating pleiotropy would be a variant that is associated with both an increase in low-density lipoprotein (LDL) cholesterol and an increased risk of coronary artery disease (CAD), as the increase in LDL cholesterol is a causal risk factor for developing this disease²¹. Different polymorphisms in a single gene can have different effects on different traits, some of which may be pleiotropic whereas others of which may affect only one trait. For example, different molecular polymorphisms at the *Catsup* locus in *Drosophila melanogaster* are independently associated with variation in longevity, locomotor behaviour and sensory bristle number²². Therefore, *Catsup* is a pleiotropic gene, but the individual polymorphisms do not have pleiotropic effects on all the measured traits.

Apparent pleiotropy refers to the situation whereby different molecular polymorphisms in linkage disequilibrium independently affect different traits (Fig. 1d). Apparent pleiotropy can occur not only within genes but also in multi-gene haplotypes in regions of high linkage disequilibrium, where a single variant can serve as a proxy for all variants in the haplotype. Apparent pleiotropy can disappear over time in a random mating population because linkage disequilibrium decreases

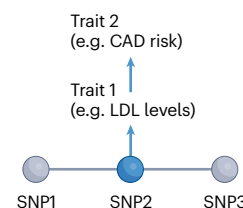
a Measuring pleiotropy



b Horizontal pleiotropy



c Mediating pleiotropy



d Apparent pleiotropy

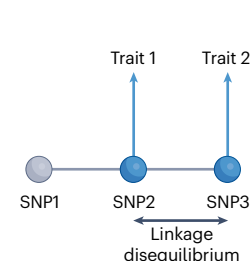


Fig. 1 | Pleiotropy for quantitative traits. **a**, The genotypes and genotypic effects for a single locus (A) with two alleles (A_1 , A_2). For a given trait, the additive effect (a) of locus A is $(\mu_{A_1A_1} - \mu_{A_2A_2})/2$, where $\mu_{A_1A_1}$ and $\mu_{A_2A_2}$ are, respectively, the mean of the trait for all individuals with genotype A_1A_1 or A_2A_2 in the population². The dominance effect (d) of locus A is $\mu_{A_1A_2} - (\mu_{A_1A_1} + \mu_{A_2A_2})/2$, where $\mu_{A_1A_2}$ is the mean of the trait for all individuals with genotype A_1A_2 in the population². The locus contributes to a quantitative trait if a and/or d have non-zero values. If multiple traits are measured, pleiotropy exists between any traits for which $a \neq 0$ and/or $d \neq 0$; here, traits 2 and 3. The genetic correlation between traits 2 and 3 depends on the frequencies of the A_1 and A_2 alleles in the population (Box 1). **b–d**, Different types of pleiotropy: true pleiotropy (panels **b**, **c**) and apparent pleiotropy (panel **d**). Horizontal pleiotropy is a form of true pleiotropy whereby the same

SNP independently affects two (or more) quantitative traits; for example, mammalian coat colour and hearing (panel **b**). Mediating (or vertical) pleiotropy is a form of true pleiotropy whereby a SNP affects one trait, which in turn affects a second trait; for example, low-density lipoprotein (LDL) levels and risk of coronary artery disease (CAD) (panel **c**). Apparent pleiotropy occurs when the traits are not caused by a common SNP but, rather, by two (or more) SNPs that are in linkage disequilibrium and each affects one of the traits. Here, SNP2 affects one trait and SNP3 affects a second trait, and there is strong linkage disequilibrium between SNP2 and SNP3 (indicated by horizontal arrow) (panel **d**). Linkage disequilibrium can be at any scale, from within a single gene to between genes. Horizontal lines denote a genomic region and circles represent SNPs. Blue circles denote SNPs affecting one or more quantitative traits (that is, $a \neq 0$ and/or $d \neq 0$).

as the number of generations of recombination increases. In the case of tightly linked variants in the same or different genes with independent effects on traits, the distinction between true and apparent pleiotropy may be moot in a particular population, but the pleiotropic effects of the polymorphism(s) will differ between populations with different patterns of linkage disequilibrium.

In addition to the classic definition of pleiotropy referring to two different quantitative traits, the concept can be extended to the same trait measured in males versus females or in different macro-environments (any environmental variable that can be reproduced across individuals)²³. Imperfect genetic correlations between males and females cause genotype by sex interactions, showing that the genetic basis of the trait is not identical in the two sexes. Similarly, imperfect genetic correlations between a trait measured across environments causes genotype by environment interactions, indicating that the genetic basis of the trait varies across environments. Genotype by sex and genotype by environment interactions are often observed in model organisms^{24–26} but their contributions to genetic architecture are typically not examined in human studies.

Quantifying pleiotropy

Quantifying pleiotropy from induced mutations. The clearest demonstration of pleiotropy is when a single mutation has been induced in a homozygous strain. In this case, the only difference between the mutant and wild-type strains is homozygosity for alternative alleles at the mutated locus. If the phenotypes of multiple quantitative traits are measured on the mutant and wild-type strains and their additive effects computed², non-zero additive effects for more than one trait indicate pleiotropy that is not confounded by linkage disequilibrium (Fig. 1a).

Collections of mutations or RNA interference (RNAi) constructs designed to target most genes in the genome are available for *Saccharomyces cerevisiae*²⁷, *Caenorhabditis elegans*²⁸, *Arabidopsis thaliana*²⁹, *D. melanogaster*^{30–32} and *Mus musculus*³³. These collections have largely

been used for forward genetic screens to identify genes affecting phenotypes of interest to specific laboratories but are also suitable for quantifying pleiotropy across studies. For example, the most straightforward organismal phenotypes to quantify in yeast are fitness and growth rate, and these have been assessed for strains from the yeast deletion collection under many environmental conditions to quantify pleiotropy from gene by environment interactions^{27,34}. Similarly, the effects of *D. melanogaster* P-element insertional mutations have been measured for multiple organismal quantitative traits^{35–40} (Fig. 2). There have also been large-scale efforts in laboratory mice to couple genome-wide mutagenesis with high throughput phenotyping to understand the pleiotropic functions of, ultimately, all genes in the mammalian genome^{33,41–45}. All of these studies have shown that pleiotropic effects on organismal-level phenotypes are ubiquitous. However, not all genes have equal levels of pleiotropy. Some mutations are highly pleiotropic and affect many traits, whereas others affect a few traits, only one trait or no traits⁴⁶. This pattern is reflected in Gene Ontology (GO) terms associated with genes curated from functional analyses of mutations by individual laboratories in each model organism community⁴⁷.

The magnitude of pleiotropy and the distribution of pleiotropic effects depends on the numbers of traits and mutations that are assessed. The ability to quantify gene expression, protein abundance and metabolites genome-wide enables large numbers of intermediate molecular phenotypes to be quantified at once and the pleiotropic effects of genetic variation on these phenotypes to be examined. Analysis of gene expression in a subset of genes from the yeast deletion collection showed that most mutations affected the expression of the deleted genes themselves (*cis*-regulatory mutations) as well as many other co-regulated genes (*trans*-regulatory mutations)^{48–50}. The number of *trans*-co-regulated genes was not uniform across the mutations and could be described with a ‘scale-free’ transcriptional network topology, whereby a few genes, when individually deleted, affected the transcription of large numbers of downstream target genes

Gene name	Quantitative trait													
	Abdominal bristle number	Stenopleural bristle number	Startle response	Olfactory behaviour	Starvation resistance	Ethanol sensitivity	+O Night bout number	Q ₁ Night bout number	+O Night sleep	Q ₁ Night sleep	+O Day bout number	Q ₁ Day bout number	+O Day sleep	Q ₁ Day sleep
6-phosphofructo-2-kinase					Q ₁									
alan shepard		♂			Q ₁									
Basigin					Q ₁									
Beadex														
bendless		♂												
b ⁿ integrin														
boudin														
brother of odd with entrails limited					Q ₁									
CG13377		♂												
CG1806														
CG33523					+O									
CG33796					+O									
CG3638														
cookie monster					+O									
desert														
E2F transcription factor	♂													
escargot														
extra macrochaetae					+O									
frizzled	♂													
Gliolectin		♀												
Glilotactin														
Heat shock RNA w/SIF amide receptor														
Hr39					+O									
IGF-II mRNA-binding protein	♀	♂			Q ₁									
Laminin A														
lilliputian					+O									
Malic enzyme														
mir-279					Q ₁									
Misexpression suppressor of ras 3					+O									
misshapen					+O									
Mo25	♂													
Multidrug resistance protein 4														
muscleblind		♀												
Myosin 31DF	♂ ♀													
no ocelli					+O									
Odorant receptor 83b														
outsiders														
pico														
Ras-related protein														
roundabout														
Rtnl1					Q ₁									
Sema-5c														
smell impaired 35A					Q ₁									
sprint														
sugarless		♀												
taiman	♂				Q ₁									
Tenascin major					Q ₁									
toucan														
Tousled-like kinase														
tramtrack	+O				+O									
Twin of m4	Q ₁													
what else														
wing blister														

Fig. 2 | Pleiotropy of P-element insertions in *Drosophila melanogaster*.

Additive effects of P-element insertions in 53 genes on 10 quantitative traits. All P-element insertions are in the same genetic background. Cell colours indicate whether the P-element increases (red cells) or decreases (blue cells) the trait value relative to the control. The intensity of the colour (dark, medium and light) denotes deviations from the mean of co-isogenic controls exceeding the 95%, 99% and 99.9% confidence intervals, respectively. Light green cells indicate decreased trait values in males and increased trait values in females for the P-element insertion compared with the control. Black cells are not significantly

different from the control, and grey cells indicate traits for which the effect of the P-element insertion was not measured. All traits were measured for males and females. Effects on all sleep traits³⁸ are given for males and females separately. Effects on abdominal and sternopleural bristle numbers³⁵, startle response³⁷, olfactory behaviour³⁶, starvation resistance³⁹ and ethanol sensitivity⁴⁰ are pooled across sexes, unless sex-specific effects for females (♀) or males (♂) are specified. This figure illustrates the ubiquity of pleiotropy, that pleiotropic effects exhibit antagonistic pleiotropy across traits and that pleiotropy can be sex-specific and even sex-antagonistic.

whereas deletion of other genes affected the transcription of fewer target genes^{49,50}. Yeast *trans*-regulatory mutations have more pleiotropic effects than *cis*-regulatory mutations^{51,52}. Similar magnitudes of pleiotropy of gene expression were found for single P-element mutations in *D. melanogaster*^{53,54}. These observations show how a mutation in a single gene can potentially affect multiple organismal phenotypes by altering expression levels of other genes and provide insight into the molecular and mechanistic basis of pleiotropy. Similar mechanistic insight can be obtained by profiling multiple mutations that affect a focal phenotype^{53,54} and determining common gene expression traits. For example, transcriptional profiling of 7 *D. melanogaster* mutations associated with increased life span revealed 553 transcripts that were co-regulated in 4 or more of the 7 genotypes⁵⁴. These transcripts were encoded by genes affecting reproduction, chemosensation, metabolism, immunity and/or defence response, and function and development of the nervous system.

Many rare diseases in humans are caused by de novo mutations that typically have pleiotropic effects⁵⁵. In recent years it has become possible to perform genome-scale forward genetic screens in human cell lines using CRISPR–Cas9 gene editing. Methods such as Perturb-Seq, which combine these large-scale mutagenesis approaches with rich single-cell transcriptional phenotyping of each CRISPR-induced perturbation, reveal widespread functional transcriptional pleiotropy at the cellular level in humans⁵⁶.

Quantifying pleiotropy from natural variation. Many induced mutations demonstrate pleiotropy at the gene level. However, because genes often contain multiple domains with different functions, it is possible that induced mutations have larger effects (for example, on all or multiple functions) and are more pleiotropic than naturally occurring polymorphisms, which have survived the sieve of natural selection and may have more subtle and specific effects. Quantitative genetics theory^{2,3} shows that correlations between different quantitative trait phenotypes in a natural population are caused by shared genetic variation affecting both traits (genetic correlation) or because the environmental circumstances experienced by an individual affect the traits in the same direction (environmental correlation) (Box 1). The genetic correlation is attributable to directional pleiotropy for all loci affecting both traits and is a genome-wide population average (Box 1). In the absence of polymorphic molecular markers, cross-trait genetic and environmental correlations are estimated from any experimental design that enables heritability to be estimated from relatives, when both traits are measured on all relatives^{2,3}. However, genetic correlations can be measured across all loci in linkage or association mapping populations without the need for related individuals, to estimate pleiotropy globally across the genome. Pleiotropy can also be assessed locally in linkage and association mapping populations to infer pleiotropy contributed by individual loci.

In model organisms, pleiotropy can be assessed at the level of individual haplotypes or SNPs by using publicly available inbred linkage and association mapping populations that are genotyped for dense polymorphic molecular markers or have their genomes fully sequenced. The linkage mapping populations are derived by crossing two or more inbred strains, followed by inbreeding (or self-fertilization) of the F₂ or later segregating generations to near homozygosity. The association mapping populations are obtained by inbreeding (or self-fertilization) of individuals collected from a natural population to near homozygosity. Recombinant inbred lines derived from biparental or multi-parental crosses of inbred strains, diverse inbred strains and heterogeneous outbred stocks are available for linkage mapping in mice⁵⁷ and rats⁵⁸; experimental results from studies using these resources to map quantitative traits, including intermediate molecular phenotypes, are curated in searchable databases that can be mined to assess pleiotropy^{59–61}. In *D. melanogaster*, the inbred, fully sequenced *Drosophila* Genetic Reference Panel (DGRP) lines^{62,63} are available for association mapping and the *D. melanogaster* Synthetic Population Resource (DSRP)⁶⁴ for linkage mapping. Panels of recombinant inbred lines for linkage mapping and wild-derived inbred lines for association mapping are also available for *C. elegans*⁶⁵ and *A. thaliana*^{66,67}. Most model organism studies designed to detect pleiotropic QTLs for organismal traits have typically found them^{24–26,68,69}. Examples include a study in *C. elegans* mapping QTLs for 24 traits upon exposure to 16 toxins which found 33 QTLs that mapped to one of three regions associated with many toxins⁶⁸. A study in *S. cerevisiae* mapping QTLs affecting fitness in 18 different growth conditions identified 449 pleiotropic QTLs⁶⁹.

In humans, many large, highly powered GWAS have been performed for quantitative traits and diseases. Although individual-level data are reported for studies using large publicly available biobanks^{70,71}, results of disease GWAS are only available as summary statistics (effect size, *P* value, standard error of associated polymorphisms)⁷², rather than at the level of individual SNPs. Moreover, most GWAS only evaluate a single phenotype. Thus, assessing pleiotropy from GWAS required new methods to be developed to estimate cross-trait genetic correlations across different studies using summary statistics^{10,11,19,73,74}. If large numbers of individuals are assessed for the same constellation of genetically correlated traits, considering all traits in a single GWAS can boost the power to detect SNP associations and to predict disease risk^{11,19}. Large-scale biobanks^{70,71} for which dense genotypes or exome or whole-genome sequencing data have been linked to deep phenotyping information via electronic health records have enabled phenome-wide association studies^{75,76}. Phenome-wide association studies evaluate associations of each molecular polymorphism with a plethora of phenotypes to uncover pleiotropy between traits at an unprecedented scale. Collectively, research based on GWAS has shown substantial pleiotropy between human quantitative traits that is sometimes unexpected,

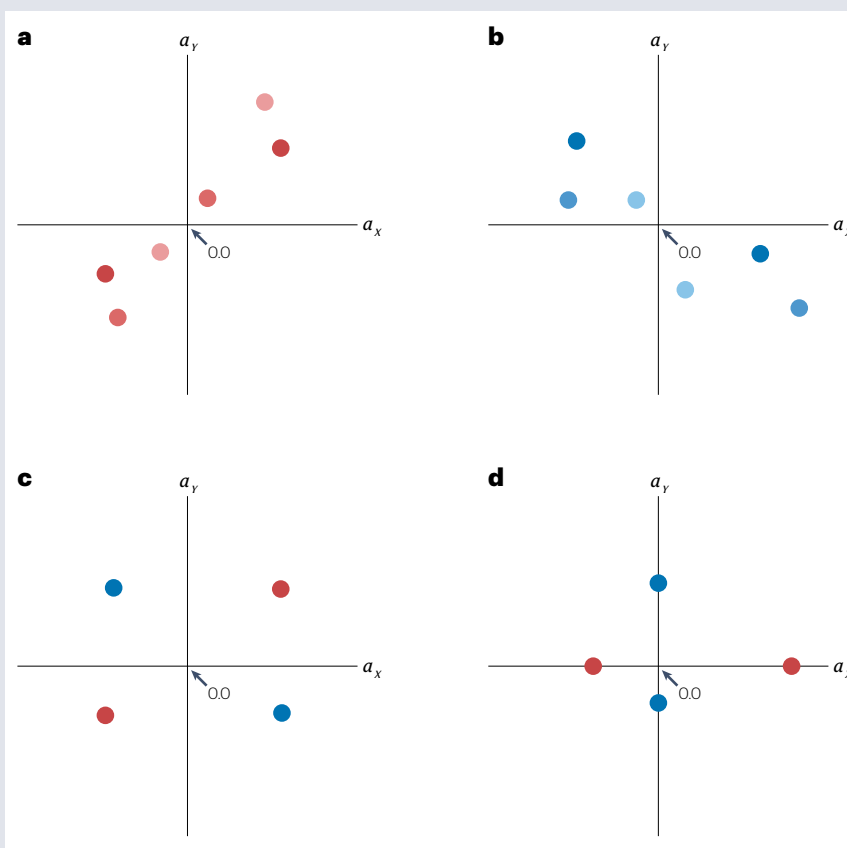
Box 1

Phenotypic, genetic and environmental correlations

The phenotypic correlation (ρ_{PXY}) between two quantitative traits X and Y can be estimated in a population where measurements of both traits on a sample of individuals are available: $\rho_{PXY} = \text{cov}_{PXY} / \sigma_{PX} \sigma_{PY}$, where cov_{PXY} is the covariance of X and Y and σ_{PX} and σ_{PY} are their phenotypic standard deviations. ρ_{PXY} can be decomposed into genetic (ρ_{GXY}) and environmental (ρ_{EXY}) correlations between traits X and Y . The cross-trait genetic correlation is defined as the correlation of breeding values between the two traits. The breeding value of an individual is determined by the mean value of its progeny when mated to random individuals in the population, expressed as a deviation from the population mean^{1–3}. Breeding values for each genotype at a locus affecting a quantitative trait are a function of allele frequencies (p and q for A_1 and A_2 alleles at locus A , respectively) and the average effects of the alleles it carries. The average effects of alleles A_1 (α_1) and A_2 (α_2) are $\alpha_1 = q[a + d(q - p)]$ and $\alpha_2 = -p[a + d(q - p)]$, and the breeding values of the A_1A_1 , A_1A_2 and A_2A_2 genotypes are $2\alpha_1$, $\alpha_1 + \alpha_2$ and $2\alpha_2$, respectively^{1–3}, where a is the additive effect and d is the dominance effect (Fig. 1a).

Non-zero values for cross-trait genetic correlations indicate pleiotropy between the traits^{1–3}. However, a genetic correlation of zero does not always mean that pleiotropy is absent. In the figure, the average effects of biallelic loci on trait X (α_{1X} , α_{2X}) are plotted against their average effects on trait Y (α_{1Y} , α_{2Y}). Both alleles of a given locus are coloured the same and each locus has a different colour. In the figure, panels **a** and **b**, the average effects of all loci are non-zero for both traits, indicating that the loci are pleiotropic and the traits have shared genetic variation. When taken across all loci, the cross-trait genetic correlations are positive in panel **a** and negative in panel **b**. In the figure, panel **c**, the average effects of both loci are non-zero and the loci are therefore pleiotropic. However, one locus has a positive

genetic correlation and the other a negative genetic correlation, which cancel out to give an overall genetic correlation of zero. Genetic correlations cannot detect this type of pleiotropy. In the figure, panel **d**, the average effect of each locus is zero for at least one of the two traits: there is no pleiotropy, and the genetic correlation is zero. The non-zero effects are owing to non-genetic (environmental) effects on both traits and non-additive genetic variation (dominance and epistasis).



and that there is variation in the sign and/or magnitude of pleiotropy between the same traits across the genome^{10,11,74,77,78}. Moreover, a large meta-analysis of 4,155 GWAS confirmed extensive pleiotropy for human quantitative traits, with 90% of GWAS loci overlapping multiple traits⁷⁹. Most of these studies mapped individual molecular polymorphisms associated with two or more traits^{73–76,78,79} and a few also estimated global genetic correlations^{77,79}. This prevalence of pleiotropy is particularly evident for psychiatric disorders. There is a shared genetic basis between risk of schizophrenia and bipolar disorder⁷⁷; between multiple psychiatric disorders and variation in calcium channel genes⁷⁸; between anorexia nervosa and schizophrenia, anorexia and obesity,

and educational attainment and several diseases⁷⁴; and between risk of schizophrenia and Parkinson disease¹⁰. The shared genetic architecture between these disorders provides new biological insight into potential common disease processes.

Once pleiotropy at a locus has been established between two or more traits, it is important to discriminate between true pleiotropy and apparent pleiotropy due to linkage disequilibrium (Fig. 1), as only the former reflects shared genetic underpinnings. Apparent pleiotropy from linkage disequilibrium is likely to be common in most model organism linkage mapping populations due to the large linkage disequilibrium blocks tagged by each molecular marker. Although the

extent of linkage disequilibrium is greatly reduced in model organism outbred populations and multi-parental mapping populations as well as human association mapping studies, linkage disequilibrium as a potential cause of pleiotropy always needs to be ruled out. Statistical co-localization methods can help to address this question^{10,74}, but the gold standard is experimental validation. First, all variants in linkage disequilibrium with the putative pleiotropic variant that could potentially cause the multiple trait associations need to be identified, followed by fine-mapping by further generations of recombination in model organisms⁸⁰ or by performing GWAS for the same traits in a population with high-density genotypes in humans⁸¹. Advances in genome editing using CRISPR-based approaches⁸² have revolutionized our ability to functionally validate pleiotropic SNP associations in model organisms using scarless allelic replacement⁸³ or base editing⁸⁴. Functional validation in humans can be done by gene editing in cell lines⁸⁵ from the relevant tissue(s) but is restricted to traits that can be measured in cells. In the future, massively parallel CRISPR-based screens^{86,86} have the potential to functionally validate pleiotropic SNP effects.

Once true pleiotropy has been confirmed, mediation analyses are used to discriminate between horizontal and mediating pleiotropy (Fig. 1b,c). One type of mediation analysis uses conditional regression models of the effect of the polymorphism on the traits, conditioning on either trait *X* or trait *Y*^{87,88}. If conditioning on trait *X* removes the effect on trait *Y*, but conditioning on trait *Y* does not remove the effect on trait *X*, then trait *X* mediates trait *Y*. With horizontal pleiotropy, conditioning the polymorphism effect on either trait will not remove the effect on the other. Classical statistical mediation analyses define the relationship between an exposure variable (*X*), a mediator variable (*M*) and an outcome variable (*Y*)^{89,90}. The effects to be estimated are the total effect of *X* on *Y*, the effect of *X* on *M*, the effect of *M* on *Y* and the effect of *X* on *Y* after accounting for the effect of *M*. The models are then used to partition the total effect of *X* on *Y* into a direct effect and a mediation effect. Mendelian randomization methods are a class of mediation analyses where *X* is a genetic polymorphism that affects trait *Y* solely via its effects on the mediator trait *X* (mediated pleiotropy) and assumes there is no horizontal pleiotropy^{19,91,92}. More recent Mendelian randomization methods that detect and correct for horizontal pleiotropy are capable of distinguishing horizontal and mediated pleiotropy. These methods show that horizontal pleiotropy is common among human quantitative traits⁹³ and was observed in ~50% of significant causal relationships⁹⁴.

Systems genetics and regulatory variation

Technological developments in genome-wide profiling of gene expression, metabolites and proteins combined with the ability to map organismal-level QTLs in linkage mapping populations catalysed the field of systems genetics^{95,96}. We focus here on the genetics of gene expression as these analyses are the most common, but the same principles and results hold for other molecular phenotypes. Early studies in experimentally tractable organisms showed that the levels of expressed transcripts are quantitative traits that vary between sexes, ages and genotypes^{97,98} and between natural populations⁹⁹. The first experiments mapping QTLs associated with variation in gene expression (expression QTLs (eQTLs)) were in yeast^{100–102}, followed by studies in mice¹⁰³, maize¹⁰⁴, *A. thaliana*¹⁰⁵, *D. melanogaster*¹⁰⁶, *C. elegans*¹⁰⁷ and human cell lines^{103,108,109}.

These pioneering studies showed that large numbers of transcripts were genetically variable, often with high heritabilities, and mapped to one or more genomic loci proximal (*cis*-eQTLs) or distal (*trans*-eQTLs)

to the gene encoding each transcript. Some *trans*-eQTLs were associated with variation in expression of large numbers of genes. *cis*-eQTLs initially encompassed many megabases to either side of the genes encoding the transcripts due to the poor resolution of small linkage mapping populations; thus, many *cis*-eQTLs identified in the early studies could have been *trans*-eQTLs. The genetically variable transcripts clustered into distinct genetically correlated transcriptional modules (Fig. 3a,b). Enrichment analysis for GO categories, Kyoto Encyclopedia of Genes and Genomes pathways, protein–protein interactions, tissue-specific expression and transcription factor binding sites¹³ provides functional insight into gene modules, and functions of non-annotated genes in these modules can be inferred by their genetic correlation with genes of known function (Fig. 3c). Furthermore, the genetically variable transcripts (or transcriptional modules) can also be genetically correlated with organismal quantitative traits (Fig. 3c), provided the transcripts are assessed in tissues that are relevant to the organismal trait of interest^{13,95,96,103}. These observations led to the hypothesis that *cis*-eQTLs could affect organismal quantitative traits by perturbing transcriptional (and other) molecular phenotypes that, in turn, affect disease liability, thus giving mechanistic insights into the biological basis of disease which was not possible from mapping QTLs associated with the organismal phenotypes. Systems genetic analyses model the pleiotropic relationships between the genetic polymorphisms associated with the organismal quantitative trait, the genetic polymorphisms associated with *cis*-eQTLs and the gene expression traits associated with the organismal quantitative trait to infer causal relationships between genetic variation, networks of molecular phenotypes and organismal quantitative traits^{13,87,88,110}.

The hypothesis that molecular genetic polymorphisms affect variation in organismal quantitative traits via effects on gene regulation is supported by human studies showing that SNPs associated with organismal quantitative traits are enriched for eQTLs^{111–113} and deoxyribonuclease I (DNase I) hypersensitive sites¹¹⁴, and that SNPs in DNase I hypersensitivity sites¹¹⁵ and other functional annotations^{114,116} explain more heritability than SNPs in coding regions across many common diseases. The Genotype-Tissue Expression project has collected genotype and whole-genome gene expression data from post-mortem human samples of 54 non-diseased tissues from nearly 1,000 individuals, and mapped eQTLs for each tissue¹¹⁷. This reference transcriptome database has been used to predict gene expression in human populations for which genetic polymorphisms and organismal phenotypes have been assessed to enable transcriptome-wide association studies to detect pleiotropy between gene expression and the organismal phenotypes^{118,119}. The eQTLGen Consortium, a large-scale effort to map eQTLs in blood from more than 30,000 individuals, showed that nearly 90% of genes had eQTLs, the expression of which was associated with more than 1,200 quantitative traits¹²⁰.

The ubiquity of eQTLs, the large numbers of QTLs in non-coding and, presumably, regulatory genomic regions and the association of variation in gene expression with variation in quantitative trait phenotypes led to the proposal that the genetic basis of human quantitative traits is ‘omnigenic’, such that many loci indirectly affect many traits via highly interconnected gene regulatory networks driven by *trans*-eQTLs with small effects in disease-relevant cells and tissues^{121,122}. This model and the systems genetics analyses rely strongly on the assumption that most organismal trait QTLs correspond to eQTLs and that variation in gene expression in relevant tissues is pleiotropically correlated with variation in organismal quantitative traits. These associations have not been observed in *D. melanogaster*, where organismal trait QTLs

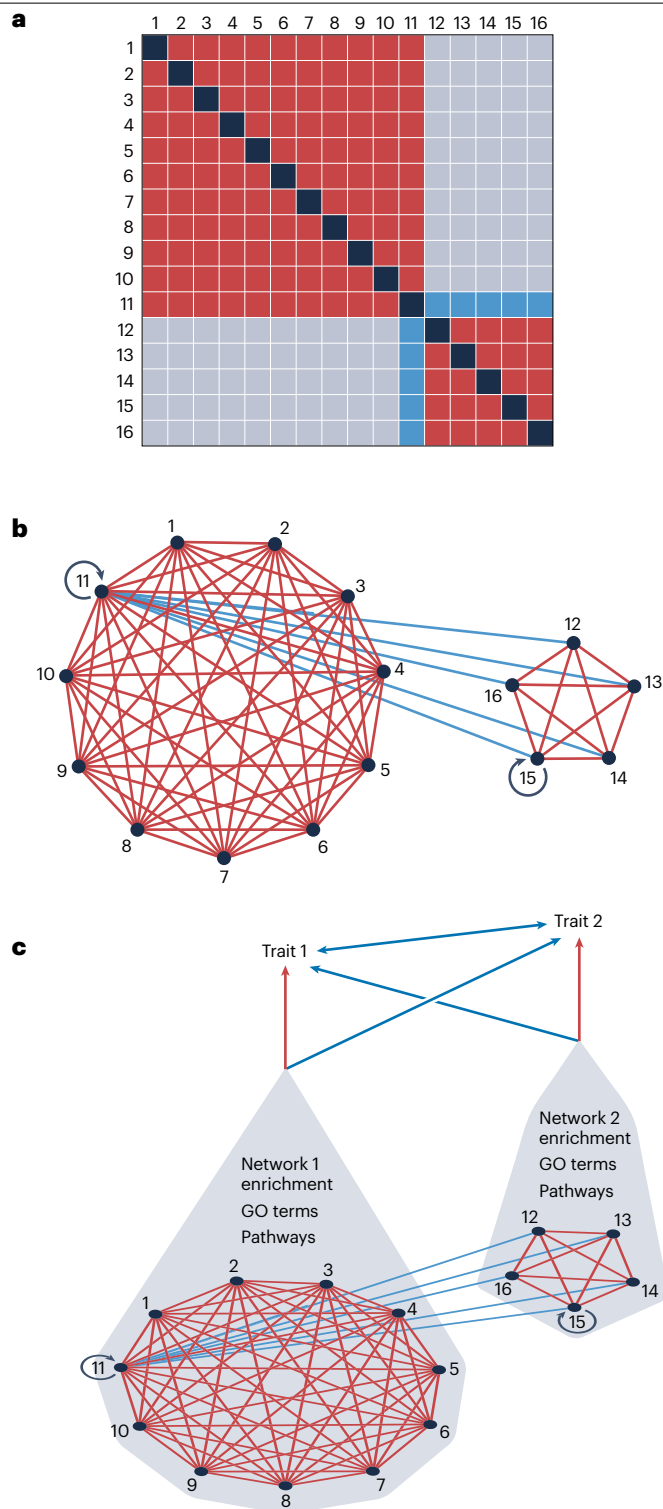


Fig. 3 | Systems genetic analysis of pleiotropy. **a**, Modular organization of gene expression correlations for 16 pleiotropic genes, showing two distinct networks of genes 1–11 and 12–16. Red cells depict positive correlations, blue cells negative correlations and grey cells no significant correlations of pairwise gene expression. Gene 11 is negatively correlated with all genes in the smaller network. **b**, Network view of the same data. Genes are indicated as nodes and pairwise correlations as edges between nodes (red edges for positive correlations, blue edges for negative correlations). Arrows indicate that genes 11 and 15 are associated with *cis*-expression quantitative trait loci (*cis*-eQTLs). The *cis*-eQTL for gene 11 is a *trans*-eQTL for the other genes in both the large and small networks, and the *cis*-eQTL for gene 15 is a *trans*-eQTL for the other genes in the small network. Mendelian randomization or mediation analyses will give the extent to which the pleiotropic effects of the *cis*-eQTLs are horizontal or mediating. **c**, Pleiotropic effects of gene expression networks on quantitative traits. The large module is positively correlated (red arrow) with trait 1 and negatively correlated (blue arrow) with trait 2, and the small module is positively correlated with trait 2 and negatively correlated with trait 1. Traits 1 and 2 are negatively correlated. The module enrichment for Gene Ontology (GO) terms and pathways provides mechanistic insight about the traits.

in *D. melanogaster*, giving greater mapping precision than for humans and thus separating QTLs from eQTLs. More recently, human studies that fine-mapped QTLs and eQTLs showed that the overlap between QTLs and eQTLs is much smaller than previously thought^{125–127}. Furthermore, formal tests show that only 11% of disease heritability is mediated by levels of gene expression¹²⁸. It is likely that the assays of baseline gene expression from bulk tissues described above are not as relevant to disease status as are eQTLs inferred from single-cell expression data¹²⁹.

Pleiotropy and evolution

A major focus of evolutionary quantitative genetics is to understand the mechanisms by which genetic variation for quantitative traits is maintained, which depends on the relationship between the trait and fitness. Most quantitative traits are thought to be under stabilizing natural selection, whereby intermediate phenotypes have the highest fitness and extreme phenotypes are deleterious⁷. Quantitative traits that are fitness components are subject to directional natural selection for trait phenotypes associated with increased fitness, and purifying natural selection for trait phenotypes that decrease fitness. Neutral quantitative traits are those for which there is no association between trait values and fitness. It is exceedingly difficult to quantify fitness. However, both purifying and stabilizing selection cause a negative correlation between the effect size of variants associated with the trait and their minor allele frequency. This phenomenon is a hallmark of association studies in humans^{130,131} and model organisms⁶², suggesting that most quantitative traits are under natural selection and not neutral. Thus, there is pleiotropy between effects on quantitative traits and fitness.

Stabilizing selection, directional selection and purifying selection reduce genetic variation; therefore, models in which loss of genetic variation from selection is balanced by a gain of genetic variation from the input of new spontaneous mutations (mutation–selection balance (MSB)) have been explored. Early MSB models considered that stabilizing selection is directly on the trait of interest ('real' stabilizing selection)¹³². However, these models cannot be applied to many independent traits without incurring significant genetic load¹³³, and they assume many new mutations are beneficial. Models of 'apparent' stabilizing selection^{133,134} explicitly invoke pleiotropic effects of new mutations on quantitative traits and fitness. These models assume that all pleiotropic effects on other traits are subsumed into the fitness

are not enriched for eQTLs and neither eQTLs nor gene expression levels are associated with quantitative traits more than is expected by chance^{123,124}. This could be because sample sizes were small, because gene expression was profiled in whole animals and not relevant tissues or because linkage disequilibrium decays rapidly with physical distance

effect, that mutations have a bivariate distribution of effects on both the trait and fitness, and that trait effects can be positive or negative, but fitness effects are always deleterious. The appearance of stabilizing selection is generated because extreme phenotypes have more mutations. However, alleles maintained under an apparent stabilizing selection model have large effects on the quantitative trait and are only mildly deleterious. These early models made assumptions about numbers and allele frequencies of loci affecting quantitative traits and their relationship to fitness, which were unknown. A more recent model of MSB under stabilizing selection¹³⁵ explicitly accounts for the effective number of pleiotropic traits and shows that MSB can maintain genetic variation for quantitative traits for a range of minor allele frequencies.

Antagonistic pleiotropy¹² is a form of balancing selection that can maintain variation for life history traits. Antagonistic pleiotropy between variants with beneficial effects on fitness early in life but detrimental effects on fitness late in life will cause senescence because the force of natural selection declines with age^{12,136}. These arguments can be extended to variants that have antagonistic pleiotropy effects in males and females, or between different macro-environments. Antagonistic pleiotropy will thus constrain response to natural selection while maintaining alleles at intermediate frequencies. Antagonistic pleiotropy is frequently observed in model organisms. In *Escherichia coli*, mutations in the *hfq* gene have antagonistic pleiotropy effects on fitness under different growth rate conditions¹³⁷. In *S. cerevisiae*, 32 of 49 long-lived mutations tested had reduced fitness compared with the wild type¹³⁸, and competitive fitness assays between the yeast deletion collection mutations and the wild type under six growth conditions uncovered substantial antagonistic pleiotropy¹³⁹. The *age-1* allele in *C. elegans* confers an increased life span and resistance to pathogens and temperature stress relative to the wild-type allele, but has antagonistic pleiotropy effects on fertility, with reduced fertility early in life and increased fertility late in life relative to the wild-type allele¹⁴⁰. In *D. melanogaster*, P-element mutations show antagonistic pleiotropy across many traits (Fig. 2). P-element mutations associated with increased life span have deleterious effects on resistance to starvation stress⁵⁴, and mutations and genetic constructs with large effects on life span are often sterile¹². Further, P-element mutations often exhibit antagonistic pleiotropy between males and females^{41–45}. Antagonistic pleiotropy has been described for *D. melanogaster* natural polymorphisms associated with different quantitative traits^{24,141} and for the same variants evaluated in different macro-environments¹⁴¹. In humans, there is emerging evidence for antagonistic pleiotropy between alleles associated with protective effects on disease early in life and increased risk of disease late in life¹⁴², between alleles associated with early reproduction and late-age disease and between alleles that are protective for one disease but risk factors for another disease^{143,144}.

Pleiotropy is critical in models of adaptation. If pleiotropy is universal such that most polymorphisms will have some pleiotropic effects¹⁴⁵ or that all polymorphisms affect all traits (the infinitesimal model carried to the extreme), then pleiotropy will constrain evolution because it becomes likely that a polymorphism will have a deleterious effect on at least one trait. Pleiotropy introduces a ‘cost of complexity’, such that new mutations are less likely to be beneficial in complex organisms than simple organisms, thus impeding evolutionary advances^{146,147}.

Epistasis

Epistasis was first described in the context of crosses between individuals who are heterozygous for large-effect alleles at two loci affecting the same trait^{148,149}. In such crosses, a departure of the observed

phenotypic ratios from those expected from segregation of unlinked genotypes infers the presence of epistasis (Fig. 4a,b), which can mask the expected phenotype or result in a new phenotype. Epistasis between polymorphisms with large effects has been utilized to infer the order of genes in biochemical pathways and gene–gene interaction networks in developmental pathways. For quantitative traits, epistasis is generalized to refer to any non-additive interaction between two loci affecting the same trait¹ (that is, the effects of the two-locus genotypes cannot be predicted from the average effects of each locus separately) (Fig. 4 and Box 2). Epistasis causes context-specific effects, such that the additive effect of one locus depends on the genotype of the other locus. The effect of the interacting locus can change the magnitude and/or the direction of the additive effect of the focal locus on the trait¹⁵⁰ (Fig. 4c,d). Epistatic interactions can be ‘suppressing’, where the effect of the double homozygote genotype is less than expected given the effects of the two single homozygote genotypes (sometimes called ‘less than additive’); or ‘enhancing’, where the effect of the double homozygote genotype is greater than expected given the effects of the two single homozygote genotypes (sometimes called ‘more than additive’). Knowledge of epistatic interactions is critical for predicting quantitative trait phenotypes from individual genotypes with implications for precision medicine and agriculture^{16,151–154}.

Quantifying epistasis

Quantifying epistasis from induced mutations. Epistatic interactions for quantitative traits can be assessed directly using large collections of mutations or RNAi constructs and wild-type alleles at two (or more) loci that have been generated in a common homozygous genetic background; the only genetic variation is therefore at the loci of interest. Each test for digenic epistasis requires all nine two-locus genotypes (Fig. 5a) and estimates of quantitative trait phenotypes for each. A two-way analysis of variance is then performed to estimate the main effects of genotype 1, genotype 2 and the genotype 1 × genotype 2 interaction. A significant interaction term indicates the presence of epistasis. These data can then be used to estimate additive (*a*) and dominance (*d*) effects for each locus and the epistatic interaction (*i*) for each genotype (Fig. 5a). If only homozygous effects are considered, these analyses are performed for the four double-homozygous genotypes. Such studies were pioneered at a genome-wide scale in *S. cerevisiae*, where the generation of haploid double mutations can be automated and high throughput assays for growth rate, a proxy for fitness, are possible^{155–157}. As generating and evaluating all possible 11.5 million two-locus epistatic interactions ($\sim n^2$ pairs for *n* mutations) from the 4,800 viable haploid yeast deletions is experimentally intractable, these studies chose a set of query mutations belonging to a functional category of interest to screen the entire set of deletion mutations. Digenic epistatic interactions observed in yeast have scale-free interaction network topologies that tend to connect genes with similar biological functions. Screens for trigenic epistatic interactions in yeast found that these higher-order interactions occurred between functionally related as well as more distantly related functional categories, and that the trigenic interaction network was 100 times larger than the global digenic network¹⁵⁸. Global interaction networks with similar properties were found using cell-based RNAi screens in *C. elegans*^{159,160}, *D. melanogaster*¹⁶¹ and mammalian¹⁶² and human¹⁶³ cell lines.

Epistatic interactions are also observed in more focused screens to identify epistatic modifiers of known genes or processes. Examples include epistatic interactions among tubulin mutations affecting growth of neurites in *C. elegans*¹⁶⁴; epistatic modifiers of the

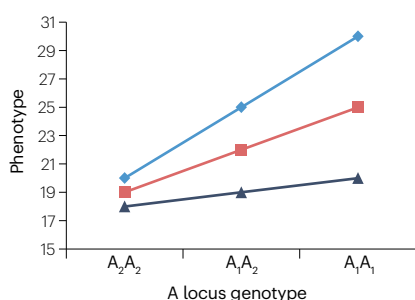
a No epistasis: 9 : 3 : 3 : 1 phenotypic ratio

	A ₁ A ₁ (1/4)	A ₁ A ₂ (2/4)	A ₂ A ₂ (1/4)
B ₁ B ₁ (1/4)	A ₁ A ₁ B ₁ B ₁ (1/16)	A ₁ A ₂ B ₁ B ₁ (2/16)	A ₂ A ₂ B ₁ B ₁ (1/16)
B ₁ B ₂ (2/4)	A ₁ A ₁ B ₁ B ₂ (2/16)	A ₁ A ₂ B ₁ B ₂ (4/16)	A ₂ A ₂ B ₁ B ₂ (2/16)
B ₂ B ₂ (1/4)	A ₁ A ₁ B ₂ B ₂ (1/16)	A ₁ A ₂ B ₂ B ₂ (2/16)	A ₂ A ₂ B ₂ B ₂ (1/16)

b Epistasis: 9 : 3 : 4 phenotypic ratio

	A ₁ A ₁ (1/4)	A ₁ A ₂ (2/4)	A ₂ A ₂ (1/4)
B ₁ B ₁ (1/4)	A ₁ A ₁ B ₁ B ₁ (1/16)	A ₁ A ₂ B ₁ B ₁ (2/16)	A ₂ A ₂ B ₁ B ₁ (1/16)
B ₁ B ₂ (2/4)	A ₁ A ₁ B ₁ B ₂ (2/16)	A ₁ A ₂ B ₁ B ₂ (4/16)	A ₂ A ₂ B ₁ B ₂ (2/16)
B ₂ B ₂ (1/4)	A ₁ A ₁ B ₂ B ₂ (1/16)	A ₁ A ₂ B ₂ B ₂ (2/16)	A ₂ A ₂ B ₂ B ₂ (1/16)

c Variance epistasis



d Sign epistasis

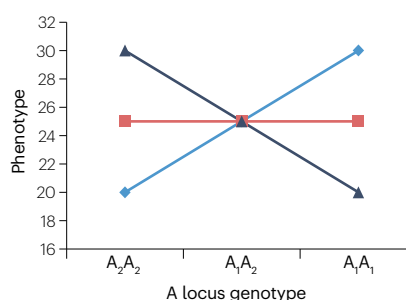


Fig. 4 | Epistasis for Mendelian and quantitative trait loci. **a**, Phenotypes and phenotypic ratios for two Mendelian loci (A and B), each with two alleles (A₁, A₂ and B₁, B₂), when the A₁ and B₁ alleles are dominant and there is no epistasis. The nine genotypes and their expected Mendelian segregation ratios from crossing two A₁A₂B₁B₂ individuals (9:3:3:1) are given in parentheses for each of the one-locus and two-locus genotypes. **b**, The same 9 two-locus genotypes when there is recessive epistasis. Rather than the expected ratio, the phenotypes are present in a 9:3:4 ratio for blue, red and black, respectively. The B₂B₂ genotype is epistatic to all genotypes at the A locus. **c,d**, Examples of epistasis for a quantitative trait.

In each panel, the phenotype of the quantitative trait (y axis) is plotted against the three genotypes at the A locus genotype (x axis), separately for the B₁B₁ (blue), B₁B₂ (red) and B₂B₂ (black) genotypes. Variance epistasis: the difference between the slope gradients shows that the additive effect at the A locus is greater in the B₁B₁ genetic background than in the B₂B₂ genetic background. There is no dominance as the mean phenotype of the A₁A₂ genotype is the same as the average of the A₁A₁ and A₂A₂ genotypes for each of the B locus genotypes (panel c). Sign epistasis: the difference in the direction of the slopes shows that the additive effect of the A locus changes sign between the B₁B₁ and B₂B₂ genetic backgrounds (panel d).

Apc^{Min} colorectal cancer model¹⁶⁵ and retinal dysplasia model¹⁶⁶ in mice; epistatic interactions among *D. melanogaster* P-element mutations affecting intermediary metabolism¹⁶⁷, aggressive¹⁶⁸ and locomotor⁴² behaviour, and life span⁵⁴; and epistasis between *A. thaliana* transcription factors affecting growth and defence¹⁶⁹. Applications of advanced gene editing methods reveal epistasis between mutations within single genes^{170–172}.

Quantifying epistasis from natural variation. Classical quantitative genetics theory^{1,2} enables us to make inferences about the fraction of the phenotypic variance for a quantitative trait in a population that is attributable to genetic variance and to environmental variance. In the absence of detailed knowledge of the additive (*a*), dominance (*d*) and epistatic (*i*) effects for each of *n* genotyped loci causing genetic variation for the trait in the population, the theory splits the genetic variance into additive genetic variance (σ_A^2), dominance genetic variance (σ_D^2) and epistatic (σ_I^2) genetic variance components, the magnitude of each of which depends on *a*, *d* and/or *i* and allele frequencies (Box 2). Notably, *a*, *d* and *i* effects do not translate directly to σ_A^2 , σ_D^2 and σ_I^2 variance components^{2,3,173–175}; classical quantitative genetics theory confounds effects and allele frequencies such that inferences about the former cannot be made from estimates of the latter (Box 2). Thus, the magnitude of epistatic effects (sometimes called physiological or

biological epistasis)^{176,177} cannot be inferred from epistatic variance component estimates (sometimes called statistical epistasis)^{176,177} — it is possible to have large epistatic effects but undetectable epistatic variance when minor allele frequencies are low^{173–175} (Fig. 5).

All variance components depend on allele frequencies at the loci affecting the trait and are usually greatest when alleles are at intermediate frequencies. Two populations with the same estimates of *a*, *d* and *i* for all loci affecting a quantitative trait will have different variance components (and mean trait values) when the frequency of the minor allele at all loci is low to when both alleles at all loci are at intermediate frequencies^{2,3,173–175} (Fig. 5). Furthermore, genetic variance components are estimated in random mating populations by regressions or correlations of trait phenotypes between common relatives: offspring and their parents, full and half-siblings, and, in humans, fraternal and identical twins^{2,3}. σ_A^2 has the largest contribution to genetic variance and components of σ_I^2 the smallest for all types of common relatives except monozygotic twins, for which all variance components contribute equally. There are more variance components contributing to the resemblance between relatives than can be estimated for any one type of common relatives; therefore, epistatic variance is generally assumed to be absent. Thus, in the presence of epistatic variance, estimates of σ_A^2 are biased upwards by σ_I^2 for all types of common relatives, and markedly so for monozygotic twins.

Epistatic effects can be directly estimated between QTLs in linkage mapping populations and have been found in model organisms and agricultural crops. One of the first examples of pervasive epistasis was from crosses among genotypes in which short genomic regions from one species of tomato were introgressed into the homozygous genetic background of another tomato species¹⁷⁸. Approximately 30% of the tested QTL interactions showed less than additive epistasis, a phenomenon that was replicated in a larger study with introgressions of different tomato species¹⁷⁹. The first experiments to map QTLs in *D. melanogaster* using molecular markers also showed epistasis for numbers of sensory bristles on different body parts^{180,181}, which was surprising because these were considered to be quintessentially additive traits in flies based on variance component estimates. There are many other examples of epistasis between QTLs for organismal traits in yeast^{69,182–186}, nematode worms¹⁸⁷, *D. melanogaster*^{188,189}, mice¹⁹⁰ and chickens^{191,192}. Epistasis is also common between QTLs affecting molecular quantitative traits in yeast¹⁹³ and *A. thaliana*^{194,195}. Incorporating QTL by QTL interactions improves phenotypic prediction in association mapping populations¹⁶. Epistasis between QTLs can be as large as the QTL main effects and can involve QTLs with no significant main effects. However, the studies described above used recombinant inbred lines derived from crosses of two parental strains, which maximizes the

potential to detect epistatic interactions because allele frequencies at all polymorphic loci are 0.5. In humans, epistasis has been reported between candidate genes affecting susceptibility to diabetes^{196,197} and inflammatory bowel disease¹⁹⁸.

Estimating epistatic effects in genome-wide association mapping studies is challenging. First, there are many more molecular polymorphisms to test for associations and pairwise interactions (in the order of 10^6 and 10^{12} , respectively, in humans). Therefore, accounting for multiple tests using a Bonferroni correction means an interaction would need to have a P value of $<5 \times 10^{-14}$ to be declared significant, so only very large epistatic effects can be detected. Second, epistasis cannot be evaluated if the minor allele frequency at both loci is low, because sufficient individuals who are homozygous for both rare alleles need to be present to completely specify the model. Third, the focal SNPs that tag a linkage disequilibrium block are not necessarily causal. The magnitude of the additive effect estimated by an associated SNP is smaller than that of the true SNP by an amount that depends on the linkage disequilibrium between the focal and true SNPs¹⁹⁹; and any epistatic effects are even more seriously underestimated. Screening for epistatic interactions is not usually performed in human GWAS because of the low statistical power to detect them and the strong emphasis on variance explained by associated SNPs, rather than

Box 2

Effects and variance components

For two loci (A and B) affecting the same quantitative trait, the effect of any two-locus genotype can be written as the sum of the additive (a_A and a_B) and dominance (d_A and d_B) effects of the constituent genotypes, and the epistatic effects (i) from any interaction between genotypes (see the table). These effects can be estimated from knowledge of the mean phenotype of a quantitative trait for individuals in the population of each of the nine causal two-locus genotypes.

Genotype (frequency)	B_1B_1 (p_B^2)	B_1B_2 ($2p_Bq_B$)	B_2B_2 (q_B^2)
A_1A_1 (p_A^2)	$a_A + a_B + i_{A_1A_1B_1B_1}$ ($p_A^2p_B^2$)	$a_A + d_B + i_{A_1A_1B_1B_2}$ ($p_A^22p_Bq_B$)	$a_A - a_B + i_{A_1A_1B_2B_2}$ ($p_A^2q_B^2$)
A_1A_2 ($2p_Aq_A$)	$d_A + a_B + i_{A_1A_2B_1B_1}$ ($2p_Aq_Ap_B^2$)	$d_A + d_B + i_{A_1A_2B_1B_2}$ ($2p_Aq_A2p_Bq_B$)	$d_A - a_B + i_{A_1A_2B_2B_2}$ ($2p_Aq_Aq_B^2$)
A_2A_2 (q_A^2)	$-a_A + a_B + i_{A_2A_2B_1B_1}$ ($q_A^2p_B^2$)	$-a_A + d_B + i_{A_2A_2B_1B_2}$ ($q_A^22p_Bq_B$)	$-a_A - a_B + i_{A_2A_2B_2B_2}$ ($q_A^2q_B^2$)

These effects cannot be directly estimated in a random mating population for which we do not have polymorphic molecular marker information to distinguish effects at each of the many loci affecting the trait. However, the total genetic variance (σ_G^2) can be partitioned into additive (σ_A^2), dominance (σ_D^2) and epistatic (σ_I^2) variance components. Any population variance is defined as the mean of the squared deviations from the mean. The population mean for two loci in a randomly mating population is obtained by multiplying the effects for each two-locus genotype by their allele frequency and summing over all genotypes. The population mean for two unlinked loci is $\mu_{GAB} = a_A(p_A - q_A) + 2d_Ap_Aq_A + a_B(p_B - q_B) + 2d_Bp_Bq_B + i_{\dots}$, where p_A and q_A are, respectively, the frequencies of alleles A_1 and A_2 at the

A locus, and p_B and q_B are, respectively, the frequencies of alleles B_1 and B_2 at the B locus, and $i_{\dots}$ is the population average epistatic interaction obtained by multiplying the frequency of each genotype by its corresponding epistatic effect and summing overall genotypes¹⁷⁶. The average effects of each locus (α_{A1} , α_{A2} , α_{B1} , α_{B2}) are computed in the same manner as the single-locus average effects (Box 1). For example, $\alpha_{A1} = q_A[a_A + d_A(q_A - p_A)] + (p_A i_{A_1A_1A_1\dots} + q_A i_{A_1A_1A_2\dots} - i_{\dots})$ and $\alpha_{A2} = -p_A[a_A + d_A(q_A - p_A)] + (p_A i_{A_1A_1A_2\dots} + q_A i_{A_2A_2A_2\dots} - i_{\dots})$ (ref. 176). The expressions for α_{B1} and α_{B2} take the same form. Additive variance (σ_A^2) is the variance of breeding values, where the breeding value for each genotype is the sum of the average effects of the alleles it carries. The breeding value for each of the nine two-locus genotypes includes the allele frequencies of each locus, values of a and d for each locus and the population average epistatic interaction. Therefore, σ_A^2 also includes these terms. The dominance variance (σ_D^2) includes allele frequencies and terms for d for each locus and average values of i , and the epistatic variance (σ_I^2) includes allele frequencies and genotype-specific values of i (ref. 176).

One consequence of defining genetic variance components in terms of deviations from the population mean is that additive, dominance and epistatic variance components are not the same as additive, dominance and epistatic effects. A second consequence is that variance components will vary in populations with different allele frequencies. Although we want to know the actual effects, which are independent of allele frequencies, the absence of molecular marker information requires that inferences about genetic variation are drawn from the variance caused by the alleles, thereby confounding effects and frequencies.

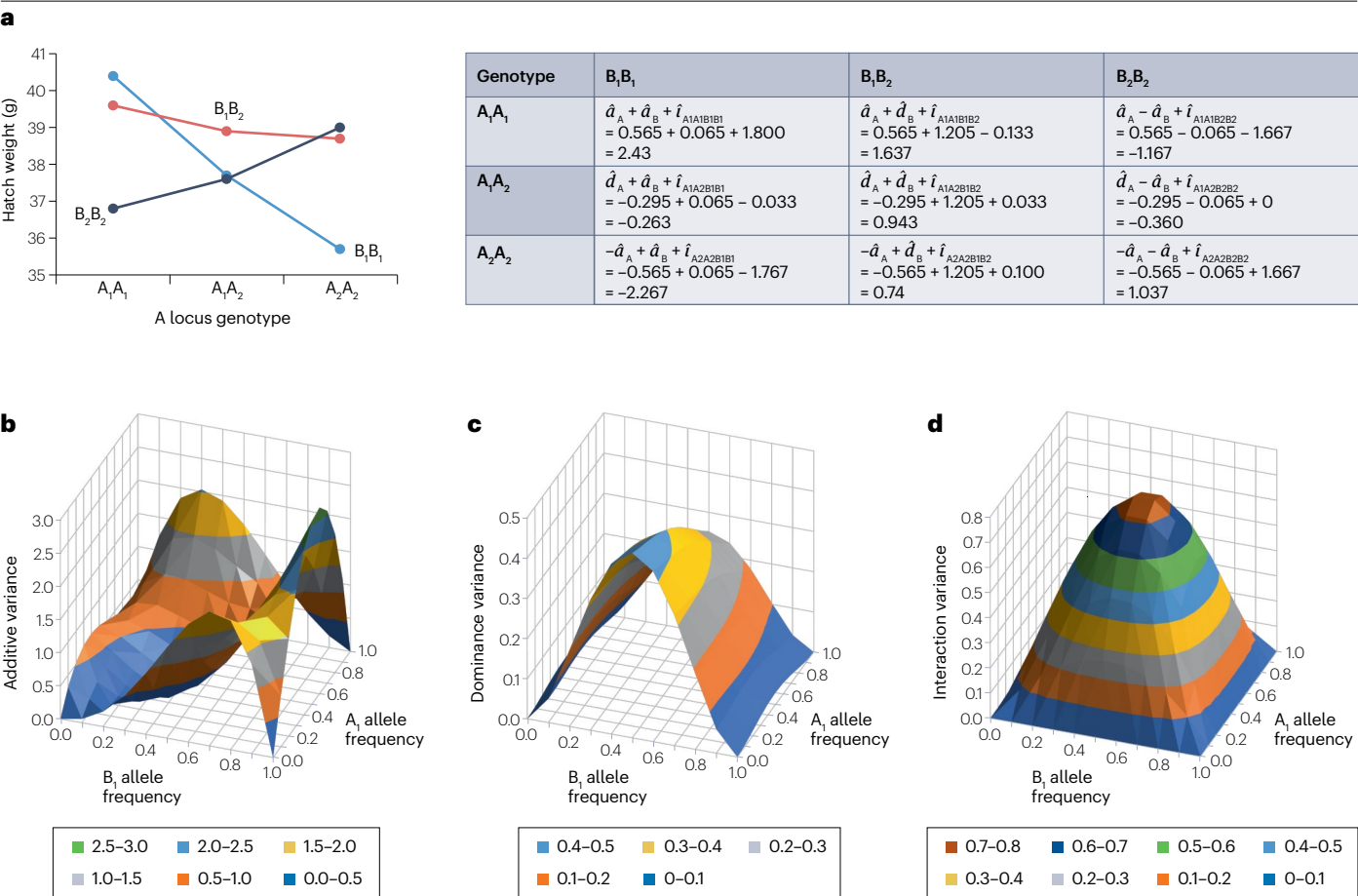


Fig. 5 | Epistatic effects and epistatic variance. An example of epistasis between quantitative trait loci (QTLs) for hatch weight of chickens in an F_2 population derived from a cross between the White Leghorn domestic chicken and the wild red jungle fowl¹⁹¹. **a**, Mean hatch weights for each of the 9 two-locus genotypes. Epistasis is evident from the changes in direction and magnitude of the effects of the A locus depending on the genotype of the B locus. A_1 , A_2 and B_1 , B_2 , are the alleles of loci A and B, respectively. The table shows estimates of the additive (a), dominance (d) and epistatic (i) effects and the total genotypic value for each genotype, using these means. Note that some dominance and epistatic effects

are much greater than the additive effects. **b–d**, Although some dominance and epistatic effects are larger than additive effects, the magnitude of the additive genetic variance (panel **b**) is greater than that of the dominance variance (panel **c**) or epistatic variance (panel **d**) for most allele frequencies. The effects (a , d , i) remain constant regardless of allele frequencies. However, the population variance components change as allele frequencies change, which is why low levels of epistatic variance do not mean epistasis can be ignored, and why no inferences about genetic architecture in terms of effects can be made from estimates of variance components.

estimating additive, dominance and epistatic effects as is done in model organisms. Using the population definition of effects means that the ‘additive’ effect of SNPs is confounded with dominance and epistatic effects, and that epistatic variance is minimized by definition and is negligible when minor allele frequencies are low^{174,175,200,201}. However, the goals of precision medicine and precision agriculture are to predict individual phenotypes from their own genotypes, which require estimates of effects, not variance contributed, of SNPs and pairs of SNPs.

Methods to prioritize pairs of SNPs to evaluate for epistatic interactions should reduce the multiple testing penalty. In the presence of epistasis, the additive effect of a focal polymorphism will change when the frequency of the interacting polymorphism changes^{14,202,203}. This suggests a way to test for epistasis either by deliberately perturbing allele frequencies between two association mapping populations or by taking advantage of random allele frequency differentiation among closely related populations. Large, trait-specific epistatic interaction

networks were identified in *D. melanogaster*^{202,203} using the former approach. Most epistatic interactions in these studies^{202,203} were such that the homozygotes for the two minor alleles were associated with the most extreme phenotype, with no phenotypic differences between the other two-locus genotypes. This suggests that screening for variance QTLs (that is, SNPs for which there is a difference in variance of the trait between the genotypes) (Fig. 6) may be a useful approach to identify SNPs that possibly interact with other SNPs in the genome, thus reducing the numbers of pairwise SNPs to test for association²⁰⁴. Screens for variance eQTLs affecting genome-wide gene expression have been successful in *D. melanogaster*¹²⁵ and humans^{205,206}, and variance eQTLs interacted epistatically with other eQTLs^{125,206}. Identifying pairs of SNPs with long-range linkage disequilibrium possibly due to epistatic selection²⁰⁷ is another strategy for reducing the number of tests for epistasis.

In contrast to the underestimation of epistatic effects in association mapping populations, epistasis is usually found when gene

editing is used to engineer naturally occurring SNPs into a common genetic background. In yeast, the fitness effects of more than 1,800 naturally polymorphic variants were assessed in 4 different genetic backgrounds²⁰⁸. A total of 24% of these SNPs only affected fitness in one genetic background, indicating epistasis. In human cells, individual and combinatorial editing of three genes showed strong epistatic interactions affecting cellular and transcriptomic phenotypes²⁰⁹.

Quantifying epistasis from natural variation and induced mutations.

Another approach to reduce the multiple testing penalty for detecting epistasis is to screen genome-wide for interactions between a focal SNP or mutant allele and all other polymorphisms. Most rare human diseases have variable phenotypic manifestations of disease severity, age of onset and progression^{210,211}. In addition, population sequencing shows that unaffected individuals can have genotypes that are known to cause rare diseases^{212,213}. Moreover, polygenic background affects disease risk for heterozygous carriers of rare monogenic disease alleles in natural populations^{214,215}. This variation in penetrance and expressivity is likely attributable to epistasis. Identifying genetic modifiers of rare human diseases is challenging due to low statistical power to formally test for interactions with small numbers of patients but is important as modifier alleles that suppress the disease phenotype may lead to new therapeutic targets.

Epistatic interactions between mutations and genetic background are also observed in model organisms. Seven yeast deletion mutations had variable fitness effects when crossed to genotypes from a QTL mapping population²¹⁶, and evaluation of the effects of deletion mutations in four yeast genetic backgrounds showed that ~19% had background-dependent fitness effects²¹⁷. Similarly, there is considerable evidence for variable phenotypic effects of the same mutation in different *D. melanogaster* genetic backgrounds^{15,218–220}.

The DGRP and other *D. melanogaster* inbred lines with full genome sequences can be used to map naturally occurring epistatic modifiers of mutations. This can be done by crossing genotypes with a wild-type or mutant allele for the focal polymorphism in otherwise

identical genetic backgrounds to the DGRP lines, phenotyping the F_1 offspring for a relevant quantitative trait, and performing a GWAS on the difference in the trait mean between the wild-type and mutant alleles^{221,222}. If the mutation or RNAi knockdown of the focal gene has an effect that lies well outside the range of natural variation, it is only necessary to cross the mutant allele to the DGRP lines and perform the GWAS analysis on the trait phenotype in the F_1 progeny^{223–226}. Genetic background effects are significant for all such studies published to date, reflecting the ubiquity of epistasis. If the focal mutation is in a *D. melanogaster* orthologue of a human gene associated with a rare disease, human orthologues of the *D. melanogaster* modifier genes can be tested for epistatic effects in humans.

Epistasis and evolution

There are many ways in which epistasis can contribute to evolution, including by maintaining genetic variation in natural populations, affecting long-term response to natural selection, reducing the burden of deleterious mutations and promoting speciation^{227,228}. Suppressing epistasis, whereby naturally occurring variants ameliorate mutational effects, is often observed in natural populations^{15,63,210–226}. Suppressing epistasis buffers the effects of new mutations and causes genetic canalization, whereby cryptic genetic variation¹⁵ persists in natural populations that can be revealed in the presence of a de-canalizing mutation^{15,229}. This gives the population the ability to evolve under circumstances in which the previously neutral genetic variation becomes adaptive or deleterious^{218,230–232}. Epistatic variance depends on allele frequencies, and natural selection changes allele frequencies. It is therefore possible that epistatic variance caused by alleles at intermediate frequency in an equilibrium population can become additive genetic variance as allele frequencies change in a population under selection, thereby accelerating the selection response in later generations^{233–236}. Epistatic variance for fitness is also the basis of Wright's rugged adaptive landscapes that are a central tenet of his shifting balance theory of evolution²³⁷. Epistasis is the foundation for Dobzhansky–Muller incompatibilities required for speciation^{238,239}, whereby alleles that are neutral or beneficial in

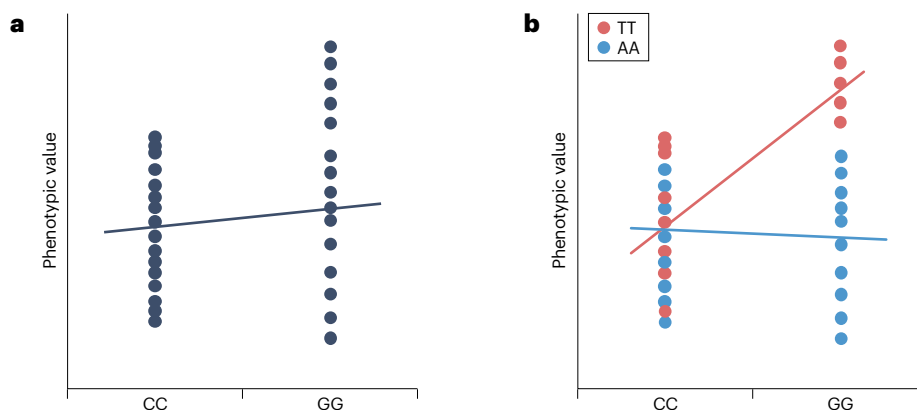


Fig. 6 | Variance quantitative trait loci and epistasis. a, Homozygous variants at a locus affecting the variance of a quantitative trait phenotype. Black line connects the mean phenotype of the two genotypes, which are not significantly different. However, the variance among individuals with the GG genotype is greater than the variance among individuals with the CC genotype; this can be detected by an analysis of variance heterogeneity. **b**, Homozygous variants at a second locus (indicated by blue and red circles). Blue line connects the means of the CCAA and GGAA genotypes, which are not significantly different. Red line

connects the means of the CCTT and GGTT genotypes, which are significantly different. The different slopes of these lines indicate epistasis. The TT and AA genotypes have the same mean phenotype in the CC genetic background, but different mean phenotypes in the GG genetic background. Performing association mapping analyses for heterogeneity of variance among genotypes at a focal locus can help to prioritize variants to include in analyses of two-locus epistasis, thus reducing the multiple testing penalty incurred by genome-wide epistasis screens.

Glossary

Additive effects

One half of the difference in mean phenotypes associated with homozygous genotypes at a biallelic locus affecting a quantitative trait.

Additive genetic variance

The variance of breeding values.

Deoxyribonuclease I (DNase I) hypersensitivity sites

Accessible nucleosome-free chromatin regions sensitive to cleavage by the DNase I enzyme; these regions are inferred to be involved in gene regulation.

Dominance effects

The difference between the mean phenotype of the heterozygous genotype and the mean phenotype of the two homozygous genotypes at a biallelic locus affecting a quantitative trait

Fitness

The contribution of offspring to the next generation, or reproductive success (viability and fertility) of an individual.

Genetic architecture

The loci, genomic locations and additive, dominance, epistatic and pleiotropic effects and allele frequencies of causal variants affecting a quantitative trait.

Haplotypes

Alleles at different loci on a chromosome that tend to be inherited together.

Heritability

The ratio of the variance of additive genetic variance (narrow sense heritability) or the total genetic variance (broad sense heritability) to the total phenotypic variance.

P-element

A *Drosophila* transposable element that can be mobilized to new genomic locations by simple genetic crosses.

Shifting balance theory of evolution

A model of evolution proposed by Sewall Wright that assumes widespread epistasis for fitness, natural selection, genetic drift and differentiation among subpopulations, and migration.

the genetic backgrounds of two genetically divergent populations are highly deleterious in the hybrids between the populations.

Pleiotropy and epistasis

Pleiotropy and epistasis are orthogonal concepts. However, genetically correlated gene expression networks and epistatic genetic interaction networks both have similar scale-free network architectures, raising the possibility that pleiotropy and epistasis could be functionally related. For example, if epistatic interactions between mutations affect one trait in a model organism, we can determine experimentally whether there is epistasis between the mutations for other traits that show pleiotropy with the focal trait (epistatic pleiotropy). Alternatively, if we know the pleiotropic effects of a single mutation on organismal phenotypes and genome-wide gene expression (or other genetically correlated molecular networks), we can query whether mutations in the genes encoding the gene expression traits exhibit epistasis for the organismal trait(s) (pleiotropic epistasis).

Studies in *D. melanogaster* show that there may be reciprocal relationships between pleiotropy and epistasis. Single P-element mutations affecting olfactory behaviour and fitness-related traits have pervasive pleiotropic effects on genome-wide gene expression, and mutations in genes encoding the pleiotropic gene expression traits do interact epistatically with the focal P-element mutations to affect the organismal traits^{53,240}. Furthermore, P-element mutations that exhibit

epistasis for locomotor⁴² or aggressive¹⁶⁸ behaviour also exhibit epistasis for brain morphology^{42,168} and genome-wide gene expression¹⁶⁸. The epistatic networks are distinct but partially overlapping for the different pleiotropic traits.

Combined Analysis of Pleiotropy and Epistasis (CAPE) is a formal method to infer and interpret epistatic interactions among mutations that utilizes the pleiotropic effects of mutations on multiple organismal traits and gene expression traits^{241,242}. Application of CAPE to epistatic mutations affecting yeast mating and pheromone response and genome-wide gene expression identified co-regulated transcriptional modules in common between traits and specific to each trait²⁴¹. Application of CAPE to RNAi single and double knockdown constructs for 93 *D. melanogaster* genes with pleiotropic effects on cell number, nuclear area and nuclear intensity identified enhancing and suppressing epistatic interactions that were consistent across the three phenotypes and that were not found by analysis of single phenotypes²⁴². CAPE has also been used to derive an interaction network of module QTLs derived from genetically correlated gene expression traits in mouse kidney affected by *trans*-eQTLs from an F₂ linkage mapping population²⁴³ and to infer epistatic interactions between QTLs in a multi-parental mapping population of mice affecting pleiotropic organismal metabolic traits and liver gene expression²⁴⁴.

Another way in which pleiotropy and epistasis can be functionally related is to identify QTLs for which the correlation between two traits varies between the genotypes of a marker locus^{245,246}. Such relationship QTLs (rQTLs) exhibit genetic variation in pleiotropy, possibly caused by epistasis between the rQTLs and other loci. Epistatic interactions with rQTLs have been observed for mouse morphological traits^{245,246} and in humans for metabolic traits and coronary heart disease associated with variation at the *APOE* locus²⁴⁷. Identification of rQTLs in human GWAS may increase the power to detect epistatic interactions.

Practical implications of pleiotropy and epistasis

Observations of pervasive pleiotropy emphasize the complexity of the genotype–phenotype map and highlight the importance of quantifying genetic variation for high-dimensional organismal phenotypes (phenomes) to fully understand pleiotropy²⁴⁸. Phenomes include molecular traits such as RNA abundance, chromatin state, protein abundance and isoforms, and metabolites; organismal-level traits quantifying aspects of physiology, morphology (including high-resolution imaging), behaviour and fitness-related traits; and individual microbiomes. Phenomes are not static, so quantification must account for variation between cells and across development in a range of environments. Plant²⁴⁹ and animal²⁵⁰ breeders view the future of crop and animal improvement by integrating increasingly complex phenotypes in breeding programmes. Crop and animal improvement programmes may also transfer favourable alleles from one genotype to another. This could have unintended effects if these alleles have deleterious pleiotropic effects on other phenotypes.

The full extent of pleiotropy is unknown in humans and will require systematic integration of multi-omics, lifestyle, biomedical imaging, clinical measurements and electronic health records on a large scale and over time²⁵¹. Pleiotropic relationships can implicate predictive biomarkers of disease, leading to better disease prevention and diagnosis and clinical tests. Shared genetic aetiology between diseases can indicate that both diseases may respond similarly to drug treatments. However, the converse is also possible if the correlation between diseases is negative: drugs may then have beneficial effects for one disease but detrimental effects for the other¹⁹.

A major goal of precision agriculture and precision medicine is accurate prediction of phenotypes from genotypes. Additive models will provide good prediction accuracy in most cases. For example, prediction accuracy of an additive model for human height is very high²⁵², and application of additive models in animal and plant breeding is responsible for improving production trait phenotypes in agricultural species. Therefore, the additive model dominates empirical breeding programmes and human complex trait genetics. However, epistatic effects are found in model organisms with the experimental tractability to detect genetic interactions when they are present. Epistasis is also thought to be the mechanism accounting for heterogeneous clinical symptoms of patients with the same mutation for rare human diseases. Knowledge of epistatic interactions can be important in agricultural breeding programmes designed to simultaneously introgress multiple loci with individually favourable effects into the same genetic background. These efforts will not have the desired outcome if the loci exhibit suppressing epistasis^{178,179}. Simulations show that using an additive prediction model when there is widespread epistasis will have low accuracy, whereas accounting for epistasis in the prediction model greatly increases accuracy²⁵³. To illustrate this point, consider two loci (A and B) each with two alleles (A_1, A_2 and B_1, B_2) and that only the $A_2A_2B_2B_2$ genotype causes a disease. If A_2 and B_2 are rare, the population average effects of these loci will be small, the genetic variance will be predominantly additive and the A and B loci will not predict disease liability. However, if the true effects (a, d, i) were used, the effect of the $A_2A_2B_2B_2$ genotype will be large and predictive ability will be excellent²⁵⁴.

Conclusions and perspectives

Pleiotropy and epistasis are fundamental features of the genetic architecture of quantitative traits. Pleiotropy is pervasive in model organisms and humans for organismal quantitative traits and highly dimensional molecular traits, such as genome-wide levels of gene expression traits, metabolites and proteins. The molecular traits cluster into distinct pleiotropic modules enriched for common functional annotations. The pleiotropic associations between mutations and naturally segregating molecular variants with these modules provide the molecular basis of pleiotropy. Deep phenotyping of the same molecular polymorphisms can reveal unexpected shared genetic bases of different common diseases and quantitative traits. Pleiotropic effects of new mutations on quantitative traits and fitness can explain the maintenance of quantitative genetic variation by MSB, and antagonistic pleiotropy between fitness components can constrain selection response and maintain genetic variation for fitness.

Epistasis is an important feature of the genetic architecture of quantitative traits in model organisms but there is little evidence for epistasis for complex traits in human populations. This could be attributable to the ability to estimate epistatic effects directly in model organisms and the difficulty in doing so in humans; and/or to the different parameterizations of epistasis used in humans (variance components) and model organisms (effects). The nature of epistatic interactions inferred from model organism studies suggests several future avenues to explore to increase the power to detect epistasis affecting human quantitative traits. These include screening for epistatic effects between variants in genes with similar biological functions associated with the trait; assessing heterogeneity of variance between SNP genotypes to identify focal variants for subsequent tests of epistatic interactions; and leveraging the functional relationship between pleiotropy and epistasis.

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Author contributions

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Opinion

Polygenic Adaptation: Integrating Population Genetics and Gene Regulatory Networks

Maud Fagny^{1,*} and Frédéric Austerlitz¹

The adaptation of populations to local environments often relies on the selection of optimal values for polygenic traits. Here, we first summarize the results obtained from different quantitative genetics and population genetics models, about the genetic architecture of polygenic traits and their response to directional selection. We then highlight the contribution of systems biology to the understanding of the molecular bases of polygenic traits and the evolution of gene regulatory networks involved in these traits. Finally, we discuss the need for a unifying framework merging the fields of population genetics, quantitative genetics and systems biology to better understand the molecular bases of polygenic traits adaptation.

Genetic Architecture of Adaptive Polygenic Traits

Many adaptive traits (i.e., traits involved in adaptation to local environments) are **polygenic traits** (see [Glossary](#)). Also called quantitative or complex traits, they are determined by multiple genes and regulatory loci. Population genetics studies have helped improve our understanding of the effects of **directional selection** on polygenic traits [1–4], which depends upon the trait's underlying genetic architecture. Since Fisher's seminal work on phenotypic traits [5], quantitative genetics researches have revealed a continuum of genetic architectures for traits in many organisms, including yeasts, insects, worms, plants, and mammals [6–8]. These architectures range from the Mendelian model, where a trait is determined by a single gene, to the infinitesimal model [9], where a seemingly infinite number of loci determine a trait. In between, many traits present an oligogenic or polygenic determinism: flowering time in plants, ethanol tolerance in yeasts, or lipid traits in humans [10–13]. Many of these traits are considered to be under directional selection [14–16].

Independently, systems biology studies initiated a switch in the perception of molecular bases of polygenic traits, from a gene-first to an interaction-first model [17]. Quantitative genetics represents the molecular bases of polygenic phenotypes as a collection of independent loci, each coding for a fraction of the phenotype, with potentially some interactions (1A). Conversely, systems biology focuses primarily on these interactions. In particular, **gene regulatory networks (GRNs)** represent the regulatory relationships between genes, their products, and their regulators (1C). Because they integrate and organize information from different levels (genomics, epigenomics, transcriptomics, proteomics, ...), they allow new information on the molecular bases of polygenic traits to be obtained [18]. These methods can now be applied to most species, as long as the required data are available (assembled genomes, gene expression data, and/or proteomic data).

A model describing the genetic architecture of polygenic traits has been recently proposed that integrates information from GRN and quantitative genetics: the omnigenic model [19,20] (Figure 1B). It describes the genetic contribution to a trait as a combination of direct effects

Highlights

Quantitative genetics models have highlighted the diversity of genetic architectures underlying polygenic traits. This diversity has an impact on how the traits respond to directional selection and on its molecular signatures on the genome.

Genome-wide scans for selection have revealed examples of polygenic selection driving local adaptation of populations in several species. This polygenic selection disproportionately targets regulatory regions, hinting for an important role of gene regulatory networks in evolution.

Gene regulatory network inference helps identifying and grouping together genes and regulatory elements that participate to the same biological processes. It also helps discovering how the structure of regulatory networks can put constraints on which genes and regulatory regions can be leveraged by polygenic selection.

The introduction of gene regulatory network information in the omnigenic model highlights the pervasive pleiotropy in the genome. The general interconnection between all genes within the regulatory network might strongly limit the action of selection.

¹UMR7206 Eco-Anthropologie, Muséum National d'Histoire Naturelle, Centre National de la Recherche Scientifique, Université de Paris, Paris, France

*Correspondence: maud.fagny@mnhn.fr (M. Fagny).



from core genes and indirect effects from peripheral genes acting in *trans*. This model shows how a modification in a *trans* gene can be amplified by the GRN, up to the point where *trans* genes explain a majority of trait variance. However, this model postulates that most of the genome participates in each trait, potentially severely limiting the action of directional selection on polygenic traits [19]. In the next sections, we will summarize the different models developed to explore the genetic and molecular bases of polygenic traits and their adaptation. We will discuss how they affect our perception of polygenic trait evolution and their advantages and limits. Finally, we will propose development of a new framework integrating their main aspects in order to improve our understanding of polygenic trait adaptation and better detect its signatures (see Figure 2).

What Do We Learn from Quantitative and Population Genetics Models?

Several population genetics models were developed to investigate polygenic trait selection and its molecular signatures at the genomic level. These models are generally additive: the effects of each allele on the phenotype are independent of each other, but dominance and **epistatic interactions** can also be considered. They showed that several selection regimes allow the phenotypic optimum to be reached [1], from selective sweeps, where mutations with relatively large effects increase rapidly in frequency, to the polygenic selection model, where each locus undergoes only small frequency changes, while allelic associations build up among loci.

Three main factors can affect the selection model: the proportion of variance explained by each locus, the mutation rate, and the shape of the selective function [21]. Models assuming an additive trait coded by many loci show that the occurrence of sweeps will depend on the number of loci involved in the trait: if low, each locus will respond more strongly in terms of allelic frequency changes [1]. Moreover, if, among these loci, some are major loci with a high contribution to genetic variance, they are more likely to be affected by strong allelic frequency shifts [2]. This result was confirmed by another approach modeling a locus coding for a given trait, while all other loci were treated as background genetic variance [3]. Finally, a more complex model including epistatic interactions showed that the probability of observing a sweep at a locus involved in a quantitative trait depends mainly on the population-scaled background mutation rate Θ_{bg} (i.e., the product of the effective size of the population by the total mutation rate of all background loci) [4]. Low Θ_{bg} yield selective sweeps at the focal locus, while high values yield small allele frequency shifts. Several partial selective sweeps will be observed for intermediate values. Similarly, another study showed that sweeps will occur only for low mutation rates of large-effect mutations [22].

Detecting Polygenic Selection Signatures: What Are We Missing?

Based on the theoretical results discussed earlier, several methods were developed for detecting polygenic selection (Box 1). These methods aim at detecting enrichment of positive selection signals among loci affecting the same biological function or trait. They use three categories of data to group mutations: gene functional annotations, regulatory interactions among genes, and genome-wide association studies (GWAS) results. The shortcomings of these approaches are extensively discussed in recent articles [23–26] including the technical caveats and issues with their interpretation. We wish here to point out limitations linked with the genetic architecture of these traits.

Shortcomings in polygenic selection tests interpretation can indeed be explained by the lack of knowledge of the ‘true’ genetic architecture of traits. In approaches based on functional annotations of genes or regulatory interactions, a considerable part of polygenic trait architecture is ignored, because these approaches often target genes, ignoring regulatory regions. Yet, genome-wide scans detect polygenic selection signatures in regulatory regions, through an

Glossary

Core-score: proportion of modularity explained by a node in a module.

Measures the importance of the node in the module structure, its local centrality.

Degree centrality: number of edges adjacent to a node.

Directional selection: a mode of selection that favors an extreme version of a trait over all others. In the case of a polygenic trait, it can lead to the increase in frequency of beneficial alleles at several independent loci.

Edges: in gene regulatory networks, edges represent relationships of activation or inhibition between nodes.

Epistatic interaction: interaction between two or more loci leading to nonadditive effects. In the case of two epistatic loci, the effect of a mutation at one of the loci depends on the presence or absence of mutations in the other locus.

Gene regulatory network (GRN): network representation that aims at representing the influence of transcription factors or other regulators over genes.

Modularity: measures the network division in modules or communities. A network has a high modularity when there is a high edge density (numerous connections) between nodes from the same module and sparse edges or connections between nodes from different modules.

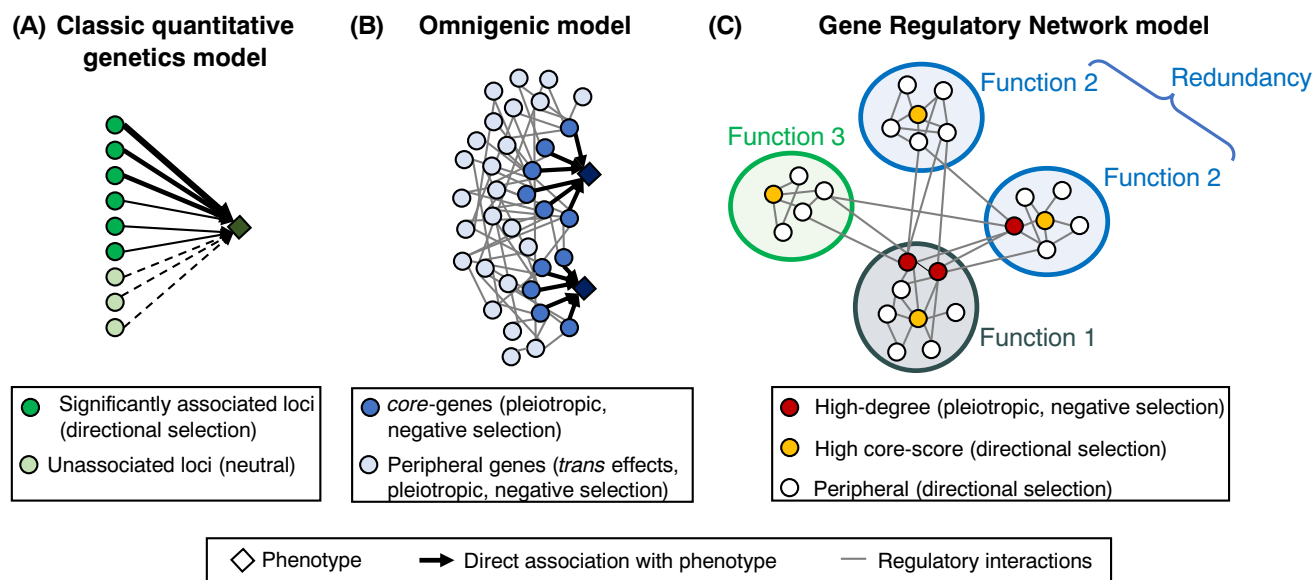
Nodes: in gene regulatory networks, nodes represent elements of the network, whether genes or proteins or other types of messengers.

Pleiotropic: property of a gene or genomic locus that determines several seemingly unrelated traits.

Polygenic trait: also called quantitative or complex trait, a polygenic trait is determined by several genes and/or regulatory loci. Different classes of polygenic traits are described, based on the number of loci underlying them: at one extremity of the spectrum, oligogenic traits are determined by a handful of genes and loci, while at the other extremity, virtually all the genome contributes to omnigenic traits.

Robustness: property of gene regulatory networks that makes them able to withstand mutations, without modifying the phenotype and the fitness of the individuals.

Stabilizing selection: a mode of selection that favors an intermediate version of a trait.



Trends in Genetics

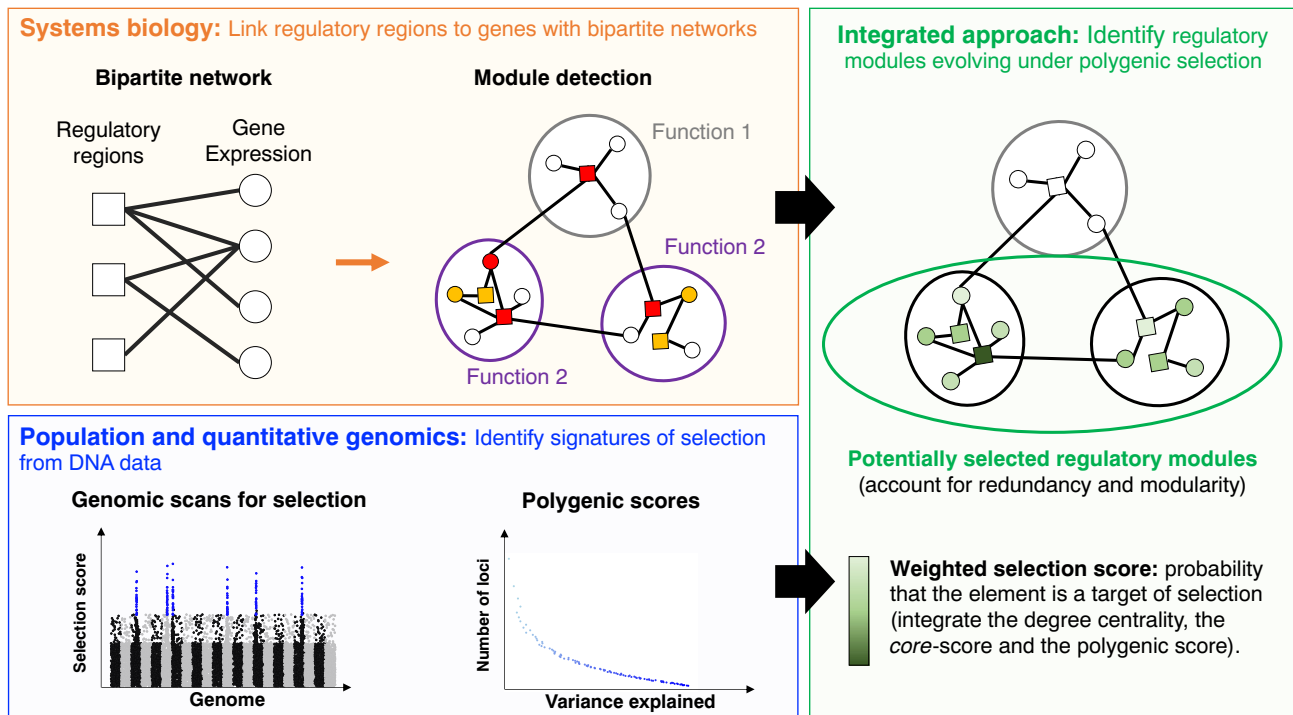
Figure 1. Different Models of Polygenic Trait Genetic Architecture and Their Consequence on Polygenic Selection. (A) Classic quantitative genetics model. Traits (diamond) are studied in isolation and genetic loci (circles) are associated with each trait using quantitative trait loci or genome-wide association studies approaches. Only loci that are significantly associated with the studied trait are potential selection targets (dark green), other loci are considered as evolving under neutrality (pale green). (B) The omnigenic model. The traits (diamonds) are explained by genes (circles) that can have a direct effect on the trait (*core*-genes in dark blue) or act in *trans* (peripheral genes in pale blue). All genes are highly pleiotropic, susceptible to affect all traits, and thus are probably evolving under strong negative selection. (C) Modular structure of a gene regulatory network. Traits are defined by biological functions. Each function corresponds to one or several regulatory modules (large ovals). Within each module, we find elements (genes, proteins, or regulatory elements represented as small circles) that can be high-degree pleiotropic elements that are likely to evolve under negative selection (red), high core-score elements that are likely to be major targets of directional selection (yellow), and peripheral elements that can also be targeted by directional selection (white).

enrichment in positive selection signals among *cis*-regulatory elements and expression quantitative trait loci (QTLs) [27,28]. GWAS-based approaches partially solve this issue by scanning the whole genome to find associations and thus include regulatory regions.

However, these approaches show several other limits. First, a good part of the genetic bases of traits is usually unknown, the famous ‘missing heritability’ [19], limiting our understanding of their evolution. Second, epistatic interactions are also often ignored, while they constitute a fundamental component of polygenic trait architecture, constraining their evolution [29]. Indeed, while overlooked epistatic interactions may not explain a lot of the missing heritability, they can either facilitate selection in the case of positive epistatic interactions or constrain it in the case of negative interactions, thus leading to complex effects of epistasis on the response to selection [30]. Third, potential interactions between traits, in particular through **pleiotropic** genes and regulatory loci, are often discarded, while they can strongly constrain evolutionary paths [31]. However, pleiotropy can also explain why an apparently deleterious trait might have been selected in the past (e.g., autoimmune diseases and response to pathogens in humans [32]). Recent publications highlight the importance of studying the adaptive landscape topography and the exceptional characteristics of GRNs in terms of epistatic interactions, pleiotropy, and **robustness** [33,34], to better understand the evolution of traits and detect signatures of polygenic adaptation [35].

Embracing the Complexity: Contributions from Systems Biology

From the systems biology point of view, understanding the molecular bases of polygenic selection therefore consists of deciphering how regulatory interactions evolve under constraints. The accumulation of data now allows inferring GRNs and testing hypotheses on their evolution. The



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Figure 2. Proposition for a New Integrated Framework to Detect Signatures of Polygenic Selection. The first panel (orange) describes the important features obtained from bipartite gene regulatory network (GRN) containing two types of elements: the regulatory elements or sequences (squares) and the genes (circles). The modularity structure is revealed by the white ovals outlined in colors (gray and purple), the redundancy is shown with two modules (in purple) corresponding to function 2. The high-degree centrality elements are in red and the high core-score elements are in yellow. The second panel (blue) describes important information obtained from population and quantitative genetics approaches, including locus-specific selection scores and polygenic scores. The third panel (green) describes our proposed model integrating information from the two first panels. We propose, for all elements in the regulatory module(s) that are relevant for the studied trait, to compute a weighted selection score that would include the degree centrality, the core-score, and the polygenic scores.

different scenarios of GRN evolution are summarized for animals [36] and plants [37] and many cases of selection on GRNs are described [28,38,39].

In parallel, GRN evolution models were developed [40], starting with matrix models simulating how each gene regulates each other and itself [41,42]. In these models, mutations affect the regulatory relationships and selection occurs through an optimal level of expression for each gene. The analysis of GRNs and their evolution, together with the simulations of GRN evolutionary models, led to hypotheses about how polygenic selection acts and how GRN structure might

Box 1. Methods to Detect Polygenic Selection

Methods to detect genomic signatures of polygenic selection events have mainly focused on combining signatures of positive selection across sets of mutations defined *a priori* as likely candidates for adaptive pressures. The first approach consists of investigating whether known pathways or gene sets are particularly enriched in selection signals [14], with many loci exhibiting a high level of genetic differentiation among populations. A recently developed related approach looks for local subnetworks under strong differentiation [58]. The second approach consists of testing whether a known trait has been submitted to divergent selection [59–61]. These methods are based on previously performed genome-wide association studies (GWAS) that detected loci associated with the traits and estimated their additive effect on these traits. This allows the polygenic score of the genotyped individuals in the studied populations to be computed; this can then be compared with its expectation under genetic drift only. Because they rely on already existing GWAS studies, these methods cannot be used without this *a priori* information on the genome.

facilitate or constrain it. Four GRN features are important in this context: robustness, pleiotropy, **modularity**, and their scale-free property (Figure 1 and Box 2).

GRN robustness emerges under **stabilizing selection**. In such a scenario, GRNs evolve toward a state where single mutations become less likely to reduce individual fitness [42]. Robustness is, however, not a constraint for polygenic selection, as it promotes the accumulation of hidden loss-of-function mutations, which can then accelerate adaptation in changing environments [43].

Robustness results in part from redundancy, that is, when a modification in a network **node** or **edge** does not impact gene expression because alternate nodes or edges mask its effect. Redundancy also facilitates polygenic adaptation: in an experimental evolution in *Drosophila*, phenotypic adaptation occurred in response to temperature change in ten populations. Phenotypic adaptation resulted from changes in different genetic pathways depending on the replicate [44].

Box 2. Gene Regulatory Networks (GRNs)

GRNs describe a collection of molecules that interact together to regulate the level of expression of genes, both as mRNA and proteins. These regulatory relationships can be direct or indirect: networks generated from transcription factor binding maps and gene expression level tend to reflect direct relationships, while regulatory networks generated from mRNA data only reflect indirect regulatory relationships, because directly regulate the expression of other genes.

GRNs are often represented as graphs, mathematical objects that represent the elements (nodes) and the interactions between elements (edges). They can include one type of element, such as proteins (unipartite graphs), or several types of element, such as transcription factors, regulatory mutations, and mRNAs (multipartite graphs). The study of the topology of these graphs allows better understanding of how the elements interact together and how GRNs evolve.

GRNs, whether in yeasts, insects, plants, or mammals, share some topological characteristics. Most importantly, they are scale-free. This means that few nodes (the key or master regulators) are connected to many others, while most nodes are connected only to few others (Figure 1). GRNs are also highly modular, that is, they are organized into groups of nodes that are strongly interconnected but only loosely connected to the rest of the network, thus constituting a module. The modules often correspond to biological pathways.

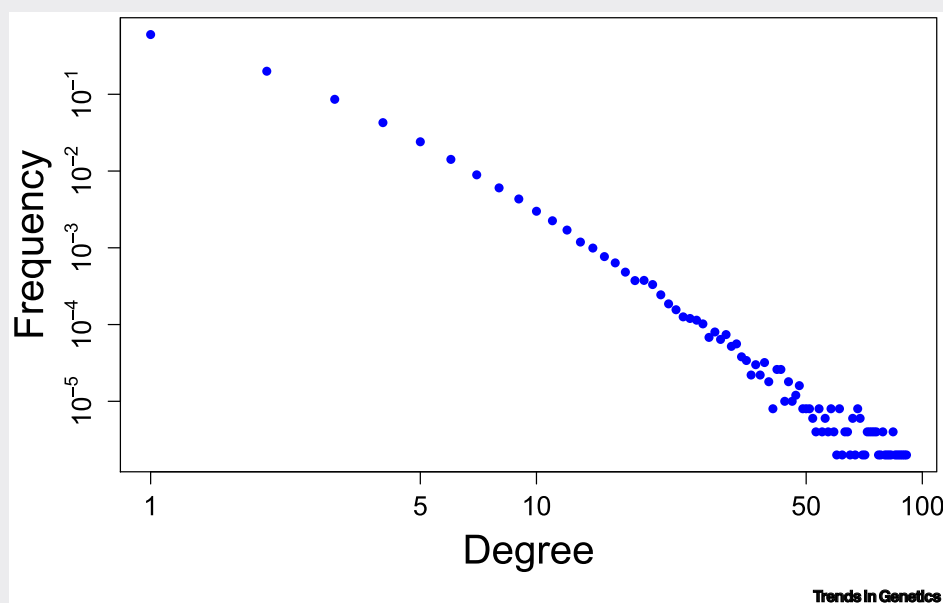


Figure 1. Expected Power Law Distribution of the Node Degree in a Gene Regulatory Network.

Redundancy thus facilitated polygenic selection by increasing the number of available targets (Figure 1B). Such selection will nevertheless be difficult to detect, precisely because it affects different pathways. Identifying cases of redundancy in GRNs and integrating them in polygenic tests, for example, by considering all alternative pathways coding for the same function, would thus help to retrieve currently missed selection signals.

Pleiotropy is another GRN feature that impacts the individual response of genes to selection. The omnigenic model proposes that each trait might be influenced by all genes [19]. Theoretically, such extreme generalized pleiotropy would severely constrain polygenic selection, because it could not modify a given trait without influencing all others. However, while this model seems adequate for some traits, such as human height, it might not be universal: for example, lipid traits such as cholesterol and triglyceride blood levels in humans [13] are best explained by a small number of loci.

Moreover, modularity can mitigate the effects of the omnigenic model on polygenic adaptation. Indeed, GRNs are highly modular, that is, made of small groups of genes strongly interconnected with each other but only loosely connected to the rest of the network, thus constituting a module. In each module, only a few genes are connected with genes from other modules. Most genes in a module are thus isolated from the rest of the network and can be leveraged by polygenic selection (Figure 1A).

Finally, GRNs tend to be scale-free, that is, most genes are connected to only few others, while few genes are connected to many others (Figure 1B), as shown on various types of networks [18,45]. In the context of highly modular scale-free networks, two measures are thus important: the **degree centrality**, which measures the centrality of a node in the whole network, and the **core-score**, which measures its centrality in its module. The degree centrality of a gene is inversely correlated with its level of genetic diversity [46,47], as confirmed by transcriptomics and proteomics data. The few highly pleiotropic nodes at the GRNs center thus evolve under strong negative selection, presenting limited variations in expression among individuals. They are not likely to be targeted by polygenic adaptation because their modification would affect numerous traits. Conversely, the vast majority of genes are at the periphery of the network, involved in one or few pathways. Changes in their regulation will, therefore, only slightly affect other pathways (Figure 1A,B). These genes include the high core-score nodes, which are central for a regulatory module but not for the whole GRN. Simulations show indeed that high core-score genes, which are more upstream in the modules, are more affected by selection [48]. This could explain the empirical observation that transcription factors evolve faster than other genes [49]. High core-score nodes could thus be preferential targets for polygenic adaptation. GRN structure offers the possibility for polygenic selection to act on many genes and gene regulatory elements, while limiting the potential side effects of selection on other pathways.

Polygenic Selection and GRNs: The Issue of *cis* -Regulatory Elements

One limitation of systems biology models is their focus on the evolution of networks made only of genes interacting with each other, thus overlooking important genomic features that would be useful to get a complete picture of how polygenic selection can impact GRNs. Indeed, most GRN models do not explicitly include *cis*-regulatory regions, such as promoters and enhancers, which can impact the expression level of several genes at once. The major role of these regions in the adaptation of complex traits has nevertheless been demonstrated. QTL and GWAS show indeed that up to 90% of variants associated with polygenic traits occur in noncoding regions [50,51]. There are evidences for polygenic selection targeting regulatory mutations [39,52], and mutations in transcription factor binding sites play a crucial role in species evolution [53].

However, integrating the effect of *cis*-regulatory sequences in polygenic adaptation models might be difficult, because the actual role of regulatory mutations on gene expression is hard to evaluate. It depends upon many factors, including chromatin activation state at the regulatory locus and availability of transcription factors in the cell [54]. Integrating such information might be easier in the future through massively parallel reporter assays, which allow investigation of the effects of regulatory regions on gene expression at the genome-wide level and assessment of their effect on fitness [55], in parallel with the development of machine-learning algorithms predicting the regulatory effect of mutations and their potential evolutionary role [56]. Such detailed data are available for some model species and should soon be available much more broadly thanks to decreasing sequencing costs. Finally, some current approaches integrate heterogeneous data using multipartite networks based on genomic and gene expression data, in order to identify not only gene–gene regulatory interactions but also interactions between regulatory regions and genes [18,57]. These approaches could allow development of new GRN evolution models directly integrating *cis*-regulatory regions.

Concluding Remarks and Future Perspectives

In order to overcome the limitations of current models of polygenic trait evolution, a new integrative evolutionary framework becomes indispensable. This new framework would take the GRN architecture into account (Figure 2). Here, we argue that, in order to achieve such a unifying framework, we must identify which genes and pathways contribute to polygenic traits, and how redundant they are, but also investigate the complex regulatory interactions that govern their expression (see Outstanding Questions and Figure 2). These features appear crucial not only to identify all loci involved in the genetic architecture of a trait, but also to assess whether these loci can realistically be targeted by polygenic selection. We propose, as done in the omnigenic model [20] to use the regulatory information given by GRNs to identify the relevant genes. However, we suggest *cis*-regulatory regions in the GRNs are explicitly modeled, as this allows information about the biological role of these regions to be retrieved [57]. In addition, we suggest that particular GRN features are investigated, focusing on redundancy and partition in regulatory modules. Finally, we suggest that the position of the different regulatory actors in the GRN are integrated in order to account for their probability of being targeted by polygenic selection.

Declaration of Interests

No interests are declared.

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Outstanding Questions

Redundancy allows for the selection of the same version of a trait through the selection of genetic variants in different pathways. Do we observe this phenomenon at the scale of the population? If so, which molecular signatures do such evolutionary processes leave on the genome and how can we detect them?

Pleiotropy is a major property of GRN actors that constrains how polygenic selection acts. Genes are often involved in several distinct pathways. What is the level of pleiotropy of the regulatory elements, in particular of the distal ones such as enhancers, and how does it constrain their evolution?

Classical population genetics approaches model the evolution of genetic sequences but ignore their complex interactions in GRNs. However, GRNs model the evolution of these regulatory relationships but not their genetic bases. Will more integrated approaches that explicitly model the molecular evolution of regulatory regions in the context of GRNs improve our understanding of the molecular bases of local adaptation?

Selection on polygenic traits yields linkage disequilibrium among genes involved in the traits, at the within- and among-population levels [1]. How do regulatory interactions affect this linkage disequilibrium? Can it be used to detect polygenic selection?

GWAS-based methods aimed at detecting polygenic selection use the additive effect of each gene on the trait. Could the interaction (dominance, epistasis) between these genes also be integrated in these approaches?

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Opinion

Polygenic Adaptation: Integrating Population Genetics and Gene Regulatory Networks

Maud Fagny^{1,*} and Frédéric Austerlitz¹

The adaptation of populations to local environments often relies on the selection of optimal values for polygenic traits. Here, we first summarize the results obtained from different quantitative genetics and population genetics models, about the genetic architecture of polygenic traits and their response to directional selection. We then highlight the contribution of systems biology to the understanding of the molecular bases of polygenic traits and the evolution of gene regulatory networks involved in these traits. Finally, we discuss the need for a unifying framework merging the fields of population genetics, quantitative genetics and systems biology to better understand the molecular bases of polygenic traits adaptation.

Genetic Architecture of Adaptive Polygenic Traits

Many adaptive traits (i.e., traits involved in adaptation to local environments) are **polygenic traits** (see [Glossary](#)). Also called quantitative or complex traits, they are determined by multiple genes and regulatory loci. Population genetics studies have helped improve our understanding of the effects of **directional selection** on polygenic traits [1–4], which depends upon the trait's underlying genetic architecture. Since Fisher's seminal work on phenotypic traits [5], quantitative genetics researches have revealed a continuum of genetic architectures for traits in many organisms, including yeasts, insects, worms, plants, and mammals [6–8]. These architectures range from the Mendelian model, where a trait is determined by a single gene, to the infinitesimal model [9], where a seemingly infinite number of loci determine a trait. In between, many traits present an oligogenic or polygenic determinism: flowering time in plants, ethanol tolerance in yeasts, or lipid traits in humans [10–13]. Many of these traits are considered to be under directional selection [14–16].

Independently, systems biology studies initiated a switch in the perception of molecular bases of polygenic traits, from a gene-first to an interaction-first model [17]. Quantitative genetics represents the molecular bases of polygenic phenotypes as a collection of independent loci, each coding for a fraction of the phenotype, with potentially some interactions (1A). Conversely, systems biology focuses primarily on these interactions. In particular, **gene regulatory networks (GRNs)** represent the regulatory relationships between genes, their products, and their regulators (1C). Because they integrate and organize information from different levels (genomics, epigenomics, transcriptomics, proteomics, ...), they allow new information on the molecular bases of polygenic traits to be obtained [18]. These methods can now be applied to most species, as long as the required data are available (assembled genomes, gene expression data, and/or proteomic data).

A model describing the genetic architecture of polygenic traits has been recently proposed that integrates information from GRN and quantitative genetics: the omnigenic model [19,20] (Figure 1B). It describes the genetic contribution to a trait as a combination of direct effects

Highlights

Quantitative genetics models have highlighted the diversity of genetic architectures underlying polygenic traits. This diversity has an impact on how the traits respond to directional selection and on its molecular signatures on the genome.

Genome-wide scans for selection have revealed examples of polygenic selection driving local adaptation of populations in several species. This polygenic selection disproportionately targets regulatory regions, hinting for an important role of gene regulatory networks in evolution.

Gene regulatory network inference helps identifying and grouping together genes and regulatory elements that participate to the same biological processes. It also helps discovering how the structure of regulatory networks can put constraints on which genes and regulatory regions can be leveraged by polygenic selection.

The introduction of gene regulatory network information in the omnigenic model highlights the pervasive pleiotropy in the genome. The general interconnection between all genes within the regulatory network might strongly limit the action of selection.

¹UMR7206 Eco-Anthropologie, Muséum National d'Histoire Naturelle, Centre National de la Recherche Scientifique, Université de Paris, Paris, France

*Correspondence: maud.fagny@mnhn.fr (M. Fagny).



from core genes and indirect effects from peripheral genes acting in *trans*. This model shows how a modification in a *trans* gene can be amplified by the GRN, up to the point where *trans* genes explain a majority of trait variance. However, this model postulates that most of the genome participates in each trait, potentially severely limiting the action of directional selection on polygenic traits [19]. In the next sections, we will summarize the different models developed to explore the genetic and molecular bases of polygenic traits and their adaptation. We will discuss how they affect our perception of polygenic trait evolution and their advantages and limits. Finally, we will propose development of a new framework integrating their main aspects in order to improve our understanding of polygenic trait adaptation and better detect its signatures (see Figure 2).

What Do We Learn from Quantitative and Population Genetics Models?

Several population genetics models were developed to investigate polygenic trait selection and its molecular signatures at the genomic level. These models are generally additive: the effects of each allele on the phenotype are independent of each other, but dominance and **epistatic interactions** can also be considered. They showed that several selection regimes allow the phenotypic optimum to be reached [1], from selective sweeps, where mutations with relatively large effects increase rapidly in frequency, to the polygenic selection model, where each locus undergoes only small frequency changes, while allelic associations build up among loci.

Three main factors can affect the selection model: the proportion of variance explained by each locus, the mutation rate, and the shape of the selective function [21]. Models assuming an additive trait coded by many loci show that the occurrence of sweeps will depend on the number of loci involved in the trait: if low, each locus will respond more strongly in terms of allelic frequency changes [1]. Moreover, if, among these loci, some are major loci with a high contribution to genetic variance, they are more likely to be affected by strong allelic frequency shifts [2]. This result was confirmed by another approach modeling a locus coding for a given trait, while all other loci were treated as background genetic variance [3]. Finally, a more complex model including epistatic interactions showed that the probability of observing a sweep at a locus involved in a quantitative trait depends mainly on the population-scaled background mutation rate Θ_{bg} (i.e., the product of the effective size of the population by the total mutation rate of all background loci) [4]. Low Θ_{bg} yield selective sweeps at the focal locus, while high values yield small allele frequency shifts. Several partial selective sweeps will be observed for intermediate values. Similarly, another study showed that sweeps will occur only for low mutation rates of large-effect mutations [22].

Detecting Polygenic Selection Signatures: What Are We Missing?

Based on the theoretical results discussed earlier, several methods were developed for detecting polygenic selection (Box 1). These methods aim at detecting enrichment of positive selection signals among loci affecting the same biological function or trait. They use three categories of data to group mutations: gene functional annotations, regulatory interactions among genes, and genome-wide association studies (GWAS) results. The shortcomings of these approaches are extensively discussed in recent articles [23–26] including the technical caveats and issues with their interpretation. We wish here to point out limitations linked with the genetic architecture of these traits.

Shortcomings in polygenic selection tests interpretation can indeed be explained by the lack of knowledge of the ‘true’ genetic architecture of traits. In approaches based on functional annotations of genes or regulatory interactions, a considerable part of polygenic trait architecture is ignored, because these approaches often target genes, ignoring regulatory regions. Yet, genome-wide scans detect polygenic selection signatures in regulatory regions, through an

Glossary

Core-score: proportion of modularity explained by a node in a module.

Measures the importance of the node in the module structure, its local centrality.

Degree centrality: number of edges adjacent to a node.

Directional selection: a mode of selection that favors an extreme version of a trait over all others. In the case of a polygenic trait, it can lead to the increase in frequency of beneficial alleles at several independent loci.

Edges: in gene regulatory networks, edges represent relationships of activation or inhibition between nodes.

Epistatic interaction: interaction between two or more loci leading to nonadditive effects. In the case of two epistatic loci, the effect of a mutation at one of the loci depends on the presence or absence of mutations in the other locus.

Gene regulatory network (GRN): network representation that aims at representing the influence of transcription factors or other regulators over genes.

Modularity: measures the network division in modules or communities. A network has a high modularity when there is a high edge density (numerous connections) between nodes from the same module and sparse edges or connections between nodes from different modules.

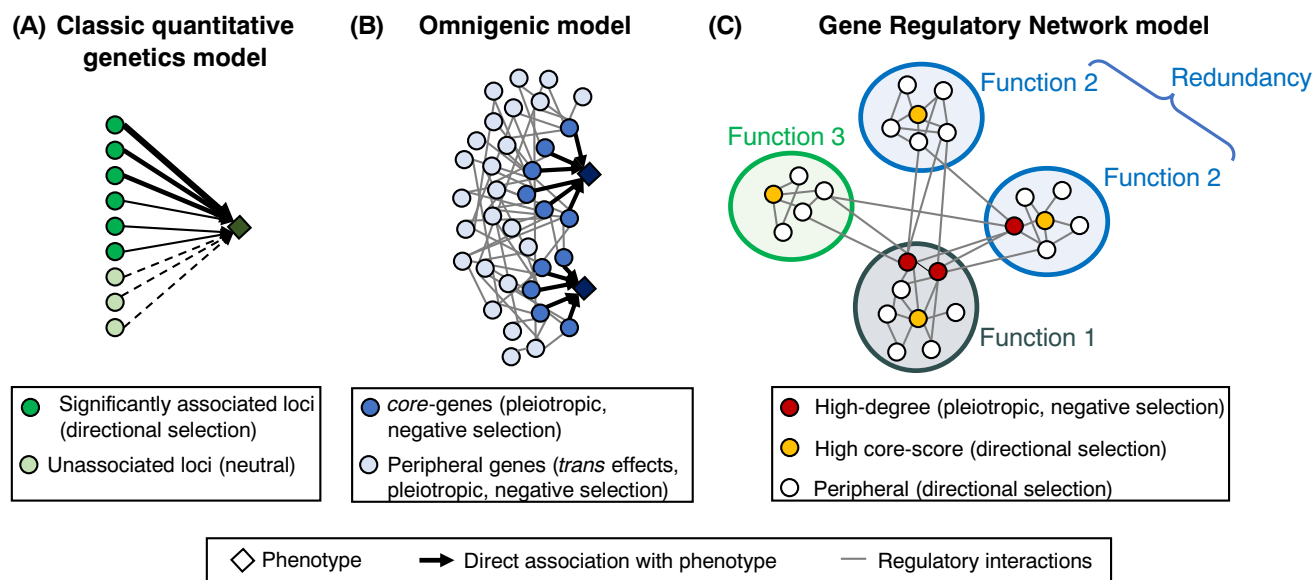
Nodes: in gene regulatory networks, nodes represent elements of the network, whether genes or proteins or other types of messengers.

Pleiotropic: property of a gene or genomic locus that determines several seemingly unrelated traits.

Polygenic trait: also called quantitative or complex trait, a polygenic trait is determined by several genes and/or regulatory loci. Different classes of polygenic traits are described, based on the number of loci underlying them: at one extremity of the spectrum, oligogenic traits are determined by a handful of genes and loci, while at the other extremity, virtually all the genome contributes to omnigenic traits.

Robustness: property of gene regulatory networks that makes them able to withstand mutations, without modifying the phenotype and the fitness of the individuals.

Stabilizing selection: a mode of selection that favors an intermediate version of a trait.



Trends in Genetics

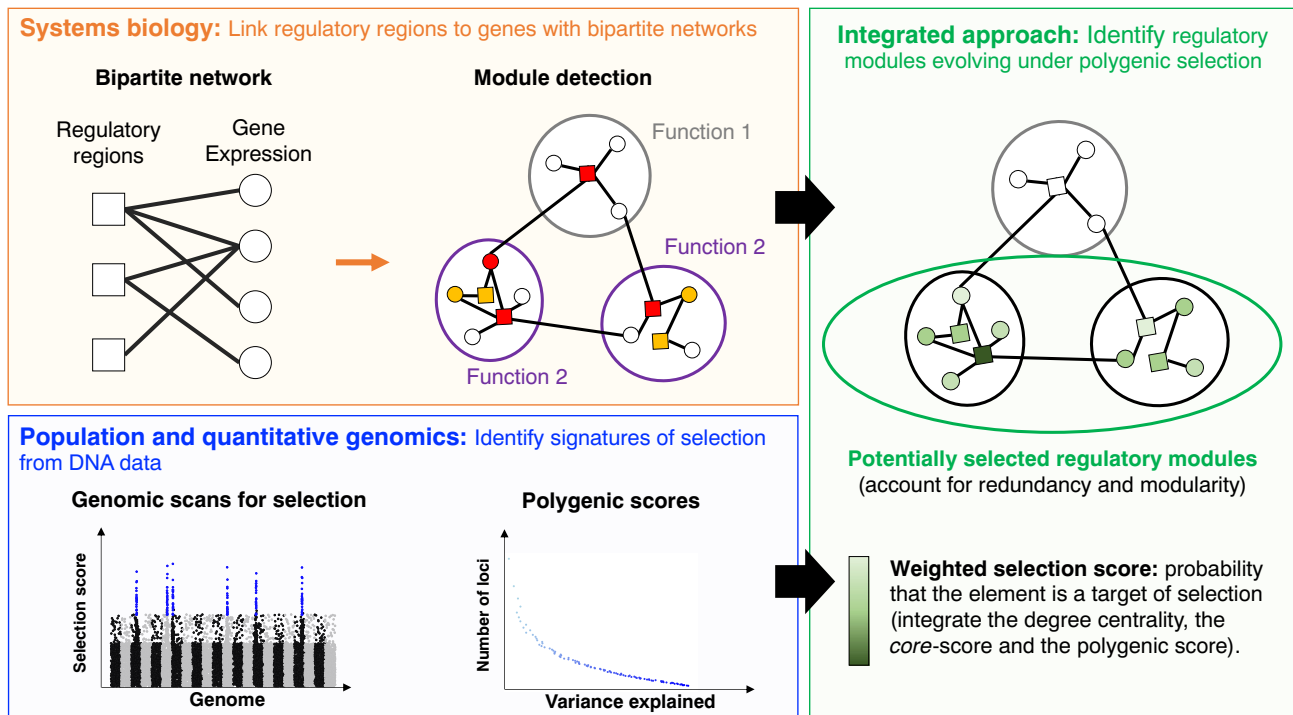
Figure 1. Different Models of Polygenic Trait Genetic Architecture and Their Consequence on Polygenic Selection. (A) Classic quantitative genetics model. Traits (diamond) are studied in isolation and genetic loci (circles) are associated with each trait using quantitative trait loci or genome-wide association studies approaches. Only loci that are significantly associated with the studied trait are potential selection targets (dark green), other loci are considered as evolving under neutrality (pale green). (B) The omnigenic model. The traits (diamonds) are explained by genes (circles) that can have a direct effect on the trait (core-genes in dark blue) or act in *trans* (peripheral genes in pale blue). All genes are highly pleiotropic, susceptible to affect all traits, and thus are probably evolving under strong negative selection. (C) Modular structure of a gene regulatory network. Traits are defined by biological functions. Each function corresponds to one or several regulatory modules (large ovals). Within each module, we find elements (genes, proteins, or regulatory elements represented as small circles) that can be high-degree pleiotropic elements that are likely to evolve under negative selection (red), high core-score elements that are likely to be major targets of directional selection (yellow), and peripheral elements that can also be targeted by directional selection (white).

enrichment in positive selection signals among *cis*-regulatory elements and expression quantitative trait loci (QTLs) [27,28]. GWAS-based approaches partially solve this issue by scanning the whole genome to find associations and thus include regulatory regions.

However, these approaches show several other limits. First, a good part of the genetic bases of traits is usually unknown, the famous ‘missing heritability’ [19], limiting our understanding of their evolution. Second, epistatic interactions are also often ignored, while they constitute a fundamental component of polygenic trait architecture, constraining their evolution [29]. Indeed, while overlooked epistatic interactions may not explain a lot of the missing heritability, they can either facilitate selection in the case of positive epistatic interactions or constrain it in the case of negative interactions, thus leading to complex effects of epistasis on the response to selection [30]. Third, potential interactions between traits, in particular through **pleiotropic** genes and regulatory loci, are often discarded, while they can strongly constrain evolutionary paths [31]. However, pleiotropy can also explain why an apparently deleterious trait might have been selected in the past (e.g., autoimmune diseases and response to pathogens in humans [32]). Recent publications highlight the importance of studying the adaptive landscape topography and the exceptional characteristics of GRNs in terms of epistatic interactions, pleiotropy, and **robustness** [33,34], to better understand the evolution of traits and detect signatures of polygenic adaptation [35].

Embracing the Complexity: Contributions from Systems Biology

From the systems biology point of view, understanding the molecular bases of polygenic selection therefore consists of deciphering how regulatory interactions evolve under constraints. The accumulation of data now allows inferring GRNs and testing hypotheses on their evolution. The



Trends in Genetics

Figure 2. Proposition for a New Integrated Framework to Detect Signatures of Polygenic Selection. The first panel (orange) describes the important features obtained from bipartite gene regulatory network (GRN) containing two types of elements: the regulatory elements or sequences (squares) and the genes (circles). The modularity structure is revealed by the white ovals outlined in colors (gray and purple), the redundancy is shown with two modules (in purple) corresponding to function 2. The high-degree centrality elements are in red and the high core-score elements are in yellow. The second panel (blue) describes important information obtained from population and quantitative genetics approaches, including locus-specific selection scores and polygenic scores. The third panel (green) describes our proposed model integrating information from the two first panels. We propose, for all elements in the regulatory module(s) that are relevant for the studied trait, to compute a weighted selection score that would include the degree centrality, the core-score, and the polygenic scores.

different scenarios of GRN evolution are summarized for animals [36] and plants [37] and many cases of selection on GRNs are described [28,38,39].

In parallel, GRN evolution models were developed [40], starting with matrix models simulating how each gene regulates each other and itself [41,42]. In these models, mutations affect the regulatory relationships and selection occurs through an optimal level of expression for each gene. The analysis of GRNs and their evolution, together with the simulations of GRN evolutionary models, led to hypotheses about how polygenic selection acts and how GRN structure might

Box 1. Methods to Detect Polygenic Selection

Methods to detect genomic signatures of polygenic selection events have mainly focused on combining signatures of positive selection across sets of mutations defined *a priori* as likely candidates for adaptive pressures. The first approach consists of investigating whether known pathways or gene sets are particularly enriched in selection signals [14], with many loci exhibiting a high level of genetic differentiation among populations. A recently developed related approach looks for local subnetworks under strong differentiation [58]. The second approach consists of testing whether a known trait has been submitted to divergent selection [59–61]. These methods are based on previously performed genome-wide association studies (GWAS) that detected loci associated with the traits and estimated their additive effect on these traits. This allows the polygenic score of the genotyped individuals in the studied populations to be computed; this can then be compared with its expectation under genetic drift only. Because they rely on already existing GWAS studies, these methods cannot be used without this *a priori* information on the genome.

facilitate or constrain it. Four GRN features are important in this context: robustness, pleiotropy, **modularity**, and their scale-free property (Figure 1 and Box 2).

GRN robustness emerges under **stabilizing selection**. In such a scenario, GRNs evolve toward a state where single mutations become less likely to reduce individual fitness [42]. Robustness is, however, not a constraint for polygenic selection, as it promotes the accumulation of hidden loss-of-function mutations, which can then accelerate adaptation in changing environments [43].

Robustness results in part from redundancy, that is, when a modification in a network **node** or **edge** does not impact gene expression because alternate nodes or edges mask its effect. Redundancy also facilitates polygenic adaptation: in an experimental evolution in *Drosophila*, phenotypic adaptation occurred in response to temperature change in ten populations. Phenotypic adaptation resulted from changes in different genetic pathways depending on the replicate [44].

Box 2. Gene Regulatory Networks (GRNs)

GRNs describe a collection of molecules that interact together to regulate the level of expression of genes, both as mRNA and proteins. These regulatory relationships can be direct or indirect: networks generated from transcription factor binding maps and gene expression level tend to reflect direct relationships, while regulatory networks generated from mRNA data only reflect indirect regulatory relationships, because directly regulate the expression of other genes.

GRNs are often represented as graphs, mathematical objects that represent the elements (nodes) and the interactions between elements (edges). They can include one type of element, such as proteins (unipartite graphs), or several types of element, such as transcription factors, regulatory mutations, and mRNAs (multipartite graphs). The study of the topology of these graphs allows better understanding of how the elements interact together and how GRNs evolve.

GRNs, whether in yeasts, insects, plants, or mammals, share some topological characteristics. Most importantly, they are scale-free. This means that few nodes (the key or master regulators) are connected to many others, while most nodes are connected only to few others (Figure 1). GRNs are also highly modular, that is, they are organized into groups of nodes that are strongly interconnected but only loosely connected to the rest of the network, thus constituting a module. The modules often correspond to biological pathways.

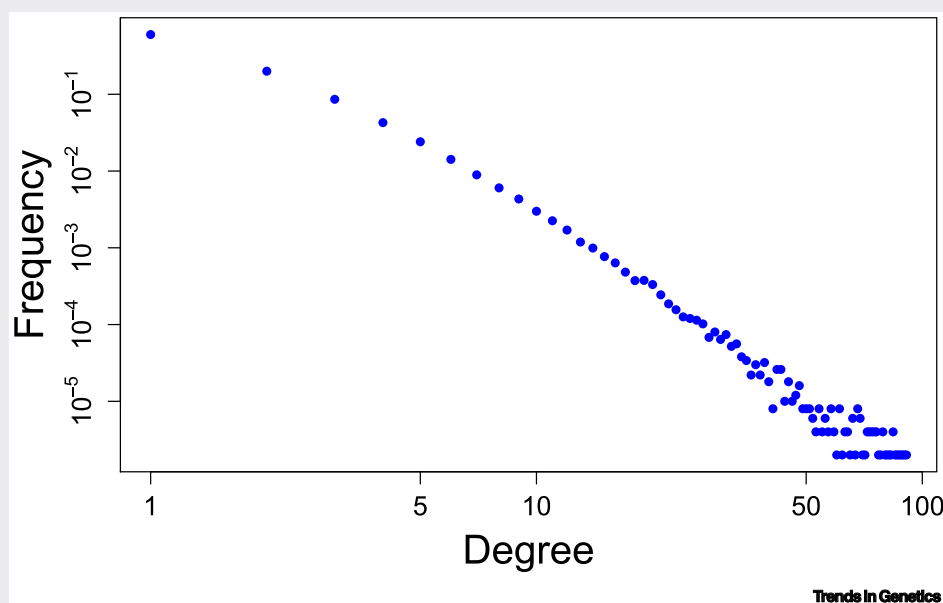


Figure 1. Expected Power Law Distribution of the Node Degree in a Gene Regulatory Network.

Redundancy thus facilitated polygenic selection by increasing the number of available targets (Figure 1B). Such selection will nevertheless be difficult to detect, precisely because it affects different pathways. Identifying cases of redundancy in GRNs and integrating them in polygenic tests, for example, by considering all alternative pathways coding for the same function, would thus help to retrieve currently missed selection signals.

Pleiotropy is another GRN feature that impacts the individual response of genes to selection. The omnigenic model proposes that each trait might be influenced by all genes [19]. Theoretically, such extreme generalized pleiotropy would severely constrain polygenic selection, because it could not modify a given trait without influencing all others. However, while this model seems adequate for some traits, such as human height, it might not be universal: for example, lipid traits such as cholesterol and triglyceride blood levels in humans [13] are best explained by a small number of loci.

Moreover, modularity can mitigate the effects of the omnigenic model on polygenic adaptation. Indeed, GRNs are highly modular, that is, made of small groups of genes strongly interconnected with each other but only loosely connected to the rest of the network, thus constituting a module. In each module, only a few genes are connected with genes from other modules. Most genes in a module are thus isolated from the rest of the network and can be leveraged by polygenic selection (Figure 1A).

Finally, GRNs tend to be scale-free, that is, most genes are connected to only few others, while few genes are connected to many others (Figure 1B), as shown on various types of networks [18,45]. In the context of highly modular scale-free networks, two measures are thus important: the **degree centrality**, which measures the centrality of a node in the whole network, and the **core-score**, which measures its centrality in its module. The degree centrality of a gene is inversely correlated with its level of genetic diversity [46,47], as confirmed by transcriptomics and proteomics data. The few highly pleiotropic nodes at the GRNs center thus evolve under strong negative selection, presenting limited variations in expression among individuals. They are not likely to be targeted by polygenic adaptation because their modification would affect numerous traits. Conversely, the vast majority of genes are at the periphery of the network, involved in one or few pathways. Changes in their regulation will, therefore, only slightly affect other pathways (Figure 1A,B). These genes include the high core-score nodes, which are central for a regulatory module but not for the whole GRN. Simulations show indeed that high core-score genes, which are more upstream in the modules, are more affected by selection [48]. This could explain the empirical observation that transcription factors evolve faster than other genes [49]. High core-score nodes could thus be preferential targets for polygenic adaptation. GRN structure offers the possibility for polygenic selection to act on many genes and gene regulatory elements, while limiting the potential side effects of selection on other pathways.

Polygenic Selection and GRNs: The Issue of *cis* -Regulatory Elements

One limitation of systems biology models is their focus on the evolution of networks made only of genes interacting with each other, thus overlooking important genomic features that would be useful to get a complete picture of how polygenic selection can impact GRNs. Indeed, most GRN models do not explicitly include *cis*-regulatory regions, such as promoters and enhancers, which can impact the expression level of several genes at once. The major role of these regions in the adaptation of complex traits has nevertheless been demonstrated. QTL and GWAS show indeed that up to 90% of variants associated with polygenic traits occur in noncoding regions [50,51]. There are evidences for polygenic selection targeting regulatory mutations [39,52], and mutations in transcription factor binding sites play a crucial role in species evolution [53].

However, integrating the effect of *cis*-regulatory sequences in polygenic adaptation models might be difficult, because the actual role of regulatory mutations on gene expression is hard to evaluate. It depends upon many factors, including chromatin activation state at the regulatory locus and availability of transcription factors in the cell [54]. Integrating such information might be easier in the future through massively parallel reporter assays, which allow investigation of the effects of regulatory regions on gene expression at the genome-wide level and assessment of their effect on fitness [55], in parallel with the development of machine-learning algorithms predicting the regulatory effect of mutations and their potential evolutionary role [56]. Such detailed data are available for some model species and should soon be available much more broadly thanks to decreasing sequencing costs. Finally, some current approaches integrate heterogeneous data using multipartite networks based on genomic and gene expression data, in order to identify not only gene–gene regulatory interactions but also interactions between regulatory regions and genes [18,57]. These approaches could allow development of new GRN evolution models directly integrating *cis*-regulatory regions.

Concluding Remarks and Future Perspectives

In order to overcome the limitations of current models of polygenic trait evolution, a new integrative evolutionary framework becomes indispensable. This new framework would take the GRN architecture into account (Figure 2). Here, we argue that, in order to achieve such a unifying framework, we must identify which genes and pathways contribute to polygenic traits, and how redundant they are, but also investigate the complex regulatory interactions that govern their expression (see Outstanding Questions and Figure 2). These features appear crucial not only to identify all loci involved in the genetic architecture of a trait, but also to assess whether these loci can realistically be targeted by polygenic selection. We propose, as done in the omnigenic model [20] to use the regulatory information given by GRNs to identify the relevant genes. However, we suggest *cis*-regulatory regions in the GRNs are explicitly modeled, as this allows information about the biological role of these regions to be retrieved [57]. In addition, we suggest that particular GRN features are investigated, focusing on redundancy and partition in regulatory modules. Finally, we suggest that the position of the different regulatory actors in the GRN are integrated in order to account for their probability of being targeted by polygenic selection.

Declaration of Interests

No interests are declared.

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Outstanding Questions

Redundancy allows for the selection of the same version of a trait through the selection of genetic variants in different pathways. Do we observe this phenomenon at the scale of the population? If so, which molecular signatures do such evolutionary processes leave on the genome and how can we detect them?

Pleiotropy is a major property of GRN actors that constrains how polygenic selection acts. Genes are often involved in several distinct pathways. What is the level of pleiotropy of the regulatory elements, in particular of the distal ones such as enhancers, and how does it constrain their evolution?

Classical population genetics approaches model the evolution of genetic sequences but ignore their complex interactions in GRNs. However, GRNs model the evolution of these regulatory relationships but not their genetic bases. Will more integrated approaches that explicitly model the molecular evolution of regulatory regions in the context of GRNs improve our understanding of the molecular bases of local adaptation?

Selection on polygenic traits yields linkage disequilibrium among genes involved in the traits, at the within- and among-population levels [1]. How do regulatory interactions affect this linkage disequilibrium? Can it be used to detect polygenic selection?

GWAS-based methods aimed at detecting polygenic selection use the additive effect of each gene on the trait. Could the interaction (dominance, epistasis) between these genes also be integrated in these approaches?

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REVIEW

What does modularity mean?

Miriam L. Zelditch¹ | Anjali Goswami²

¹Museum of Paleontology, University of Michigan, Ann Arbor, Michigan, USA

²The Natural History Museum, London, UK

Correspondence

Miriam L. Zelditch, Museum of Paleontology, University of Michigan, Ann Arbor, MI 48109, USA.
Email: zelditch@umich.edu

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Abstract

Modularity is now generally recognized as a fundamental feature of organisms, one that may have profound consequences for evolution. Modularity has recently become a major focus of research in organismal biology across multiple disciplines including genetics, developmental biology, functional morphology, population and evolutionary biology. While the wealth of new data, and also new theory, has provided exciting and novel insights, the concept of modularity has become increasingly ambiguous. That ambiguity is underlain by diverse intuitions about what modularity means, and the ambiguity is not merely about the meaning of the word—the metrics of modularity are measuring different properties and the methods for delimiting modules delimit them by different, sometimes conflicting criteria. The many definitions, metrics and methods can lead to substantial confusion not just about what modularity means as a word but also about what it means for evolution. Here we review various concepts, using graphical depictions of modules. We then review some of the metrics and methods for analyzing modularity at different levels. To place these in theoretical context, we briefly review theories about the origins and evolutionary consequences of modularity. Finally, we show how mismatches between concepts, metrics and methods can produce theoretical confusion, and how potentially illogical interpretations can be made sensible by a better match between definitions, metrics, and methods.

KEYWORDS

evolvability, modularity

1 | INTRODUCTION

Integration and modularity are widely regarded as fundamental and ubiquitous features of complex organisms (e.g., Felice et al., 2018; Melo et al., 2016; Raff, 1996; Wagner et al., 2007). What makes modularity so interesting to evolutionary biologists is that modularity may answer a question that modern evolutionary biologists only recently thought to ask: How is it that organisms can evolve? That ability had been largely taken for granted because it is self-evident that organisms *can* evolve given that they do. Surprisingly, however, evolvability cannot be taken for granted, as evident

in studies by evolutionary computer scientists who found that mutation, recombination and selection cannot always improve computer programs (Wagner & Altenberg, 1996). What is missing from that list of requirements for evolvability is the ability to generate *selectively useful* variation—the variation that enables them to improve in a stepwise fashion, optimizing each function independently without interfering with other, already optimized functions (Gerhart & Kirschner, 2007; Hansen, 2003; Riedl, 1978; Wagner & Altenberg, 1996).

Considering that we have a theory about the causes and consequences of modularity, plus a large and rapidly

increasing empirical literature, and an expanding array of methods for both quantifying modularity and delimiting modules, it may seem odd, even perverse, to now ask: “what is a module and how does that matter to evolution?” The reason for asking that pair of questions is that, as often noted, the concept is intuitively easy to grasp but difficult to define precisely (Bolker, 2000; Schlosser & Wagner, 2004; von Dassow & Munro, 1999). Being intuitively easy to grasp does not pin down the meaning of the concept because intuitions vary among the several disciplines now engaged in studies of modularity. Modularity is thus approached by researchers from different perspectives who understand modules differently, use different metrics to quantify modularity and different methods to delimit modules and work within different theoretical frameworks for relating modularity to evolutionary processes. Some differences are due to traditions within disciplines and may be largely matters of style, some are due to differences in the physical structure of the modules studied by different disciplines, but others are due to different concepts of modularity.

What complicates defining modularity precisely is that it is generally accepted that organisms are not strictly modular; modules are not actually autonomous. Rather, organisms are “nearly-decomposable” and modules are “quasi-autonomous” (Klingenberg et al., 2003; Simon, 1962; Wagner & Schlosser, 2004). Parts of an organism are not equally tightly linked to all others, but neither are they entirely independent of others. Modularity is thus a matter of degree, and assessing that degree is open to subjectivity without metrics to quantify modularity and to delimit modules that allow for a “degree” of modularity. To be scientifically useful, these need to be translatable across both systems and disciplines. The problem is not that we lack metrics to quantify modularity or methods for delimiting modules; we have many such metrics and methods (even though none may be wholly satisfactory). The more fundamental problem is that we have metrics that measure different properties as well as methods that delimit modules by different criteria. Consequently, theories and empirical results may be incomparable—incommensurate in both the mathematical and philosophical senses of that word.

The diversity of concepts and intuitions underlying them has substantial consequences for the meaning of modularity in evolutionary theory. That consequence of conceptual diversity should be obvious if we consider the question: “Does evolvability depend on modularity?” The answer to that will obviously depend on the properties of “modularity” (and “evolvability”). When “modularity” is not precisely defined, or when it is inconsistently defined, and when metrics for “degree of modularity”

quantify different properties, we cannot build upon the many studies of modularity to evaluate alternative theories about modularity and what it means for evolution.

When it comes to the measurement of modularity and the related methods for delimiting modules, there are many technical issues, but the three most basic questions are: (1) How well does that metric match our concept (and intuitions) about what modularity is? (2) How suitable is that metric given the physical properties of our system? (3) Does the most intuitively sensible metric also make sense in light of the theory we are testing? Ideally, we would begin with an overview of the concepts and then survey the metrics, associating each metric with the concept it most effectively captures. We could then understand the evolutionary consequences of modularity in context of each metric (and concept). But concepts are difficult to articulate precisely across disciplines when they seem self-evident within one discipline, but are utterly foreign in another. There are also intuitions about modularity that have not necessarily been formalized into concepts. In all these situations, words can get in the way. Concrete examples might seem helpful, but not when they rely on specialized knowledge of one discipline. For this reason, we begin by representing each concept visually. That may betray a bias of morphologists who are accustomed to visual information. This approach necessarily limits us to concepts that are easily portrayed, and some crucial metaphors are not easily conveyed visually such as a modular computer program. Hence, we supplement visualizations of concepts with descriptions of some common metaphors. After presenting these diverse intuitions, we summarize metrics of modularity, and methods for delimiting modules. Our objective is not to criticize them but rather to display their variety. We then summarize theories about the evolutionary implications of modularity and finally consider how mismatches between concepts, metrics and methods can lead to illogical reasoning.

2 | DEFINING MODULARITY AND INTEGRATION

The concepts of integration and modularity are usually regarded as hierarchically related: integration characterizes the relationship among components within a module, and quasi-autonomy characterizes the relationship among modules (Simon, 1962; Wagner & Altenberg, 1996). That intuition underlies all the concepts of modularity. Another hierarchy is central to studies of integration and modularity: the three levels at which they are manifest: (1) within individuals,

(2) within populations, and (3) within lineages. Distinguishing levels is important because the causes (and consequences) of integration/modularity differ by level. That traditional hierarchy, however, oversimplifies current conceptions of modularity in that modularity is now recognized at another level, the one most crucial to current theories of modularity: the genotype-phenotype map (Hallgrímsson et al., 2009; Hansen et al., 2007; Wagner & Altenberg, 1996), which bridges the levels of individual and populations.

2.1 | Levels of modularity and integration

The classification of modularity by levels is generally accepted, and, with the exception of the genotype-phenotype map, is unambiguous except in unusual cases (e.g., clones). Within individuals, integration is customarily ascribed to developmental and/or functional interdependencies among traits (Bissell & Diggle, 2010; Cheverud, 1982; Wagner, 1996) and modularity is thus due to their developmental and/or functional independence. An individual's fitness depends on the relationship among interdependent traits. Understanding the developmental and functional structure of integration and modularity is the key to explaining integration of the genotype-phenotype map and population.

At the levels of the genotype-phenotype map, that is, the mapping of the genotype onto the phenotype, and the population, integration is due primarily to shared genetic effects (i.e., pleiotropy), although, at the population level it can also be due to linkage-disequilibrium (Cheverud, 1982; Lande, 1980). The genotype-phenotype Pleiotropy, in this context, encompasses what Atchley and Hall (1991) termed “epigenetic pleiotropy” as well as “intrinsic genetic pleiotropy.” Intrinsic genetic pleiotropy refers to genes that are expressed in two or more cell populations, whereas epigenetic pleiotropy refers to the effect of a gene expressed in one cell population on other populations of cells, such as via tissue-interactions or the cascading effects of genes expressed early in development on processes occurring later. For example, a gene expressed in muscles can affect the development of the bone on which it inserts, with the result that the bone's phenotype is co-inherited with the muscle's phenotype. The distinction between the genotype-phenotype map and the population is important because variability (and other variational properties of phenotypes) are determined by genetic architecture, that is, the structure of that map. Traits can be variable in the sense of being capable of variation even if they do not happen to vary within some populations. Similarly, they can be integrated even if, in the absence of variation, there is no realized covariation

(Hallgrímsson et al., 2009). A modular structure of the genotype-phenotype map and of variation, could be due not only to modular pleiotropy but also to antagonistic pleiotropy if the positive and negative pleiotropic effects cancel each other out (Leamy et al., 1999; Mezey et al., 2000; Pavlicev & Hansen, 2011). Integration at the level of the genotype-phenotype map and populations is commonly termed “genetic” and “phenotypic” integration/modularity. In the case of populations, genetic and phenotypic integration refer to the components of variation; in practice, phenotypic integration/modularity often serves as a proxy for genetic integration/modularity. In their formulation as covariance/correlation matrices, these matrices at the population level are usually referred to as **G** and **P** for the genetic and phenotypic matrices, respectively.

At the level of lineages and clades, integration and modularity refer to the evolutionary relationships among traits. Here we distinguish integration within a lineage; by “lineage” we mean a sequence of ancestral/descendant populations, whereas a clade is a set of lineages that branched. The distinction is important because traits that might be integrated within a lineage might not be regularly correlated across all branches. In either case, evolutionary integration refers to correlated evolutionary changes and evolutionary modularity to their independent evolutionary changes. Traits that are evolutionarily integrated coevolve whereas those that are evolutionarily modular evolve independently. At this level, integration is due primarily to pleiotropy, in the sense defined above, including epigenetic pleiotropy, coselection, or correlational selection (Arnold et al., 2001; Cheverud, 1996; Felsenstein, 1988; Hansen & Martins, 1996; Zeng, 1988). Evolutionary modularity can therefore be due to genetic modularity or to natural selection countering integration.

Given that the causes of integration and modularity, as well as the concepts about them, differ across levels, we divide this section about intuitions according to these levels.

2.2 | Intuitions of modularity: Visual representations of concepts

In this section, we introduce graphical representations of modularity, all of which appear in the literature. The mapping of these graphical depictions onto underlying processes, the relationships among them, and even the degree of overlap among them, could be contested. Readers of this paper, like the authors of the figures we present, may view them in different and even conflicting ways—the figures depict how people literally see modularity, but how those figures are seen may also differ among readers. Later we will discuss the relationships

between these perceptions of modularity and how they inform the measurement of modularity and the evolutionary origins and consequences of modularity.

2.2.1 | Modularity at the level of an individual

Modularity of an individual can be represented as a network (Figure 1a), especially if there are physical connections among the components or information flows through the network, or the output of one module is the input of another. The same diagram can describe neural networks, metabolic pathways, or networks of gene regulation and protein interactions (e.g., Clune et al., 2013; Davidson & Levine, 2008). A related diagram, explicitly lacking directionality, is an undirected graph (Figure 1b), which can convey that traits depend on some common (unobserved) cause extrinsic to them, such as a local growth factor. They have also been used to represent the structural (topological) modularity of complex skeletal forms based on the criterion of whether bones articulate with each other (e.g., Esteve-Altava et al., 2013). One convenience of these related visualizations of modularity is that modularity is often defined in terms of graphs: a modular graph has subgraphs, with dense connections among nodes within a subgraph and sparse connections between subgraphs.

Developmental modules are often represented less abstractly, depicting their physical location and spatial extent on the organism, and they often contain information about the genetic specification of modules (Figure 1c,d). For example, Figure 1c shows regions of the embryo and their genetic specification (after Raff & Sly, 2000); modules, in this case, correspond to territories—the specific regions shown in the figure. The properties ascribed to modules follow from that conception of territories: (1) autonomy, (2) discrete organization defined by expression of specific genes, and (3) occupation of specific physical sites. It is worth noting that developmental modules may influence morphological evolution even when they do not result in correlations in adult phenotype. More broadly, not all concepts of modularity induce phenotypic correlations at any stage, although many, perhaps even most, will. Figure 1d represents a model for the specification of gnathostome jaws (after Depew & Compagnucci, 2008; Fish et al., 2011). It depicts signaling gradients derived from the Hinge (black) or Caps (red), the coordination between modular units that need to be co-registered (purple, two-headed arrows), and signaling factors that establish the potential for modular domains by their balance and/or thresholds levels. The net effect is to

establish polarity along an axis, and thus a coordinate system that spans the delimited anatomical regions. Despite the differences between these two representations of developmental modules, one obvious common feature is the spatial sense of “modularity.” In both, modularity is the net effect of multiple spatially localized processes.

Functional modules are frequently represented by anatomical figures that delineate components of a module; they typically designate structural components of some mechanism such as thermoregulation (Figure 1e) (after Kingsolver & Wiernasz, 1987). In this case of physiological integration, the figure shows adaptively interrelated traits, whose coordination within the individual enhances performance. Unlike the other diagrams, this one depicts spatially dispersed modules. One representation of functional modules emphasizes an analogy to an engineered mechanical system (Figure 1f), highlighting functional roles of the modules: “tool,” “engine,” and “amplifier” of a power-amplified system (after Claverie et al., 2011). Another schematic emphasizes mechanical linkages, conveying how one biomechanical parameter (kinematic transmission [KT]) changes according to changes in the geometry (joint shifts) (Figure 1g) (after Baumgart & Anderson, 2018). Here, the KT of linkages that are highly integrated with each other, to the exclusion of others, are functional modules. When these represent cases in which the geometry of the structures matter, they can be distinguished from the more abstract network shown in Figure 1a.

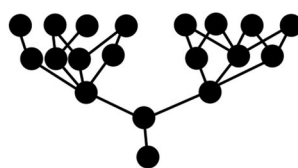
Figure 1h is an enormously influential hypothesis for vertebrate cranial modules, and for integration/modularity more generally (after Cheverud, 1996). Each color-coded region is simultaneously a developmental *and* functional module, not developmental *or* functional module. One reason for that inseparability is that development is regarded as serving a (dynamic) function (see Section 2.2.2) and the other is that, under the hypothesis of the “imitatory epigenotype” (Riedl, 1977, 1978), functionally interdependent traits evolve to become genetically/developmentally integrated. This diagram depicts the functional matrix hypothesis (Moss & Salentijn, 1969), according to which each module serves a primary functional role, such as vision or olfaction, with the cranial bones functioning to protect/support the associated soft tissues (and spaces) on which they depend developmentally.

There are visually obvious differences among these figures of integration that reflect the intuitions underlying them. One difference is whether modules are

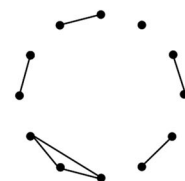
FIGURE 1 Representations of modularity within an individual. (a) Networks (after Clune et al., 2013); (b) undirected graph; (c) gene-expression territories (after Raff & Sly, 2000); (d) signaling domains that specify positional information (after Fish et al., 2011); (e) anatomically localized functional units of a physiological system (after Kingsolver and Wiernasz 1991); (f) mechanical (power-amplified) system (after Claverie et al., 2011); (g) components of a mechanical linkage system (after Baumgart & Anderson, 2018); (h) developmental and functionally integrated complexes (after Cheverud, 1996)

Developmental

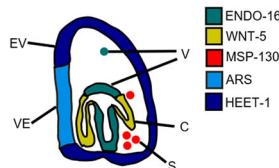
(a) Network



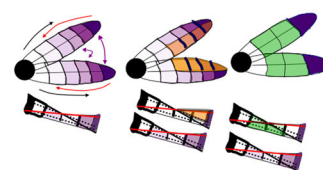
(b) (Undirected) Graph



(c) Territories

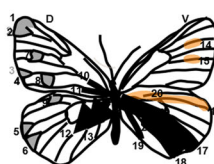


(d) Signaling modules

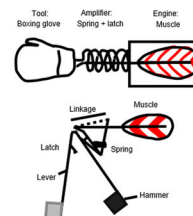


Functional

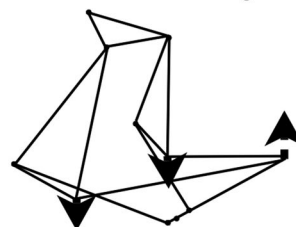
(e) Integrated Measurements



(f) Mechanical System

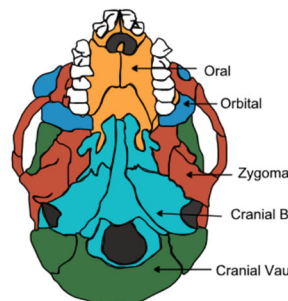


(g) Mechanical Linkages



Developmental/Functional

(h) Functional Matrices



anatomically grounded. They may be anatomically localized and spatially discrete, with clearly marked boundaries (Figure 1c,e,h), or have a more gradient-like pattern (Figure 1d), or be spatially dispersed (Figure 1f). They may be structures that serve a common role, or serve

different roles but jointly affect performance, or be causally independent. Even if all express the idea that a system can be decomposed, they differ in what they mean by a “system” and the lines along which it is decomposed.

2.2.2 | A note on the relationship between “developmental” and “functional” modules

One important element of the modern theory is the characterization of modules as traits that collectively serve a functional role. This characterization of modularity may seem unnecessarily restrictive in that it incorporates a hypothesis about functional modularity within the defining clause of modularity itself. Yet, eliminating that clause from the definition of modularity distorts its theoretical significance—it is precisely because traits collectively serve a functional role that modularity is linked to evolvability (e.g., Wagner & Altenberg, 1996). That reference to a “functional” role may seem to omit development, but development is encompassed within it because, in this theoretical context, developmental processes serve a functional role and function is a developmental process. The idea of “functional coupling” extends to developmental processes in that what it means is that the optimal value for one trait depends on the phenotype of another. A classic example of that functional coupling of developmental processes, given by Cheverud (1984), is neural tube formation, which is at a central functional position in development because subsequent developmental processes (and viability) depend on the prior closure of the anterior and posterior neuropores. Hence there is strong stabilizing selection on neural tube formation and its related causal mechanisms, making this a paradigm of internal stabilizing selection. Not only do developmental processes serve a functional role, but also functional processes serve a developmental role, exemplified by the necessity of muscle development and contraction for the maintenance and growth of associated skeletal structures, such as mammalian mandibular processes on which muscles insert (Anthwal et al., 2015; Herring & Lakars, 1981; Rot-Nikcevic et al., 2006).

Developmental biologists and functional morphologists (and physiologists) may find this equation between developmental and functional interdependences counterintuitive or even just wrong. Indeed, there are theories of modularity that focus entirely on developmental causes and omit any considerations of functional roles (or fitness), instead concentrating on the classes of developmental processes that integrate or individuate traits (e.g., Klingenberg, 2005, 2008). In that case, functional coupling is clearly not considered a defining clause of “modularity,” or even an anticipated characteristic of it. Whether, and how the distinction between “developmental” and “functional” modularity is made has important consequences for the debates about the evolutionary origins and consequences of modularity.

2.3 | Genotype-phenotype map

The first three diagrams presenting the genotype-phenotype map (Figure 2) do not depict modular mappings, but each one makes a point about the importance of the map. The first one (Figure 2a, after Lewontin, 1974) was intended to highlight the problem facing evolutionary biologists: to bridge the gap between population geneticists and biometrical geneticists. Because the first group worked with allele frequencies, and the second with measurements of phenotypes, that gap could not be bridged without a map that related genotype to phenotype; the figure shows the mapping from genotype onto phenotype, with selection acting to modify the phenotype, which, in turn, modifies the genotype. Figure 2b (after Alberch, 1991), highlights the role of development as the mapping function, as does the third (Figure 2c, after Milocco & Salazar-Ciudad, 2020), but the main point made by this one is the nonlinearity of the map (and its consequences for quantitative-genetic theory). The other three represent explicitly modular maps. These show how genetic effects are structured within an individual and how variation would be structured should there be allelic variation in the genes that have pleiotropic effects. The three are conceptually, not just visually different. The first (Figure 2d) represents the same idea depicted in Figure 1h, that each character complex (C1, C2) serves a primary functional role (F1, F2) and that genes (G1–G6) which have pleiotropic effects typically affect traits restricted to a single functional module (enclosed in an ellipse) (after Wagner, 1996). That is the concept of modularity used in the seminal paper that links modularity to evolvability (Wagner & Altenberg, 1996): “By modularity we mean a genotype–phenotype map in which there are few pleiotropic effects among characters serving different functions.”

Figure 2e depicts modules *on* an organism (after Klingenberg, 2005). That embodiment conveys two ideas. One is that modules are spatially coherent, bounded and discrete; the spacing of points in the figure represents the spacing of anatomical loci (landmarks) on the organism. A second idea is that two classes of processes integrate modules. One is the network of developmental interactions along pathways, such as signaling interactions, which are shown by arrows between landmarks within modules. This corresponds closely to the idea of “epigenetic pleiotropy” introduced above. The second idea is that pleiotropic effects of genes can affect two or more modules because the same gene affects two or more pathways in parallel; these are represented by arrows between modules, and this corresponds closely to the idea of “intrinsic genetic pleiotropy” also introduced above. In the case of direct interactions, even stochastic

Genotype-Phenotype Map

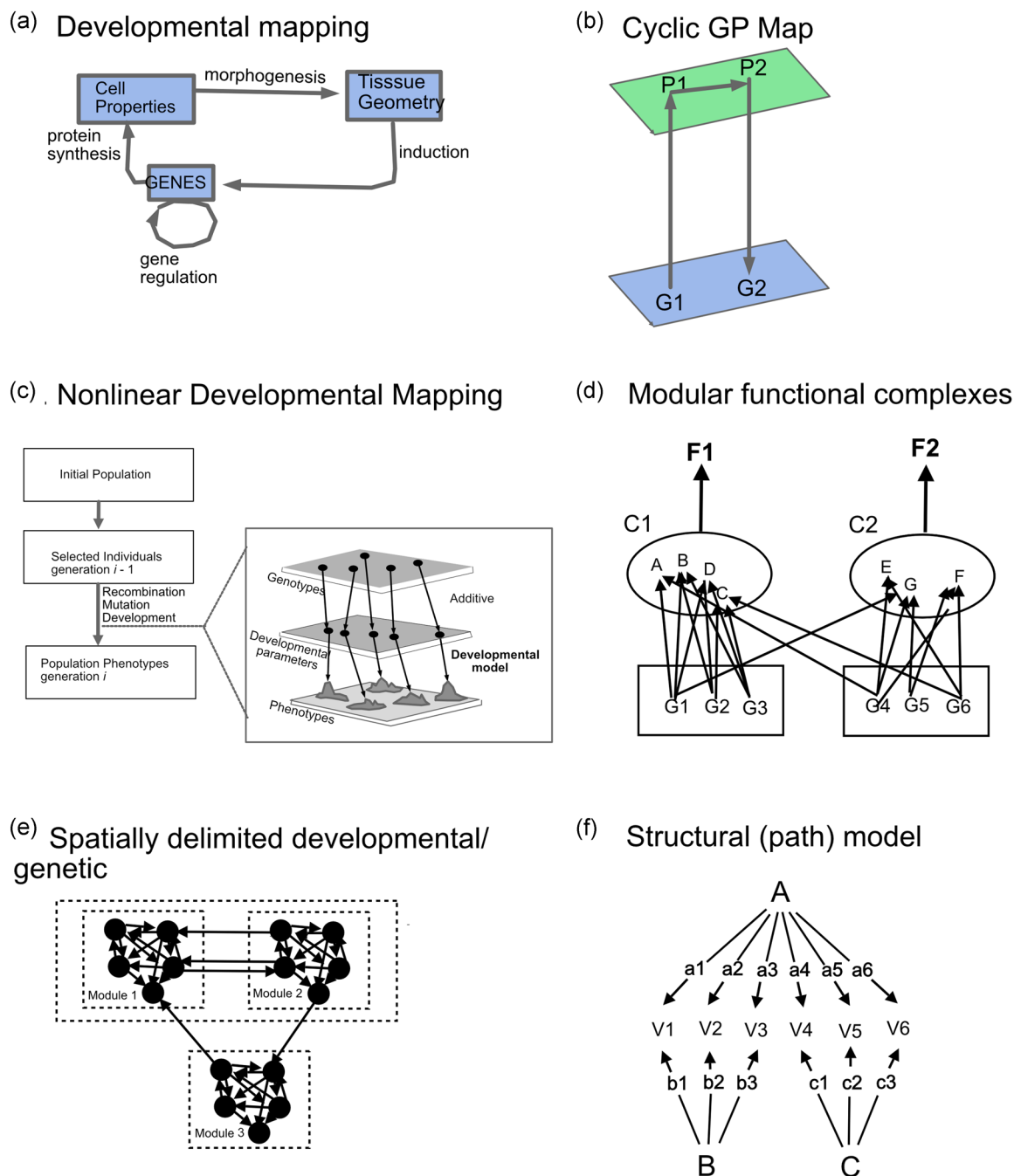


FIGURE 2 Representations of modularity bridging individual and population levels: Genotype-Phenotype Maps. (a) Mapping from allele frequencies to quantified phenotype traits (after Lewontin, 1974); (b) development as the mapping function (after Alberch, 1991); (c) A nonlinear mapping (after Mllocco & Salazar-Ciudad, 2020); (d) A map, showing that genetic effects (G1–G6) are restricted to (a–f) within complexes (C1–C2) that each serve a functional role (F1–F2) (after Wagner, 1996); (e) Modules shown in anatomical position on an organism (after Klingenberg, 2005). (f) A path diagram depicting relationships between observed and latent variables (after Mitteroecker & Bookstein, 2007)

variation arising upstream will have downstream effects along pathways, but parallel variation arises from allelic variation in the genes (or environmental factors) affecting two or more modules. Consequently, the covariance

of stochastic perturbations that produce fluctuating asymmetries can distinguish between the two types of processes if measured in populations (Klingenberg et al., 2003; Klingenberg, 2008, 2009).

Figure 2f is both an idealized version of a genotype-phenotype map and a representation of a statistical model that is explicit on how to measure integration and modularity given observations of variation (after Mitteroecker & Bookstein, 2007; Mitteroecker et al., 2012). What this figure highlights is that integration within modules is due to a common factor affecting the traits within a module, and, as a path diagram, that local modules are independent of each other; integration between modules is due to a general factor (A), such as a growth factor. This is a more general model than the other two because it does not stipulate that the traits within a module are functionally related, nor that they are integrated by a particular class of processes, but it is also less general in that it specifies a method for estimating the coefficients given observations of integration/modularity within populations (path analysis).

2.4 | Variational modularity

The distinction between the preceding diagrams and the representation of variational modularity (Figure 3) is that the genotype-phenotype maps depict how variation *could* be structured, whereas the diagrams of variational modularity show how statistical covariances/correlations *are* structured. All are based on the idea that integration is manifest at the population level by statistical covariance/correlations (Berg, 1960; Olson & Miller, 1958), or partial correlations (Terentjev, 1931). Each presents the results of a statistical method for analyzing the structure of covariances/correlations. Understanding the mathematical details of the method is necessary for discerning how the results are obtained, and each presumes an intuition about the expected structure of modularity, not just how that ought to be quantified. Figure 3a shows a cluster analysis of measured variables (V1–V12); modules correspond to clusters of variables and a high degree of modularity is expected to result in high correlations within clusters and little to no correlation between them. Cluster analysis can induce clustering, even when it is not present in the data, but the point here is not the limitation of the method but rather the perception of what modularity ought to look like. The network plots (Figure 3b,c) show more complex relationships among the variables. If these correspond to an intuition about modularity, then modularity is not conceptualized as a simple hierarchical structure (and a method capable of detecting patterns that are more complex would be preferred over one that tries to maximize hierarchical structure). Figure 3b show which variables are correlated, with line weights adding information about the magnitude of the correlation (Baranzelli et al., 2014), Figure 3c shows a clearly nonhierarchical structure in

which some variables belong to more than one cluster. Figure 3d shows a graphical model; in this one, the lines that connect the variables (“edges”) show which are correlated controlling for all other variables. In the absence of an edge, the variables are conditionally independent. Figure 3e (after Parsons et al., 2012), shows modules on the organism, with ellipses surrounding the anatomical points that comprise a module. These may be strictly mutually orthogonal subspaces, or partially overlap, with some landmarks belonging to more than one module (Marquez, 2008). The final figure, Figure 3f, shows the results of a partial least squares analysis; the axes of the plot are the paired axes that maximize the covariance between two sets of variables; the correlation between the scores on those axes measures the strength of integration between them.

2.5 | Evolutionary modularity

Evolutionary modularity is usually represented by the same diagrams that depict individual or variational modularity, such as cluster analysis (Monteiro et al., 2005), path diagrams (Monteiro & Nogueira, 2010), networks (Bardua et al., 2019; Fabre et al., 2020; Felice & Goswami, 2018; Marshall et al., 2019; Watanabe et al., 2019; Zelditch et al., 2008), or anatomical figures with color-coded modules (Goswami & Polly, 2010; Larouche et al., 2018). Obviously, the interpretations of these plots differ because they show traits that coevolve, rather than covary, but without depicting the modules on a phylogeny, that is not evident in the pictures. The one visual representation that clearly distinguishes this level of modularity from the others is shown in Figure 4, which depicts the genetic covariance matrices for two traits (X and Y) within the ancestral population (the yellow ellipse) and descendant populations (the gray ellipses). This diagram depicts the expectations under a hypothesis of random genetic drift hence it shows the impact of genetic integration on the evolutionary integration produced by that process. Under random genetic drift, 95% of the evolutionary divergence among the mean values of X and Y is within the larger orange ellipse (after Melo et al., 2016), presuming that the populations of the same size and time since divergence.

3 | HOW MODULAR IS MODULARITY?

Unless modularity is regarded as a binary concept, meaning that some structure is either modular or not, modularity is a matter of degree. Therefore, we need a

Variational modularity

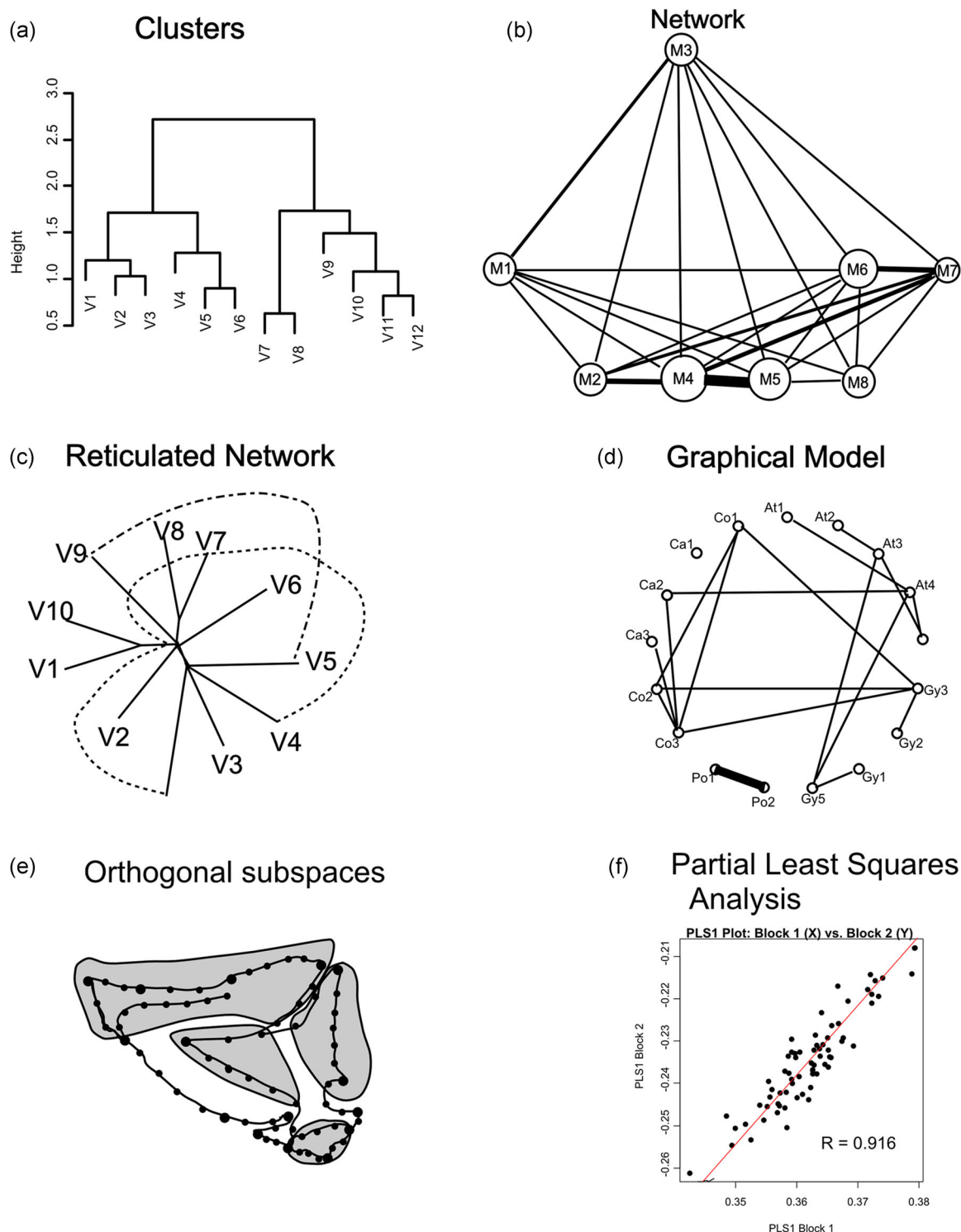


FIGURE 3 Representation of variational modularity. (a) Modules as clusters of traits (after Cheverud, 1982; Terentjev, 1931). (b) Interactions within and among modules represented as a network; the size of the circle (representing a module) is related to the strength of the correlation within the module; the line weights between modules is related to the strength of the correlation between modules (after Felice et al., 2019). (c) Modules represented by a reticulated network depicting a nonhierarchical structure of integration between morphological complexes; the dotted lines indicate the traits that belong to more than one cluster (after Zelditch et al., 2008). (d) Graphical models depicting conditional independence relationships (after Magwene, 2001). (e) Modules represented as orthogonal subspaces (after Marquez, 2008). (f) Integration represented by the correlation between scores on the axis of maximal covariance between sets of traits

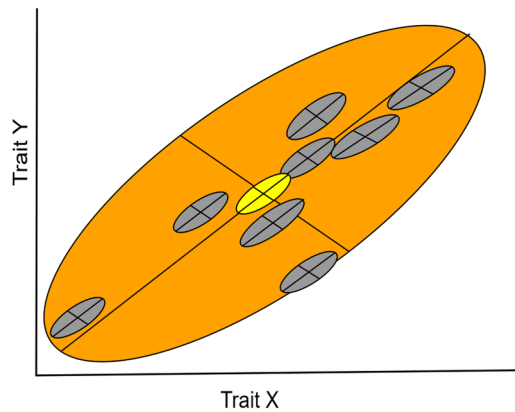


FIGURE 4 Macroevolution: expected distribution of two traits within the ancestral (yellow ellipse) and descendent (grey ellipses) populations, depicting the impact of genetic integration on evolutionary integration under a hypothesis of random drift (after Melo et al., 2016)

measure of how modular a structure is, as well as some criterion for deciding if that degree suffices to conclude that the structure is too highly integrated to qualify as “modular.” That metric is also needed if we want to determine whether a high degree of modularity elevates evolvability (or elevates some property).

Methods that analyze modularity at the individual level are usually discipline-specific, and at this level, modularity is rarely quantified. For example, if the idea of modularity is that a module is a domain of gene-expression, the analysis of modularity is done by mapping those domains. If instead the idea of modularity is a subfunction served by part of a functional complex, the analysis is done by decomposing functions into subfunctions. In contrast, at the population level, modularity usually *is* quantified; hence most metrics summarized in this section are for variational modularity. The same methods are also applicable to evolutionary modularity by replacing the population covariance matrix by an evolutionary covariance matrix. That substitution of one matrix by the other obviously does (or should) affect the interpretation of the results, but most metrics for modularity do not presume a particular set of causes, so the metrics apply equally well to variational and evolutionary modularity. The metrics, if not the causes or theoretical meaning of modularity, generalize from one level to the next.

3.1 | Analyzing modularity and integration: Metrics and methods

In this summary of metrics and methods, we focus first on metrics, then on methods for delimiting modules. Metrics and methods are not easily separated because the

metric often serves as a test statistic used when delimiting modules—a hypothesis is supported when it yields a high value of modularity (in contrast to methods that assess the relative fit of competing models of modularity). We organize this section according to the type of data analyzed, because the metrics for one type may be conceptually related to a metric for another type of data structure, but the formulae differ, and this section summarizes formulae.

3.1.1 | Network modularity

One metric for modularity of a network (Q) is called precisely that, “modularity” (Clauset et al., 2004; Newman, 2006). This is calculated from the edges between vertices of a network, considering only the presence or absence of the edge. What is measured is the fraction of the edges that lie within the given modules, minus the fraction expected if the edges are randomly distributed. For the simplest case of two modules:

$$Q = \frac{1}{(2m)} \sum_{vw} \left[A_{vw} - \frac{k_v k_w}{2m} \right] \frac{1 + s_v s_w}{2}, \quad (1)$$

where v and w are vertices belonging to two modules, $s_v = 1$ if the vertex is within module v and -1 if it is not, n is the number of vertices, m is the number of edges, and $\mathbf{A}$ is the adjacency matrix (whose elements are 0 when there is no edge between the vertices and 1 when there is). The expected number of edges between v and w is:

$$\frac{k_v k_w}{2m}, \quad (2)$$

where k_v is the degree of v and k_w is the degree of w (i.e., the number of edges incident to the vertex), and the difference between the actual and expected number of edges between v and w is:

$$A_{vw} - \frac{k_v k_w}{2m}. \quad (3)$$

Q is calculated by summing over all pairs of nodes and ranges from -1 to 1 . This metric thus quantifies modularity by determining whether the number of edges within a module is significantly more (and, equivalently, whether those between are significantly less) than expected by chance.

This measure of modularity presupposes that there is no meaning to the edges, aside from their presence or absence. That is, the presence or absence of an edge is all that is taken into account. However, when there are

weights associated with edges, Q can be calculated from those weights. Q can be used to quantify modularity for networks through which information (or signals) flow. It can also be used for undirected graphs, and variants of it have been used for a very different case, “anatomical network analysis (ANA).” In this approach, the notion of adjacency or connectedness is taken literally in that “connectivity modules” refer to physically connected groups of bones (Esteve-Altava et al., 2013).

3.1.2 | Metrics of integration and parcellation for counts

Because modularity is usually characterized by both the internal integration of the modules and the independence between them, modularity can be quantified by a pair of metrics, one measuring the internal integration of modules and the other the separation between them. These two metrics (Mezey et al., 2000) thus determine whether the putative modules have (1) a higher than average level of integration due to by pleiotropic effects of genes among the traits of a set, and (2) a higher than average level of independence from other trait sets. For trait loci (QTLs) that affect a set of traits (T) within the first module, T_1 , with k traits, the integration statistic, M_I is the total number of traits within T_I affected by m QTL that those QTLs could affect (which, for k traits, is km).

$$M_I = \frac{\sum_{j=1}^m (\text{number of traits in } T_1 \text{ affected by QTL } j) - m}{m(k-1)}, \quad (4)$$

M_I ranges from 0 to 1. The highest value is for the case in which all the QTL that have at least one effect on one trait within T_1 have effects on all of them. The lowest is for the case in which each QTL that has an effect on a trait within T_1 affects only one of them.

The parcellation statistic, M_P , quantifies the non-overlap of the effects of the QTLs. For each QTL j , the observed number of traits affected in T_1 and T_2 is symbolized by O_{j1} and O_{j2} . The expected number of affected traits is calculated as $E_{j1} = p_1 S_j$, $E_{j2} = p_2 S_j$ where S_j is the total number of effects of QTL j and p_1 and p_2 are the relative frequencies with which a trait in T_1 or T_2 is affected by any QTL in the total sample.

$$M_P = \sum_{j=1}^n \sum_{i=1}^2 \frac{(O_{ji} - E_{ji})^2}{E_{ji}}. \quad (5)$$

This measures the degree to which the QTL preferentially affect traits within a single module, either T_1 or T_2 . The larger the value, the less likely that a mutation will affect both T_1 and T_2 .

Modularity of T_1 is inferred if both M_I and M_P exceed the value expected for a randomly chosen set of traits. Similarly, modularity of T_2 is inferred the null hypothesis can be rejected for both M_I and M_P for the traits within T_2 . It is therefore possible that only one of the two metrics might be statistically significant, or that both are significant only for one of the two or more complexes. In one case, both M_I and M_P are statistically significant for one trait complex, the ascending ramus of the mouse mandible but only M_P is for the other complex, the alveolar region (Mezey et al., 2000).

That method has been criticized for its treatment of modules as discrete (Roseman et al., 2009). An alternative is to allow for a clumpy or continuous distributions of relationships among traits; QTLs that have a significantly narrower distribution of effects than expected by chance are considered locally pleiotropic, and those with either randomly or more widely distributed effects than expected by chance are considered globally pleiotropic. Local pleiotropy is then viewed as the continuous analogue of modular pleiotropy.

3.2 | Metrics of integration and modularity based on covariance/correlation matrices

Correlations/covariances have been the focus of most quantitative studies of integration ever since three tremendously influential works argued that morphological integration could be quantified by a statistical correlation and, importantly, that correlations between functionally and developmentally related traits exceed those between developmentally and functionally unrelated traits. Terentjev (1931) analyzed correlations and partial correlations, detecting hierarchically structured groups of conditionally independent traits (correlation pleiades”). Berg (1960) and Olson and Miller (1958) analyzed correlation matrices. Berg’s analysis of the ecological significance of correlation pleiades continues to stimulate research (e.g., Armbruster et al., 1999; Conner & Sterling, 1996; Hansen et al., 2007). Most of the metrics and methods can be extended to analyses of evolutionary correlations. However, analyses of evolutionary correlations require an additional step—estimating the evolutionary covariance/correlation matrix—and that requires having a model of the evolutionary processes that generated the evolutionary covariances/correlations.

Several methods compare integration among traits belonging to the same module to the integration between traits belonging to different modules. We therefore begin this section with a metric for integration, and then present methods for measuring modularity. We conclude it with a measure of another important property, conditional evolvability, which measures the ability to respond to directional selection on a trait when the other(s) are under stabilizing selection.

3.2.1 | A metric of integration: The variance/standard deviation of eigenvalues

The most commonly used metric for overall integration for multivariate data compares the integration estimated from the data to the maximum possible for that data set. In the most highly integrated case, all variables will be highly correlated and the correlation matrix will have one dominant axis (the first principal component). All others will be zero. In the least integrated case, no variables are correlated and all axes will be equal in length. The variation in the lengths of those axes is thus a measure of the strength of integration. More precisely, that is measured by the variance of the eigenvalues relative to the maximum possible variance (Pavlicev et al., 2009; Wagner, 1984b):

$$Var_{rel}(\lambda) = \frac{Var(\lambda)}{Var_{max}(\lambda)} = \frac{Var(\lambda)}{N-1}, \quad (6)$$

where λ refers to an eigenvalue and N to the number of traits. The reason for scaling the measure of integration is that the maximum for the metric depends on the size of the correlation matrix (i.e., the number of measured variables) and scaling by the maximum allows for comparisons across datasets with different numbers of traits.

The relative standard deviation of the eigenvalues is the square root of that variance, and it is generally preferred because it better reflects the unequal allocation of the variance evident in the variation among correlation coefficients (Pavlicev et al., 2009; Wagner, 1984b):

$$SD_{rel}(\lambda) = \frac{\sqrt{Var(\lambda)}}{\sqrt{N-1}}. \quad (7)$$

For covariance matrices, the eigenvalues are extracted from the covariance matrix and scaled by:

$$\frac{p}{p-1} \sum \lambda \sum \lambda, \quad (8)$$

where p is the number of variables and $\sum \lambda$ is the sum of the eigenvalues (i.e., the trace of the covariance matrix) (Haber, 2011; Van Valen, 1974).

3.2.2 | Covariance between modules: Partial least squares (PLS)

PLS is a method for analyzing the relationship between two (or more) blocks of variables (Sampson et al., 1989; Streissguth et al., 1993), introduced as a method for analyzing the relationship between shape and ecological variables, and for analyzing the relationship between two blocks of shape variables (Rohlf & Corti, 2000). The analysis is done by a singular value decomposition of the matrix S :

$$S = \begin{bmatrix} S_1 & S_{12} \\ S_{21} & S_2 \end{bmatrix}, \quad (9)$$

where S_1 is the covariance matrix for one block, S_2 is the covariance matrix for the other block, and S_{12} is the covariance matrix of the variables of the two sets (S_{21} is the transpose of S_{12}), extracting the axes of greatest covariance between the blocks, the (paired) singular axes multivariate traits, one per block. The correlation between the scores on the axes, r_{pls} , is a measure of the strength of integration, and the standardized value of that correlation, $Z_{r,pls}$, can be used as a test statistic when comparing the strength of integration across populations or hypotheses (Adams & Collyer, 2016).

3.2.3 | Covariances within relative to between modules

Several metrics measure the strength of integration within developmental/functional modules relative to that between modules. The first metric introduced is straightforward: the ratio of the average correlation (or squared correlation) of traits within the same module to the average for traits within different modules, as used by (Cheverud, 1982) to confirm the presence of discrete functional sets, or F -sets, in the primate cranium. Another is the trace correlation (Klingenberg et al., 2003), replaced by Escoffier's RV coefficient (Klingenberg, 2009). The RV coefficient is a ratio of the degree of covariation between blocks of variables, that is, modules, relative to the covariation within those blocks. In the notation of Equation (9) above,

$$RV = \frac{trace(S_{12}S_{21})}{\sqrt{trace(S_1S_1)trace(S_2S_2)}}. \quad (10)$$

The trace is the sum of the diagonals of the matrix (i.e., the sum of the variances over all the variables). The RV coefficient ranges from 0 to 1.

The RV coefficient is sensitive to the sample size and the number of variables, making it difficult to compare across datasets. An alternative metric is the covariance ratio (CR) (Adams, 2016) is:

$$CR = \sqrt{\frac{\text{trace}(S_{12}S_{21})}{\sqrt{\text{trace}(S_1^*S_1^*)\text{trace}(S_2^*S_2^*)}}}, \quad (11)$$

where S_1^* and S_2^* are the covariance matrices for the modules, with the diagonals of those matrices replaced by zeros. So the CR is the ratio of the overall covariation between modules to the overall covariation within modules. For random sets of variables, $CR = 1$ (because, on average, the covariation should be the same within and between blocks for random data). If there is greater covariation within modules than between them, $CR < 1$, and if the covariation is greater between than within modules, $CR > 1$. When there are more than two modules, the CR is an average of the pairwise CR values. The standardized effect size, the Z_{CR} is calculated from an empirically generated sampling distribution; this is the test statistic for comparing the strength of the modular signal across populations, and also for comparing alternative hypotheses of modularity (Adams & Collyer, 2019).

3.2.4 | Conditional evolvability

What may be evident from the metrics that quantify the strength of integration within and between modules is that neither captures the idea central to the theory of modularity—"quasi-independence." One metric that does capture that is not a metric of modularity per se. Rather it is a metric of "conditional evolvability, which is arguably the property of modularity most relevant to evolvability. Modularity is thought to enhance evolvability by allowing each function to be optimized independently without interfering with other, already optimized functions. Conditional evolvability (c) measures the ability of a trait (or complex of traits) to evolve in the direction of selection *when other traits are under stabilizing selection*. Conditional evolvability is thus the property that modularity enhances:

$$c(\beta) = (\beta'G^{-1}\beta)^{-1}, \quad (12)$$

where β is the selection gradient and G is the genetic variance covariance matrix (replaced by the phenotypic

covariance matrix when only that can be measured). Conditional evolvability is specific to a particular direction, the direction of the selection gradient, but it is possible to get a more general measure, an average conditional evolvability. The formula is in Hansen and Houle (2008); for high-dimensional data, such as shape, the average conditional evolvability converges on the harmonic mean of the eigenvalues of G .

A high correlation between two or more blocks may be entirely due to effect of some factor that whose variance generates the covariance between the two blocks that are otherwise independent of each other. Should that common factor be under stabilizing selection, limiting its variation, it will not generate covariance between the two blocks, hence the two blocks could be independent of each other. The effect of that common factor can be removed (regressing it out separately for each block), with the modules being blocks of variables that then exhibit a zero cross-block covariance structure, although in an observed sample, they would not be exactly zero (Mitteroecker & Bookstein, 2007) (for algebraic details of the scaling procedure, see Mitteroecker & Bookstein, 2008)

3.3 | Methods for evaluating hypotheses about the boundaries between modules

Here we present methods that test a priori hypotheses about the boundaries between modules, given data in the form of covariance/correlation matrices. The same methods can be used for testing hypotheses about variational and evolutionary modularity except that studies of evolutionary modularity are about correlated evolutionary changes (phylogenetic covariances/correlations) and therefore require analyzing those changes given a phylogeny and an evolutionary model.

Two kinds of hypotheses are commonly entertained. The first is based on a theory of the optimal covariance structure in which adaptively interdependent traits are correlated and independent of adaptively unrelated traits. In this case, there might be only one hypothesis considered worth testing—that the data fit a hypothesis predicting that modularity enables organisms to improve in a stepwise fashion without interfering with the adaptations already attained by other functional complexes (Armbruster et al., 1999; Cheverud, 1982; Cordoba et al., 2015). Given that hypothesis, the objective of the empirical analysis is to determine whether (or how closely) the observed structure matches the optimal one. The other class of studies is more concerned about the causal processes that generate the covariances and there may be many possibilities to entertain because

modularity is thought to arise from a variety of causes, resulting in multiple candidate hypotheses (Marquez, 2008; Monteiro et al., 2005; Zelditch et al., 2009). For example, there might be multiple competing hypotheses of either developmental or functional modularity or competing hypotheses of developmental and functional modularity, and the objective is to determine which model fits best (Herrera et al., 2002; Parsons et al., 2012).

An important distinction between hypotheses of optimality and hypotheses about causal processes becomes obvious when the data do not fit the model. If the data do not fit the model of the optimal pattern, the conclusion is either that the structure of integration/modularity is not optimal or that the model is incorrect about the optimal structure. In contrast, if the data do not fit the causal theory, the conclusion is usually that either that that particular process does not explain the structure of variation or that many processes structure covariation, and because they overlap in their modular boundaries, the relationship between developmental and variational modularity can be obscure (e.g., Hallgrímsson et al., 2009).

3.3.1 | The matrix correlation between the observed and expected correlation/covariance matrices

This method compares two matrices, the expected matrix, that is, the one predicted by the hypothesis, to the observed, using a matrix correlation as a test statistic (e.g., Cheverud, 1995; Cheverud et al., 1989; Hallgrímsson et al., 2004). The hypothesis is usually phrased rather crudely, by a correlation matrix predicting correlations of 1 for pairs of traits that are predicted to be within the same module and 0 for pairs of traits are predicted to be within different modules. The similarity between that and the observed matrix is measured by a matrix correlation (calculated between the upper or lower triangular matrix, omitting the diagonal). The statistical significance of that correlation is typically tested by randomly permuting the elements of one matrix and estimating the correlation between the two matrices at each iteration (Dietz, 1983). The null hypothesis is that the two matrices are no more similar than expected by chance; rejecting that null does not mean that the two matrices are highly similar. Nor does it mean that traits that are predicted to belong to different modules are independent of each other. Rather it means that the correlations among traits that are predicted to belong to the same module are higher, on average than for those that are predicted to belong to different modules.

3.3.2 | Comparing RV coefficients or covariance ratios across models

As well as testing for the presence of modularity, the RV coefficient provides a test statistic for hypotheses delimiting modules because, if the hypothesis regarding the boundary between modules is correct, there will be few interactions between the modules, but strong ones within them, hence, the RV coefficient will be low, lower than between randomly partitioned blocks of data. The position of the randomly placed boundary between modules will cross modular boundaries and therefore contain variables from the same module (Klingenberg, 2009). A hypothesis of modularity is therefore tested by comparing the RV for the data to the distribution of RV coefficients from the randomly partitioned data, counting the number of times when the RV from the randomly partitioned data exceeds the observed value. The statistical significance of the hypothesis is then determined from the proportion of cases in which the RV equals or exceeds the value obtained from the data.

The logic, if not the statistical approach, is similar when using the CR as a test statistic because, again, if the hypothesis regarding the boundary between modules is correct, the CR will be lower than expected from randomly distributed data. For purposes of comparing samples or hypotheses, the sample or model with the lowest Z_{CR} indicates a stronger modular signal than the other(s). Thus, the hypothesis yielding the statistically significantly lowest Z_{CR} is taken as the one that best describes the modular structure of the data.

3.3.3 | Evaluating models of modules as subspaces (MINT)

This method, introduced by Marquez (2008), begins with a hypothesis that assigns each variable, such as an anatomical point, to a single module; like other methods for delimiting modules, this one presumes that the modules are nonoverlapping (no variable belongs to more than one module). The data are then copied as many times as there are modules, for example, if there are three modules, there will be three copies of the data, with all the extra-modular variables assigned a value of zero. The three copies are then combined (by rows) so there are as many rows as there are individuals, times the number of modules. This method was developed for geometric morphometric data, and the resultant hypothesis cannot be realistic for those data because the Procrustes superimposition induces covariances. Consequently, the data matrix is superimposed, and the expected covariance matrix is estimated from this extended data set. Then the

fit of the data to the hypothesis is evaluated and models are ranked. After this initial analysis, modules from different hypotheses can be combined yielding more complex models that are not strictly modular in that some variables will belong to more than one module. In such cases, the expected variance of the duplicated landmarks must be adjusted accordingly (so that the variance of the landmark is distributed equally to each module, dividing it by the square root of the number of modules in which it appears). There are currently two implementations of the method (Marquez, 2014; Melo et al., 2015), although they are not strictly equivalent.

3.3.4 | Evaluating modularity with maximum likelihood (EMMLi)

This method, introduced by Goswami and Finarelli (2016) examines the relative fit of multiple hypotheses to an observed correlation matrix. Unlike the other methods used for landmark data, this one analyzes correlations rather than covariances. It treats the observed trait correlations (r) as a set of realizations of a hypothesized “true” correlation (ρ) and uses a maximum likelihood approach to estimate the ρ values within each hypothesized module and between each module pair and an overall likelihood for each hypothesis of modularity. In addition to estimating the log-likelihood for each modular structure, it evaluates whether modules have similar or different within-module ρ values and whether all between-module correlations have similar or different ρ values, meaning that up to four different parameterizations of a single model of modularity are possible. These hypotheses of modularity can then be compared using model selection but, as currently implemented, the small-sample corrected Akaike Information Criterion may favor overparameterized models (Adams & Collyer, 2019; Bardua et al., 2019; Fabre et al., 2020; Marshall et al., 2019; Watanabe et al., 2019). Alternative criteria (e.g., Extended Information Criterion, Bayesian Extended Information Criterion) perform better in simulations across a variety of conditions but support the same results as AIC in empirical analyses. The estimated ρ values provide information on the structure of correlations across a data set that can be used to construct alternative models that have not been considered in the initial analysis. Determining how much within- and between-module ρ values need to differ to delineate modules is an important consideration. To address this, model selection should be followed by further consideration of the supported model both to identify patterns that may not have been considered and to identify sets of modules that may have relatively high correlations

with another module or modules, even if not as high as the within-module correlations, a nuance that could be missed when reporting only the best supported model.

3.3.5 | Correlations between distance matrices of multivariate variables

This method, introduced by Monteiro et al. (2005), differs strikingly from the others in that the data consists of the pairwise distances between individuals, not the morphometric variables. Like other methods, this one subdivides the landmarks into modules, but the next step is to calculate the pairwise morphological distance matrix (among individuals) for each module. Given those distance matrices, the integration between them is estimated by a matrix correlation. This correlation thus says how well the position of an individual's shape relative to all others' shapes for one module predicts that individual's shape relative to all others' shapes for another module. This method does not measure the correlations *within* a module, and therefore does not test a hypothesis that those are higher than the correlations between them—it simply tests for correlations between proposed modules. If the integration within them is to be tested, the putative modules can be divided into two or more sets of landmarks, and the correlations between those can then be measured. The resultant correlation matrix, which has as many elements as there are modules, can be further analyzed by methods commonly used for correlation (or partial correlation) matrices such as path analysis (Monteiro & Nogueira, 2010) and graphical modeling (Zelditch et al., 2009).

3.3.6 | Graphical modeling

This is one of the methods, like Partial Least Squares Analysis (Mitteroecker & Bookstein, 2007) and confirmatory factor analysis (Zelditch, 1987, 1988) that can be traced to path analysis (Wright, 1921). Graphical modeling represents (probabilistic) models as graphs (Edwards, 2000). Graphical models depict conditional independence relationships between variables; the variables are the nodes of the graph, the edges between them depict the variables that are correlated conditional on the other variables, which could be all the other measured variables or a subset of them. Graphical modeling was introduced as an exploratory method for analyzing integration/modularity by Magwene (2001), proposing that modularity could be detected by examining the conditional independence of each pair of variables, conditional on *all others*. The question of whether two variables are

conditionally independent to a sufficient degree is tested by the edge-exclusion deviance, an information theoretic measure of whether a particular edge can be eliminated from a model that includes it. To test hypotheses of modularity, hypotheses are formulated as (undirected) graphs and evaluated for their relative fit to the data. The method described by Magwene is an exploratory analysis, which examines the conditional independence of each pair of variables, conditional on *all the others*. As discussed above, conditional independence is a theoretically important property of modularity in that traits from different modules may be correlated only because they are both correlated with some other trait; hence controlling for that common cause reveals that the two in question are (conditionally) independent. Controlling for *all* variables, however, can eliminate the evidence of modularity because each variable is conditioned on the other variables *within* the module, not just on those outside it (Mitteroecker & Bookstein, 2007).

3.4 | Are we measuring what we want or need to measure?

All these metrics and associated methods raise at least two questions about the meaning of “modularity”: (1) how modular does a structure need to be to qualify as modular? For example, if the cross-block covariances are far from zero, would a hypothesis of modularity still seem credible; is the statistical significance of modularity sufficient for credibility regardless of how highly the modules are integrated? (2) How well do these metrics capture what we mean by modularity? For example, if we think that modularity ought to mean that modules are conditionally independent, is that what the metric and method assess? Alternatively, if we think of modularity as a purely hierarchical concept, meaning that integration is higher, on average, within modules than between them, then does our metric capture that specific meaning?

The question “how modular is modularity?” may seem odd, taken out of context, but given a measure of modularity and a statistical test, we are still left with the question of whether statistically significant modularity indicates a high enough degree of modularity to correspond with our concept of modularity. This might not seem to be a highly subjective decision given an objective measure of the degree of modularity coupled with the statistical result, but highly integrated structures are sometimes deemed modular, and conversely, weakly integrated structures are considered integrated. For example, a high value for the RV coefficient (0.87) was taken as evidence of “a degree of modularity” because it is significantly lower than the values obtained by randomly

subdividing the data into two blocks (Drake & Klingenberg, 2010). Conversely, an average correlation coefficient (0.269) was taken as evidence of integration within modules because it is higher than the average for traits from different modules (0.105) (Cheverud, 1982). That high integration between modules can be reconciled with an inference of modularity and, conversely, that weak integration within modules can be reconciled with an inference of intra-modular integration, raises important and difficult questions: (1) At what point do we conclude that integration is too high to warrant an inference of modularity? (2) How much higher must within-module integration be relative to between-module integration to conclude that traits are quasi-autonomous? (3) Should that inference depend only on a statistical result, such as the number of permutations that yield a lower (or higher) value than the observed? (4) If that inference should depend on more than the statistical test, on what else should it also depend? Any answer to that last question presupposes that modularity is not simply a matter of degree but rather must attain some minimal level of internal coherence before modules actually qualify as modules.

Several methods raise additional questions specific to them. One arises from methods that give a summary statistic, such as the average within- and between-module correlations. The question is whether any simple summary is adequate when the magnitudes of the correlations vary substantially within or between modules. We do often need a summary statistic when comparing degrees of modularity across datasets. In particular, we do often want to know whether some species are more or less modular than others (Bell et al., 2011; Machado et al., 2018; Marroig et al., 2009), and we may also want to know whether one ontogenetic stage is more modular than others (Ackermann, 2005; Goswami et al., 2012; Ivanovic et al., 2005; Zelditch & Carmichael, 1989) or whether some structures are more modular than others in particular lineages (Haber, 2015; Hansen et al., 2007; Young & Hallgrímsson, 2005). The question is whether a simple summary obscures valuable information. For example, consider two hypotheses, one claiming that a structure consists of six modules, another that it consists of just two modules, each of which combines three of the six modules into one. When we test the two hypotheses, the two-module hypothesis may yield a stronger modular signal (e.g., a significantly lower value for the Z_{CR}). But if we then look at the pairwise CRs between the six modules, we may find that the highest of all is between two of the modules that are allocated to two different modules in the two-module hypothesis. Given the lower Z_{CR} , we might select that two-module hypothesis, but that decision ignores the evidence that those two modules are not internally integrated, at least not if we think that the very

low value between one pair of six-module model ought to count as evidence against that hypothesis.

Stepping back further, we can ask: Can any simple summary statistic effectively capture the network of interactions that describe modularity and integration of complex biological systems, or do we need a novel approach? But there is another more fundamental question, especially if we routinely find that no method for delimiting modules effectively captures the complexity of our biological system: Does that mean that we need a novel approach, or does it mean that real phenotypes are not actually modular? In asking that question, we are not disputing that organisms are modular in some general sense. However, they may comprise more complex, hierarchical systems, including non-nested modules, which hinder straightforward statistical detection by any method that cannot analyze that kind of hierarchy. Such questions obviously go beyond questioning just the utility of metrics. They concern the biological reality of modularity, at least insofar as the concept is applied to particular phenotypes. We cannot expect that any metric of modularity, or any method for delimiting modules, would work well when the phenotypes are not, in fact, strictly modular. This fundamental issue may have parallels in the long-standing debates on the definition of species, and similarly, may ultimately come down to identifying the most practical and meaningful approach for a specific system or question.

In the case of modularity, we need to match our selected metric and method to the definition of modularity presupposed by the theory motivating the analysis. Thus, in the following sections, we consider the theoretical context(s) of modularity.

4 | WHAT DOES MODULARITY MEAN FOR EVOLUTION?

The idea that modularity has important consequences for evolution motivates most of the literature on this subject, both theoretical and empirical. This is a complicated subject because there are at least two types of questions, which might be related, but often are not. The first concerns the impact of genetic and/or phenotypic integration (**G** and **P** matrices) on evolution (Haber, 2016; Machado et al., 2018; Rossoni et al., 2019; Sanger et al., 2012). The second concerns the structure of evolutionary integration (Claverie & Patek, 2013; Collar et al., 2014). These two classes of studies are partly related in that **G** and **P** are among the causes of evolutionary integration, but there are other causes of correlated evolutionary changes, most obviously natural selection. Some studies link modularity at the individual

level to macroevolutionary outcomes, such as developmental modularity to diversification rates (Gerhart & Kirschner, 2007; Yang, 2001) or disparity (Fish et al., 2011). There are also studies that relate modules of gene regulatory networks to mosaic pleiotropy to types of morphological change (Carroll, 2008), and others that relate changes in anatomical modularity, to acquisitions of novel functions (Navarro-Diaz et al., 2019), and others that studies relate properties that might be related to modularity of development and function (i.e., “versatility”) to macroevolutionary outcomes, often to disparity and/or evolutionary rates (Liem & Osse, 1975; Liem, 1980; Vermeij, 1973a, 1973b).

Meshing the several theoretical frameworks employed in this large literature is challenging, assuming it is even possible. Even when it does seem possible, that meshing often requires major efforts of translation. Unless the language is very clear, and often it is not, translation can be difficult and it is further complicated by the link often made to evolvability, which, like “modularity” has acquired multiple meanings (Brown, 2014; Crother & Murray, 2019; Pigliucci, 2008). In the following sections, we discuss two of the conceptual frameworks for macroevolutionary studies, one that predates the modern theory, but which remains influential, while the other is based on evolutionary theory.

4.1 | Versatility, degrees of freedom, and decoupling

The main idea underlying the concept of “versatility” is simple: “versatility” increases with the number of morphogenetic and/or functional parameters because that increases the number of parameters that can be varied independently (Liem & Osse, 1975; Liem, 1980; Vermeij, 1973a, 1973b). That conjecture, when framed specifically in terms of function, states: “It is axiomatic that concomitant with increased potential versatility in form there will be a proliferation in the possible functional solutions not only to a given mechanical problem but also to an increasing variety of biological problems” (Liem & Osse, 1975; Liem, 1973). This idea has also been framed as the “decoupling hypothesis” (Emerson, 1988; Lauder, 1981; Schaefer & Lauder, 1996), which relates the number of independently variable parameters to the ability to evolve novel morphologies. The decoupling hypothesis is still widely cited for the proposed link between versatility (or flexibility) and morphological disparity (Frederich et al., 2014), versatility and speciation, and speciation and divergence (Rabosky & Adams, 2012; Rabosky et al., 2014; Yang, 2001), as well as to the evolution of distinct life stages in metamorphic organisms (Bonett & Blair, 2017;

Moran, 1994; Schlosser, 2005; Vucic et al., 2019). The literature is quite varied in its usage of “versatility,” variously defined as the ability to respond in different directions under diversifying selection (Lovette et al., 2002), making it nearly synonymous with flexibility (Marroig et al., 2009),

or alternatively as “lability,” meaning “more capable of evolutionary change,” which could correspond to any one (or a combination) of respondability, evolvability, and flexibility (Hansen & Houle, 2008). We do not attempt to sort out such ambiguities here (but see Box 1).

Box 1 Properties of covariance matrices

All the metrics presented in this section, except flexibility (f) are defined and discussed in more detail in Hansen and Houle (2008).

Evolvability of a trait, in response to a particular selection gradient (β), is the length of the response to selection ($\Delta\bar{z}$) in the direction of the selection gradient (β):

$$1) e(\beta) \stackrel{\text{def}}{=} \frac{\beta' G \beta}{|\beta|^2},$$

where G is the population genetic-covariance matrix (Hansen & Houle, 2008). Because G often cannot be measured, it is commonly replaced by P , the phenotypic covariance matrix. In the case of a single trait, measured on a ratio scale, evolvability reduces to the mean scaled genetic variance. In the case of a multivariate data, evolvability is the length of the response vector on the selection gradient (if that gradient is aligned with an eigenvector of G , evolvability is the eigenvalue of that vector). Evolvability thus depends on the direction of the selection gradient. It is possible to assess evolutionary potential in the more general case by calculating the average evolvability over random selection gradients.

Conditional evolvability (c) is the ability of a trait (or complex of traits) to evolve in the direction of selection *when other traits are under stabilizing selection*:

$$2) c(\beta) \stackrel{\text{def}}{=} (\beta' G^{-1} \beta)^{-1}$$

Like unconditional evolvability, conditional evolvability is specific to a particular direction. Again, it is possible to get a more general measure; for high-dimensional data (e.g., geometric shape data), the average conditional evolvability converges on the harmonic mean of the eigenvalues of G .

A metric that may provide the best estimate of the impact of modularity on the evolution of complex adaptations, is autonomy (a), which measures the proportion of evolvability that remains after conditioning on those other traits. Integration (i), measures the decrease in evolvability due to conditioning on those traits, so autonomy (a) is the proportion of the evolvability that is left:

$$3) a(\beta) \stackrel{\text{def}}{=} 1 - i(\beta) = \frac{c(\beta)}{e(\beta)},$$

When the variation along β is independent of the variation in other directions, $i = 0$ and $a = 1$. This can also be averaged over all traits. Average autonomy decreases with increasing variation of the eigenvalues (a measure of the degree of integration, see (6), above).

Two other important concepts are respondability and flexibility. Respondability (r) is the length of the evolutionary response vector:

$$4) r(\beta) \stackrel{\text{def}}{=} \left| \frac{\Delta \bar{z}}{\beta} \right|$$

“Flexibility” (f) is the ability to evolve in the direction of the selection gradient (regardless of the magnitude of the change), which is the cosine of the angle between the evolutionary response and the direction of the selection gradient (the ratio between evolvability and respondability) (Marroig et al., 2009).

4.2 | Macroevolutionary consequences of variational modularity/genetic integration

A formal theory that links modularity to evolvability is quantitative-genetic theory extended over macroevolutionary time-scales. The foundation is often called “the Lande equation,” the origin of the model of adaptive evolution in modern macroevolutionary studies (e.g., Butler et al., 2004; Hansen & Martins, 1996; Harmon et al., 2010; Houle et al., 2017): The equation describes how the mean phenotype will change under selection over a single generation. We will obviously need to get from it to a theory about evolution over vastly longer time scales, and obtain the predictions for evolutionary correlations, but the starting point of the theory is:

$$\Delta\bar{z} = \mathbf{G}\mathbf{P}^{-1}\mathbf{S} \quad (13)$$

(Lande, 1979)

This states that the change in the mean phenotype $\Delta\bar{z}$, is a function of the genetic covariance matrix, $\mathbf{G}$ and the selection differential, $\mathbf{S}$, which is the difference between the selected parents and the base population. The multivariate extension of the breeder's equation is often rewritten as:

$$\Delta\bar{z} = \mathbf{G}\beta \quad (14)$$

(Lande & Arnold, 1983)

In this expression, β is the selection gradient, $\mathbf{S}\mathbf{P}^{-1}$, which is the strength of selection acting on each trait, holding all other traits constant; therefore, it measures selection on that trait, eliminating the indirect effects of genetic correlations with other traits. What this means is that the realized evolutionary change (the response to selection) can be deflected from the selection gradient, β , the shortest path to the optimum, by $\mathbf{G}$ (Cheverud, 1982, 1984).

Considering that this equation is about the change from one generation to the next, the question is whether $\mathbf{G}$ matters for the direction and rate of evolution over macroevolutionary time scales. Two influential early studies, both of which are relevant for deriving the expectations for evolutionary correlations, concluded that $\mathbf{G}$ does not matter over the long term; so long as there is genetic variance in the direction of the optimum, populations will eventually reach it (Felsenstein, 1988; Zeng, 1988). However, others concluded that $\mathbf{G}$ can have a profound impact on adaptation over long time-scales (Burger, 1986; Price et al., 1993; Wagner, 1984a, 1988), and $\mathbf{G}$ and $\mathbf{P}$ can have strikingly counterintuitive effects (Burger, 1986; Wagner, 1988). The difference between the models is the model of the adaptive landscape. That accounts for the most counterintuitive

conclusion: that phenotypic variance *not* in the direction of selection can impede adaptation (Wagner, 1988).

The important feature of the model is that directional selection acts along a ridge on the adaptive landscape, favoring particular combinations of traits, with stabilizing selection in all directions perpendicular to that ridge. What makes this model distinctive is that selection does not first increase one trait, then another, and then another, an unrealistic model when the adaptive value of one trait depends on the phenotype of another. That dependence is often exemplified by the upper and lower tooth row lengths because those must match each other to be functional: whether selection favors a longer or shorter jaw, a mismatch between upper and lower jaws will reduce fitness (Cheverud, 1984). In such cases, traits do not contribute independently (additively) to fitness. Rather, their relationships to each other matter. This is a model for functionally coupled traits under restricted index selection, that is, selection acts to increase the mean score on a multivariate index without altering the means of traits already at an optimum. In this case, there is an upper limit on the rate of evolution, when there are more than three functionally coupled but genetically uncorrelated traits (Wagner, 1984a). Thus, genetic integration increases the rate of adaptation so long as functionally coupled traits belong to the same module, and it does not matter whether traits are interdependent in development or function because what matters is their joint impact on fitness (and the resultant adaptive topography). The reason why phenotypic variance can impede adaptation is that, as phenotypic variance increases, progressively more of it lies outside the adaptive corridor. Hence the negative effect of that variance (for traits under stabilizing selection) will, at some point, outweigh the positive effects of variation along the corridor (Wagner, 1988).

In such adaptive landscapes, deviations from the optimal pattern of genetic and phenotypic covariances restrict the ability of a population to move along the adaptive ridge (Burger, 1986; Wagner, 1988). In rugged adaptive landscapes characterized by multiple peaks, those constraints might result in populations evolving toward phenotypically remote peaks that are more accessible given the pattern of genetic and phenotypic covariances (Burger, 1986). Consequently, in these more complex, and arguably more realistic, models, the optimal structure of modularity is to have high genetic correlations among functionally coupled traits and independence of functionally unrelated traits. That is the genetic architecture expected under the imitatory epigenotype and related hypotheses (Bonner, 1988; Cheverud, 1982, 1984; Lande, 1980; Riedl, 1977, 1978; Wagner & Altenberg, 1996). The expectation is that the genetic covariance matrix ($\mathbf{G}$) will be proportional to the pattern of stabilizing selection, represented in the fitness matrix ($\mathbf{W}$), with small values for trait combinations for which variation is not tolerated

(because deviations reduce fitness) and large values for trait combinations that differ little in performance (Cheverud, 1984).

Not surprisingly, these models have inspired a large literature (e.g., Baab, 2018; Goswami et al., 2014; Grabowski et al., 2011; Haber, 2016; Marroig et al., 2009; Shirai & Marroig, 2010). Most estimate phenotypic rather than genetic covariance matrices and compare these among taxa before reconstructing selection gradients and analyzing the directions and rates of evolutionary change. One recent study estimated $\mathbf{G}$ for seven species of West Indian *Anolis* lizards and found that evolutionary change is strongly biased in the direction of the major axis of $\mathbf{G}$ for over 40 million years even though $\mathbf{G}$ itself evolved (McGlothlin et al., 2018). Several have shown that integration and modularity can have long-term effects on evolutionary rates and directions, and that $\mathbf{P}$ in lineages that exhibit high disparity have higher flexibility (and/or lower eccentricity) than less disparate clades (Haber, 2016). For example, dogs (Canidae), differ from other Carnivora in having higher integration, evolvability, flexibility (i.e., the ability to respond to selection in many directions), and allometric coefficients on the facial region (Machado et al., 2018). Those attributes were hypothesized to enable dogs to adapt rapidly and to explain why artificial selection was able to generate the great morphological variety of dog breeds. How modularity and integration affect evolutionary rates over macroevolutionary time scales remains a subject of active research—one the one hand, integration may lessen the flexibility to respond to selection in multiple directions, but on the other hand, it can facilitate responses along the major axis of $\mathbf{G}$. Because modules can evolve independently of each other, each module might evolve in different directions at different rates, leading to high disparity of the whole complex.

4.3 | Evolutionary integration and modularity

In contrast to studies that examine the long-term impact of $\mathbf{G}/\mathbf{P}$ on $\Delta\bar{z}$, others examine the modularity of $\mathbf{D}$. As mentioned above, two studies concluded that $\mathbf{G}$ does not have a long-term impact on evolutionary change. We return to these because both analyzed interpopulational integration. One modeled constant stabilizing and variable directional selection, finding that the response to selection (over the long-term) depends only on the fitness functions of stabilizing and directional selection, independent of genetic and phenotypic correlations (Zeng, 1988). The other modeled constant directional but variable stabilizing selection, a model of randomly wandering adaptive peaks (Felsenstein, 1988). What these models show is that evolutionary correlations can arise by various processes. For

example, when evolution occurs by random genetic drift, genetically correlated traits will be correlated in their evolutionary changes (Arnold et al., 2001; Lande, 1979; Revell & Harmon, 2008). Alternatively, evolutionary correlations can also arise when traits are genetically uncorrelated, but co-selected, each one being favored in a given environment independently of the others, such as Felsenstein's (1988) example of selection in cold environments favoring lighter colors, larger size and shorter limbs. Evolutionary integration can also arise by correlational selection favoring combinations of traits that jointly affect performance (Armbruster & Schwaegerle, 1996; Walker, 2007). They can even arise from random variation in the optima, which generates correlations due to the varied optima, stabilizing selection, random genetic drift, and their interactions (Hansen & Martins, 1996).

Early theoretical studies modeled populations simultaneously diverging from the same ancestral population; hence there is no phylogenetic structure in the data. In the case of populations all diverging by random genetic drift, the expected covariance structure of the mean phenotypes of the diverging populations ($\mathbf{D}$) is:

$$\mathbf{D} = \left(\frac{t}{N_e} \right) \mathbf{G}, \quad (15)$$

where t is the number of generations and N_e is the effective population size (Lande, 1979).

That model was reformulated in a phylogenetic context, initially by estimating evolutionary covariances of independent contrasts (Baker & Wilkinson, 2003; Klingenberg & Marugan-Lobon, 2013; Revell, 2007), and then by Phylogenetic Generalized Least Squares, which incorporates a matrix of phylogenetic distances among species (in units of time since divergence), usually termed the $\mathbf{C}$ matrix (Adams & Felice, 2014; Revell & Harmon, 2008). Expressed in terms of that matrix:

$$\mathbf{D} = \mathbf{R} \otimes \mathbf{C}. \quad (16)$$

In that expression, the operator $\otimes$ (Kronecker delta) multiplies each element of $\mathbf{R}$ by each element of $\mathbf{C}$. $\mathbf{R}$ is usually called the “rate matrix” because the Brownian rates (σ^2) are along the diagonal of $\mathbf{R}$ (the evolutionary covariances due to drift are on the off-diagonals). In the case of Brownian motion (BM), $\mathbf{R}$ is sufficient to describe the evolutionary dynamics of phenotypic divergence, and, as in equation 15 above, $\mathbf{R}$ is proportional to $\mathbf{G}$ (Felsenstein, 1988; Lande, 1979; Revell & Harmon, 2008; Zeng, 1988, 1989).

As of this writing, this is the only model so far developed explicitly for analyses of evolutionary integration. Consequently, nearly all empirical studies of evolutionary integration and modularity assume that phenotypic evolution can be modeled as Brownian motion, applying methods

commonly used to study integration and modularity to the Brownian rate matrix ($\mathbf{R}$), such as PLS (Alvarez et al., 2015; Bright et al., 2016; Hedrick et al., 2019; Klingenberg & Marugan-Lobon, 2013; Randau & Goswami, 2017); RV coefficient (Alvarez et al., 2015; Klingenberg & Marugan-Lobon, 2013); Covariance Ratio (Dellinger et al., 2019; Du et al., 2019; Perez et al., 2020), and EMMLi (Evans et al., 2019; Felice et al., 2019; Hedrick et al., 2019). If the Brownian model is not true, the parameter estimates, that is, the evolutionary covariances among traits and evolutionary rates, are not correctly estimated. In the absence of methods for incorporating other models of trait evolution in studies of integration and modularity, one possibility is to fit simple models, including BM, a single stationary adaptive regime, OU1 (Hansen & Martins, 1996; Hansen, 1997), an early burst (Harmon et al., 2010), and/or Pagel's lambda (Pagel, 1999) and, after evaluating the support for those models, use the evolutionary covariance matrix for that model (Clavel et al., 2019). A related method rescales branches using a variable Brownian rates model (Pagel et al., 2004), and the tree with the transformed branch lengths is then used as the phylogenetic framework for analyses of evolutionary modularity (e.g., Fabre et al., 2020; Navalon et al., 2020). Ideally methods will allow estimation of evolutionary covariances jointly with alternative evolutionary models, and we anticipate that more direct methods will soon become available with the rapidly expanding toolkit for evolutionary modeling of multivariate traits.

Studies that link evolutionary to variational modularity can reveal the macroevolutionary significance of integration and modularity, such as determining whether the genetic, developmental, and functional interactions of traits that underlie variational modularity also predict evolutionary integration (e.g., Bardua et al., 2019; Marshall et al., 2019). Some compare the evolutionary covariance structures across lineages (e.g., Collar et al., 2014; Du et al., 2019; Felice et al., 2019) or compare the strength of evolutionary integration and modularity across subsets of traits (Vidal-Garcia & Scott Keogh, 2017). Additionally, some also analyze the relationship between evolutionary rates and evolutionary covariances (Navalon et al., 2020; Perez et al., 2020) or that between evolutionary modularity/integration, disparity, and evolutionary rate (Bardua et al., 2020; Conith et al., 2020; Du et al., 2019; Evans et al., 2019; Fabre et al., 2020; Felice & Goswami, 2018; Larouche et al., 2018; Watanabe et al., 2019).

5 | MATCHING CONCEPTUAL DEFINITIONS, METRICS, MATRICES, AND THEORIES

A major difference between studies at the phenotypic and evolutionary scales is that evolutionary integration and modularity are outcomes of evolutionary processes that are

potentially constrained or facilitated by the phenotypic integration and modularity of $\mathbf{G}$ and $\mathbf{P}$. This distinction is sometimes blurred in studies of evolutionary integration, and it is important to note the critical difference in interpreting phenotypic versus evolutionary analyses of integration and modularity. Phenotypic integration and modularity are linked to evolvability, and the hypothesis that high integration can constrain evolutionary change is acting on, and can be studied at, that scale of analysis. In contrast, analyses of evolutionary integration concern the integration of evolutionary changes, rather than the effects of integration on evolutionary change. That these are easily and often confused exemplifies the issue that the multiple definitions and scales of modularity can lead to illogical reasoning when the meaning switches over the course of an argument, as in this classic fallacy of equivocation: "The end of a thing is its perfection. Death is the end of life. Therefore, death is the perfection of life." In this case, the fallacy is so obvious that it is amusing, but illogical theoretical arguments are not always so obvious (or amusing). There is no simple solution to the problem posed by a word having multiple meanings—definitions are a matter of social convention so they cannot be empirically true or false; a word means whatever we, as a community, want it to mean. But theories are not merely matters of social convention. Ambiguous language becomes a serious scientific problem when that leads to disagreements about what a theory states or what the results of a study say about that theory. We can live with disputes about whether modularity ought to mean "X" or "Y" so long as we do not confuse theories about "X" with those about "Y." What is harder to tolerate is a jumbling of "X" and "Y" so we might talk about "X" but measure "Y" and test hypotheses about "Z."

The two of us cannot demand that everyone else (or even each other) use our preferred definition of "modularity," and our preferred metric and method. At best, we can argue for the one(s) we prefer, but we cannot insist that everyone agree with either of us. Nor can we recommend any general solution to the problem of ambiguity. What we can recommend is that authors anticipate being misread and take pains to prevent it by being explicit about their definition of modularity and about the level at which they are analyzing modularity, considering whether the theory being tested is actually a theory about the level of modularity analyzed, and ensuring that the metric (and method for delimiting modules) follow logically from the definition and theory.

The diversity of perspectives on modularity is not, by itself, a problem. There are diverse perspectives on most concepts. The variety of metrics and methods for analyzing are also not a problem, although some methods and metrics may be problematic. Progress in this field, like in any other, requires diverse perspectives, debates about theories, and development of new methods. But we cannot make progress

if we regularly speak past each other. We may not agree on what “modularity” ought to mean, but we can still convey precisely what we ourselves mean and ensure that our concepts, metrics and theories match each other. Then we can evaluate how different studies contribute to our understanding of modularity, whatever it means.

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CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

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Evolvability: progress and key questions

Christophe Pélabon , Gustavo A. Agudelo-Cantero, Yimen G. Araya Ajoy, Geir H. Bolstad , Changde Cheng , Frietson Galis, Frederic Guillaume, Thomas R. Haaland , Benedikt Hallgrímsson , Thomas F. Hansen, Agnes Holstad , David Houle , Gene Hunt , Anders Isaksen, Lisandro Milocco, Naid Mubalegh, Laura Nuño de la Rosa, Steven H. Orzack, Arthur Porto, Jane M. Reid, Jacqueline L. Sztepanacz , Eivind A.B. Undheim, Cristina Villegas , Kjetil L. Voje , Craig Walling and Jonathan Wright

Christophe Pélabon (christophe.pelabon@ntnu.no), Yimen G. Araya Ajoy, Thomas R. Haaland, Agnes Holstad, Jane M. Reid, and Jonathan Wright are affiliated with the Department of Biology at the Norwegian University of Science and Technology, in Trondheim, Norway. Gustavo A. Agudelo-Cantero, Thomas F. Hansen, Anders Isaksen, Naid Mubalegh, and Eivind A. B. Undheim are affiliated with the Centre for Ecological and Evolutionary Synthesis, in the Department of Biosciences at the University of Oslo, Blindern, in Oslo, Norway. Geir H. Bolstad and Agnes Holstad are affiliated with the Norwegian Institute for Nature Research, in Trondheim, Norway. Changde Cheng is affiliated with the Department of Medicine, Stem Cell Program and with the Department of Biomedical Informatics and Data Science at the University of Alabama, in Birmingham, Alabama, in the United States. Frietson Galis is affiliated with the Naturalis Biodiversity Center, in Leiden, in the Netherlands. Frederic Guillaume is affiliated with the Organismal and Evolutionary Biology Research Program at the University of Helsinki, in Helsinki, Finland. Benedikt Hallgrímsson is affiliated with the Department of Cell Biology and Anatomy at the University of Calgary, in Calgary, Alberta, Canada. David Houle is affiliated with the Department of Biological Science at Florida State University, in Tallahassee, Florida, in the United States. Gene Hunt is affiliated with the Department of Paleobiology, National Museum of Natural History, at the Smithsonian Institution, in Washington, DC, in the United States. Lisandro Milocco is affiliated with the Department of Zoology, SciLifeLab, at Stockholm University, in Stockholm, Sweden. Naid Mubalegh is also affiliated with the Institut d'Histoire des Sciences et des Techniques at the Université Paris Panthéon-Sorbonne, in Paris, France, and with the Centro de Filosofia das Ciências da Universidade de Lisboa, at the University of Lisbon, in Lisbon, Portugal. Laura Nuño de la Rosa is affiliated with the Department of Logic and Theoretical Philosophy at Complutense University of Madrid, in Madrid, Spain. Steven H. Orzack is affiliated with the Fresh Pond Research Institute, in Cambridge, Massachusetts, in the United States. Arthur Porto is affiliated with the Florida Museum of Natural History and with the Department of Biology at the University of Florida, in Gainesville, Florida, in the United States. Jacqueline L. Sztepanacz is affiliated with the Department of Ecology and Evolutionary Biology at the University of Toronto, in Toronto, Ontario, Canada. Cristina Villegas is affiliated with the Konrad Lorenz Institute for Evolution and Cognition Research, in Klosterneuburg, Austria. Kjetil L. Voje is affiliated with the Natural History Museum at the University of Oslo, in Oslo, Norway. Craig Walling is affiliated with the Institute of Ecology and Evolution, in the School of Biological Science at the University of Edinburgh, in Edinburgh, Scotland, in the United Kingdom.

Abstract

Since the 1990s, evolutionary biologists have recognized the importance of explaining the ability of biological systems to evolve and how this ability itself evolves. This recognition of the need to explain evolvability emerged from an awareness that the kind and the amount of heritable variation available for natural selection require explanation. The concept of evolvability is now the focus of many research programs in diverse subdisciplines within evolutionary biology. In the present article, we first review and synthesise progress made in evolvability research. We then present key questions to set an agenda for future research on evolvability, identify challenges to answer these questions, and discuss opportunities to apply results from the evolvability research to conservation biology.

Keywords: evolvability, evolutionary quantitative genetics, evo-devo, paleobiology, evolutionary systems biology

Rates of evolution are highly variable within and among traits, populations, and species and at both micro- and macroevolutionary timescales (Simpson 1953, Uyeda et al. 2011). We also often observe large differences in species diversity and morphological divergence among related lineages (figure 1). For example, in angiosperms, clades displaying bilateral symmetric (zygomorphic) flowers combined with other traits that increase precise pollination, include more species than their sister clades displaying the alternative character states including radially symmetric (actinomorphic) flowers (O'Meara et al. 2016). This may be explained by differences in selective histories, as would be expected if bilateral symmetry increases the precision of pollination by directing pollinator movement and increases the likelihood of visitation by specialist pollinators, both of which would increase the speciation rate (Armbruster 2014). However, these differences in diversity among clades may also reflect differences in evolvability—that is, differences in the ability to produce and maintain variation on which natural selection acts

(Wagner and Altenberg 1996). This suggestion was already made by Simpson (1953) but only in the last 30 years has interest in the concept of evolvability become widespread.

The origins of phenotypic variation and how differences in the amount and type of variation affect the potential for evolution have interested biologists since Darwin (Crother and Murray 2019, Distin 2023). This interest has shaped research in the study of evolutionary morphology, paleobiology, and macroevolution (e.g., Simpson 1953, Olson and Miller 1954, Vermeij 1973), but this work was often overlooked in other areas of evolutionary biology. In the 1990s, however, several research programs emerged that seek to explain differences in rates of evolution, divergence, and evolutionary innovation not only through variation in natural selection but also through the production and maintenance of heritable variation (Brigandt 2015, Nuño de la Rosa 2023). Evolvability rapidly became the focus of research in diverse disciplines such as evolutionary developmental biology (evo-devo; e.g., Alberch 1991, Raff 1996, Wagner and Altenberg 1996,

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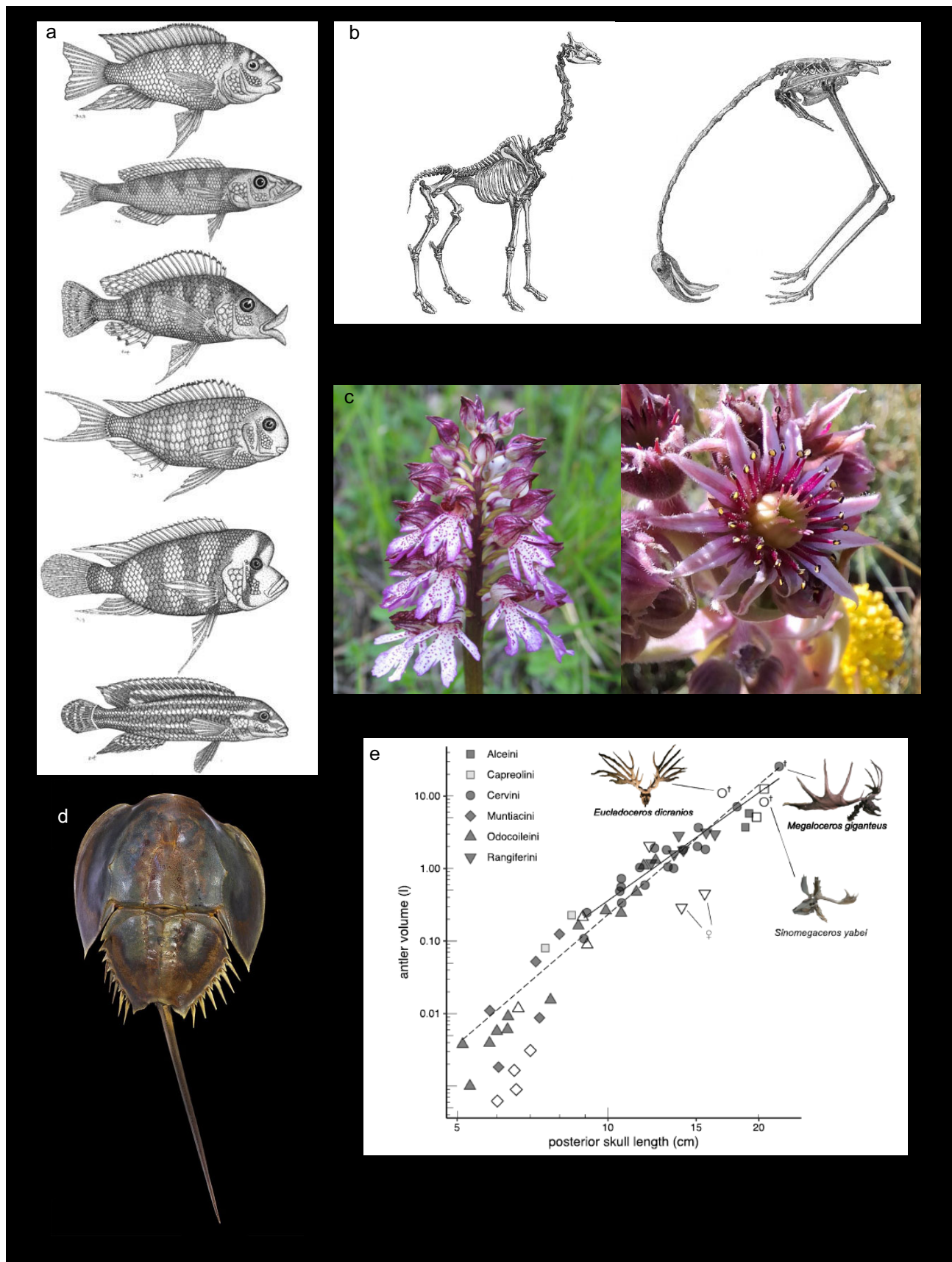


Figure 1. Example of evolutionary questions involving evolvability. (a) Species of cichlids in lake Tanganyika differ extensively in their craniofacial morphology and in their feeding regimes. Galis and Drucker (1996) suggested that the evolution of such diversity was made possible by developmental changes that allowed independent movements of the upper and lower pharyngeal jaws, thereby increasing the evolvability of the feeding apparatus.

Figure 1. (Continued) (b) Although almost all mammalian species have only seven cervical vertebrae whatever the length of their neck, bird species differ extensively in the number of cervical vertebrae; this allows the evolution of long and flexible neck, as in flamingoes. Why is the number of cervical vertebrae not evolvable in mammals? Galis (2023) suggested that the constraint in mammals results from the deleterious pleiotropic effects associated with changes in the induction of cervical vertebrae. (c) O'Meara and colleagues (2016) showed that clades with bilateral flowers (e.g., *Orchis purpurea*, Orchidaceae, left) contain many more species than clades with radial flowers (e.g., *Sempervivum montanum*, Crassulaceae, right). Whether this results from differences in the selection pressures generated by specialist or from generalist pollinators or from differences in evolvability remains unresolved (Woźniak and Sicard 2018). (d) The morphology of the horseshoe crabs (Xiphosurida) has changed little since the Jurassic. Whether this lack of evolution can be explained by natural selection or by low evolvability is unresolved (Bicknell et al. 2022). (e) The allometric relationship between antler size and body size in cervids, with the iconic *Megaloceros*, is a key example of evolutionary constraints where body size and antler size are supposedly forced to evolve along a trajectory imposed by the development (Gould 1974, Tsuboi et al. 2024). However, the low evolvability in static allometric slopes but the high evolvability in static allometric intercepts (Bolstad et al. 2015) suggest that allometric constraints alone cannot fully explain the origin of evolutionary allometry observed in deer antlers and in many other systems. Sources: (a) from Albertson and Kocher 2006; (b) from Galis 2023 (<http://creativecommons.org/licenses/by/4.0>), left reproduced from Owen (1866), and right reproduced from Evans (1900). Photographs: (c) Christophe Pélabon, (d) Didier Descouens, (e) Tsuboi et al. 2024 (<http://creativecommons.org/licenses/by/4.0>)

Time-scales	Macroevolution		Microevolution	
	←		→	
Level of variation	Among lineages		Within and among populations	
Mechanisms of evolvability	Change in mutation rate Change in mutation target size Changes in information transfer - Chromosomes - Sexual reproduction - Multicellularity		Changes in the GP map - Modularity - Integration - Robustness	
			Changes in allele frequency - Selection - Migration - Drift - Mutation Changes in allelic effects - Epistasis - G×E	
Metrics	Disparity, versatility, complexity		Modularity, integration, robustness	
Outcomes	Innovation, novelty, speciation, extinction		Change in μ and V_A Population or species divergence	
			Change in μ	
Bearer	Genome		Developmental system, GP map	
			Population, trait	

Figure 2. Evolvability at different biological and temporal scales. Top row: levels at which variation in evolvability can be studied. Second row: mechanisms affecting evolvability. Third row: evolvability measures or proxies for it at different organismal levels and timescales (V_A , additive genetic variance; I_A , mean-scaled evolvability; V_M , mutational variance). Fourth row: evolutionary events expected to be affected by evolvability, and therefore how variation in evolvability is expressed (μ , population mean). Last row: typical bearers of evolvability.

Gerhart and Kirschner 1997), evolutionary systems biology (Kauffman 1993, Wagner 1996), and evolutionary quantitative genetics (Houle 1992, Hansen et al. 2003).

Defined as the ability of organisms to produce and maintain heritable variation, evolvability depends on the ability of developmental systems, in interaction with the environment, to convert differences in DNA sequences into phenotypic variation. Over a few generations (i.e., at microevolutionary timescales), evolvability mostly depends on population-level additive genetic variance V_A (figure 2; Houle 1992), which is the component of phenotypic variance due to additive effects of alleles (i.e., directly heritable and independent of context). Over tens to hundreds of generations, V_A and, therefore, evolvability is affected by the combined effects of mutation, natural selection, migration, and genetic drift (Bürger 1999, Pélabon et al. 2023). Over longer time periods (i.e., at macroevolutionary timescales), ongoing evolution onto phenotypic variation affects the production of V_A and therefore affects the evolvability of genomes and lineages (Wagner 2014). The resulting differences in evolvability among lineages affect their ability to adapt and diverge. For example, the greater variation of the feeding apparatus of fish species in the family

Cichlidae (figure 1) than in the Centrarchidae may have been spurred by developmental changes that allowed independent movements of the upper and lower pharyngeal jaws (Galis and Drucker 1996). Over even longer time periods, changes in evolvability may result from the evolution of mutational processes and of new forms of variation. For example, the evolution of sexual reproduction and multicellularity has facilitated new modes of individual organization and information transfer that led to new sources of variation such as recombination and new patterns of variation allowed by cell differentiation and modularity (Maynard Smith and Szathmary 1995, Gerhart and Kirschner 1997).

These different perspectives illustrate the plurality of the evolvability concept and the need for integrated studies across all biological levels and timescales to generate overarching understanding of the causes and constraints on the potential for evolutionary change (figure 2). In the present article, we further demonstrate this plurality and the need for a multidisciplinary approach of the concept by reviewing progress in evolvability research since the 1990s. We then outline a series of twelve key remaining questions concerning evolvability to provide an agenda for future research.

Progress in evolvability research

Conceptualization of evolvability: Evolvability describes the potential of biological entities to evolve, and it should be understood as an intrinsic capacity or dispositional property (Wagner and Altenberg 1996, Brigandt et al. 2023). Accordingly, over long time periods evolvability depends on variability, meaning the variation that can potentially be generated by developmental systems, irrespective of the current observed variation. This distinction has contributed to focusing attention on the processes that generate and maintain heritable variation and their evolution. It has also enabled the recognition that these processes influence evolutionary possibilities in a generative manner (i.e., by generating viable phenotypes), overcoming the limiting notion of constraint by rendering the processes compatible with evolutionary explanations involving selection (Brigandt 2015). Recognizing the dispositional nature of evolvability has also clarified several conceptual issues, including distinguishing what entities are evolvable (e.g., genes, traits, genomes), the features causally contributing to their evolvability, and the characterization of ways in which evolvability contributes to evolutionary changes (figure 2; Brigandt et al. 2023, Houle and Pélabon 2023, Villegas et al. 2023).

Evo-devo and evolutionary systems biology: The concept of evolvability is central to evo-devo research, which aims to understand the mechanisms generating and structuring phenotypic variation and how these mechanisms evolve (Hendrikse et al. 2007). Accordingly, evo-devo research has revealed some core principles and complexities of the developmental mechanisms that map sequence variation onto phenotypic variation (Hallgrímsson et al. 2009, 2023). Key points include demonstrating how new phenotypic variants in many systems result from small changes in developmental mechanisms, such as changes in gene regulation (e.g., Stern 2000, Hinman and Cheattle Jarvela 2014), and how the generation of phenotypic variation can be biased in that some variants are more likely to arise than others (e.g., Oster and Alberch 1982, Cano et al. 2023).

This research has placed the genotype–phenotype map (GP map) concept at the center of the evolvability research. This concept provides a framework for studying how properties of developmental systems affect phenotypic variation and its evolution (Hallgrímsson et al. 2023, Pavlicev et al. 2023). Studies of empirical GP maps representing gene-regulatory networks, regulatory interactions between genes, have been instrumental for understanding the mechanisms underpinning crucial stages during development, whereas comparing orthologous gene-regulatory networks across species further revealed specific features about the evolution of these networks. For example, comparative studies of the mesoderm–endoderm specification in echinoderms have shown how specific motifs such as positive feedback circuitry are conserved across species and may represent constraints on the evolution of developmental systems (Levine and Davidson 2005, Hinman and Cheattle Jarvela 2014). These studies also revealed that large phenotypic changes and novelties can be generated via cooption of important motifs (for a review, see Wittkopp and Kalay 2012). These studies have advanced our understanding of the evolutionary changes in molecular and developmental processes responsible for phenotypic variation among species. However, they have so far provided limited insights into how developmental processes generate the within-population heritable variation on which natural selection acts and how these processes themselves evolve under natural selection (Houle and Jones 2021).

Theoretical and empirical studies of GP maps have also identified properties of those maps that can influence evolvability. For example, *modularity*, the organization of development into relatively autonomous modules, could increase evolvability by reducing the effective dimensionality of variation and therefore reducing evolutionary trade-offs among different parts of the organism (e.g., Wagner and Altenberg 1996, Gerhart and Kirschner 1997, Wagner 2014, but see Hansen 2003). Similarly, *robustness*, the ability of developmental systems to remain unaffected by genetic or environmental perturbations (e.g., family level characters such as scutellar bristles number in *Drosophila*, Rendel 1959, or the uniform expression of five petals in the phlox family, Huether 1968), could increase evolvability by allowing the accumulation of cryptic genetic variation that may facilitate adaptation under new conditions (Wagner 2005). Mathematical modeling of GP maps has further highlighted how specific features of gene-regulatory networks can generate modularity and robustness and thereby influence evolvability (e.g., Kauffman 1993, Nijhout et al. 2017). Recently, the integration of such models into a quantitative genetic framework has suggested how GP maps could evolve under natural selection on phenotypic traits (Milocco and Salazar-Ciudad 2020, 2022).

Evolutionary quantitative genetics: One central principle of evolutionary quantitative genetics is that the ability of a population to show an evolutionary response to selection is determined by its additive genetic variance, V_A (Lande 1979, Houle 1992). This parameter has been estimated for many kinds of complex (i.e., polygenic) traits during the last 50 years and is often standardized by the total phenotypic variance V_P to yield the heritability ($h^2 = V_A/V_P$). However, heritability should not be considered as a comparative metric of evolvability because it does not measure evolvability independently from natural selection (Hansen et al. 2003, 2024). To overcome this issue, Houle (1992) introduced mean-scaled measures of additive genetic variance (e.g., $I_A = V_A/\mu^2$, where μ is the trait mean) as a method to quantify evolvability that is independent of selection. Because mean-scaled evolvability can be interpreted as the expected proportional response to selection when selection is as strong as selection on fitness (i.e., when the mean-scaled selection gradient $\beta_\mu = 1$; Hansen et al. 2011), it provides a metric for the evolutionary potential of quantitative traits when they are exposed to selection of similar strength. Comparisons of mean-scaled evolvabilities have led to the realization that the heterogeneity in evolutionary potential among traits is mainly influenced by differences in the amount of variation generated by mutations (Houle 1998), instead of being due to patterns of selection as is often argued (Mousseau and Roff 1987). Accordingly, life-history traits are more genetically variable than morphological traits, despite being more closely related to fitness (table 1; Houle 1992, Hansen and Pélabon 2021).

Beyond simply the magnitude of V_A in any trait, Lande and Arnold (1983) highlighted the influence of genetic correlations on the evolution of complex characters. Their framework later led to the development of multivariate measures of evolvability such as *conditional evolvability*, which quantifies the remaining evolvability of a trait when other genetically correlated traits are (hypothetically) not allowed to change. Conditional evolvability therefore quantifies the evolutionary consequences of pleiotropy (i.e., when a single locus affects several traits) and operationalizes the concept of modularity within a quantitative genetic perspective (Hansen and Houle 2008).

Another important insight for evolvability in evolutionary quantitative genetics stems from the recognition that patterns

Table 1. Median evolvabilities and heritabilities for different trait categories.

Trait type	Trait	n	Evolvability I_A				Heritability h^2	
			Median (%)	t_2 (gen)	25th percentile (%)	75th percentile (%)	n	Median
Morphology		1665	0.23	301	0.05	0.82	1560	0.37
	Size length	1025	0.10	715	0.03	0.47	929	0.40
	Size volume or mass	231	0.50	139	0.17	1.5	223	0.31
	Size count	333	0.45	154	0.21	0.98	332	0.36
	Shape	50	1.19	58	0.18	5.4	50	0.27
Growth		255	0.65	107	0.22	1.9	250	0.26
Pattern		18	0.84	83	0.50	3.3	18	0.36
Life history		474	0.86	81	0.05	3.8	470	0.18
	Fertility	182	2.28	30	0.54	8.1	182	0.25
	Dev. time	157	0.18	385	0.03	0.77	153	0.25
	Longevity	33	0.81	86	0.06	1.5	33	0.11
	Fitness	102	1.28	54	0.001	4.9	102	0.02
Physiology		175	1.45	48	0.38	3.6	170	0.20
Behavior		69	1.95	36	0.24	8.1	59	0.26
	Nonsexual	39	1.93	36	0.26	7.8	29	0.34
	Sex. selected	30	2.28	30	0.16	9.2	30	0.25
Total		2656	0.37	187	0.06	1.6	2527	0.31

Note: For ease of interpretation of the median evolvability, the column t_2 (gen) reports the time in generation it takes to double a trait under unit directional selection. This is calculated as $t_2 = \ln(2)/I_A$, where I_A is the mean-scaled evolvability. For example, an evolvability of 0.10% as in size length means that the trait will change by 0.10% per generation when selection is as strong as selection on fitness, and it will take 715 generations to double the trait. Abbreviation: n, the number of traits in each category. Source: From Hansen and Pélabon (2021).

of gene interaction can modify evolvability under directional selection. Specifically, epistasis occurs when genetic changes at loci influencing a trait modify the effects of genetic changes at other loci also influencing the trait. Because net effects are expected to be small when epistatic interactions cancel each other, with some increasing and some decreasing allelic effects, the role of epistasis in driving evolutionary dynamics has often been considered negligible. However, if there is *directional epistasis*, meaning that genetic changes influencing the trait systematically increase (or decrease) the effects of changes at other loci, genetic variation and therefore evolvability changes as the trait evolves (figure 3; Carter et al. 2005, Hansen et al. 2006). Such directional epistasis can be conceptualized as the population genetical manifestation of nonlinear GP maps. Directional epistasis may arise from any sources of nonlinearity, including genetic, developmental, or physiological feedback mechanisms; enzyme or transcription factor saturation; constraints on phenotypic expression produced by hard ceilings (e.g., figure 3; Morgan et al. 2020); or dichotomous or other forms of discrete phenotypic outcome (Dempster and Lerner 1950, Acker et al. 2023). A recent meta-analysis showed that directional epistasis is common and often strong, even in morphological traits that are continuously distributed on the phenotypic scale (Bourg et al. 2024). Consequently, epistasis can no longer be ignored and may in fact be a key force driving the dynamics of evolvability. Indeed, directional epistasis implies that evolvability changes as trait means evolve, violating the assumption of constant genetic variance with respect to mean phenotype that underpins foundational models of microevolutionary change (Lande 1979). By modifying the phenotypic impact of mutations and allele substitutions and, consequently, by modifying additive genetic variance, directional epistasis also provides an explanation for the evolution of robustness and evolvability (Hansen et al. 2006, Hansen and Wagner 2023).

Paleobiology: Although the modern synthesis postulates that macroevolution can be explained by microevolutionary processes, leading twentieth century biologists and paleontologists (including Eldredge and Gould) argued that processes causing microevo-

lutionary changes within populations are not sufficient to explain macroevolution occurring among populations or among species (Futuyma 2015). Using observations such as the discontinuity of evolutionary patterns in the fossil record, they argued that macroevolution should be explained by evolution occurring at different biological levels and involving different processes (e.g., speciation, extinction, species sorting). They also questioned the ultimate power of selection for shaping evolution and introduced the concept of developmental constraints to explain patterns of evolutionary stasis (figure 1; Raup 1966, Gould 1980). Preceding the emergence of the field of evo-devo, these authors emphasized the importance of development for producing the variation on which selection could act, including concepts such as allometry (e.g., Gould 1974), versatility (Vermeij 1973), and phenotypic integration (Olson and Miller 1954).

A key hypothesis that emerged from the paleobiology research program is that genetically more variable populations and species should evolve more or faster. In the first effort to quantitatively test this hypothesis, Simpson (1953) failed to observe the predicted relationship. However, subsequent studies have reported positive correlations between within-population phenotypic variation or additive genetic variance and rates of evolutionary divergence (e.g., Kluge and Kerfoot 1973, Hunt 2007, Bolstad et al. 2014, Holstad et al. 2024). By revealing the influence of within-population evolvability on evolutionary patterns at timescales up to hundreds of thousands of years, these studies reestablish a link between micro- and macroevolution but also raise new questions concerning the evolutionary mechanisms that can generate such a link.

Key research questions about evolvability

The rich background briefly summarized above highlights the broad multidisciplinary interest in evolvability that has recently regained integrative momentum. This research activity yields a range of key questions that now need to be addressed to advance our overarching understanding of the causes and the consequences of variation in evolvability.

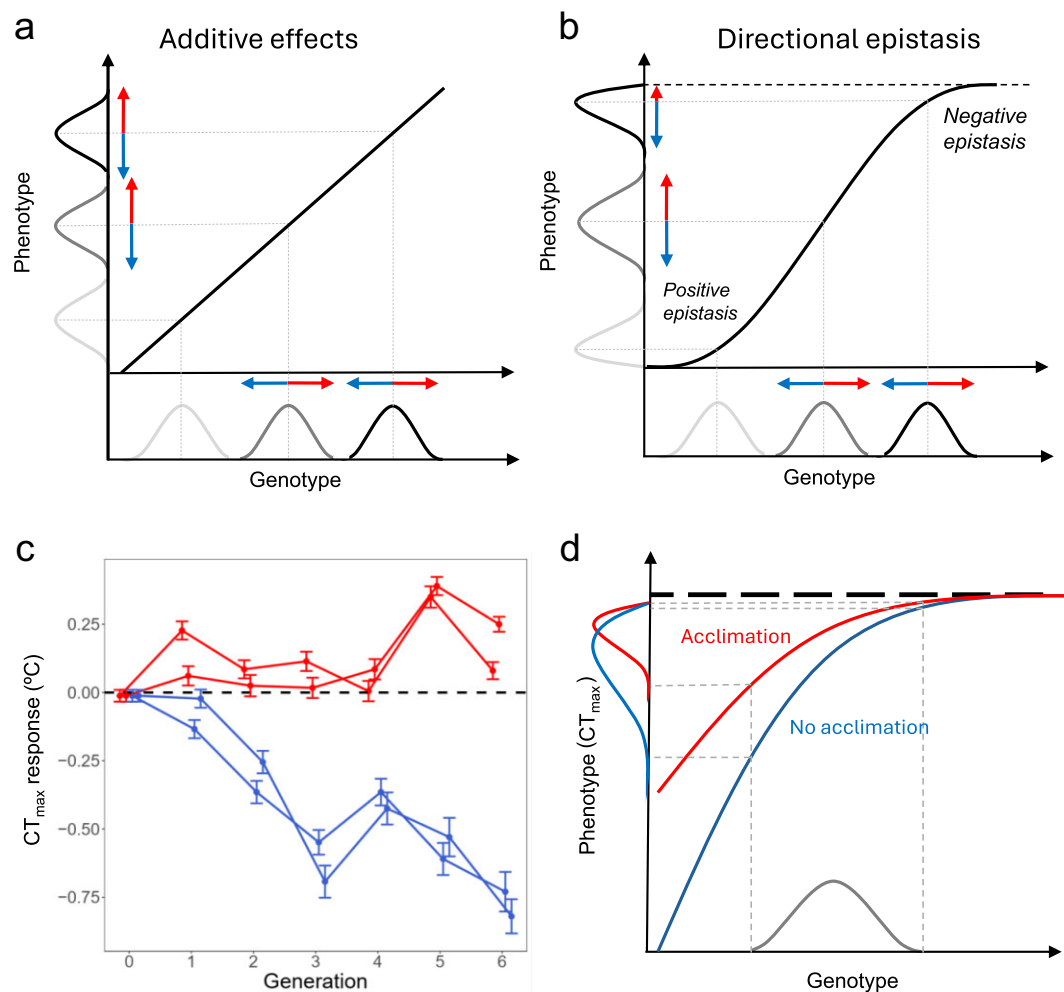


Figure 3. Linear and nonlinear genotype–phenotype maps (GP map), directional epistasis and evolvability. (a) A linear GP map where the black line represents the function that is mapping genetic variation, represented by bell curves on the *x*-axis, onto heritable phenotypic variation represented by bell curves on the *y*-axis. In this GP map, a genetic change in the DNA sequence that increases (the red arrows toward right) or decreases (the blue arrows toward left) the trait has always the same effect on the phenotype whatever the direction of the change and the genetic background in which it occurs (different genotypes on the *x*-axis correspond to different genetic background). With a such GP map, evolvability, which depends on heritable variation, remains constant when the trait evolves. (b) A nonlinear GP map where the phenotypic effect of a given genetic change depends on the direction of the change and the genetic background in which it occurs. For example, in the right half of the curve, the effect of a genetic change that increases the trait value diminishes when the trait increases (the red arrows on the *y*-axis decrease when the trait mean is moving toward its maximum value represented by the dashed line). This implies that genetic changes that increase the trait will decrease the effect of other genetic changes that also increase the trait, a mechanism referred to as negative directional epistasis. With directional epistasis, evolvability changes with the trait mean. (c) A possible example of negative directional epistasis in the artificial selection on maximum heat tolerance (CT_{max}) in zebrafish. The red lines and symbols (above the dash line) represent the response of the lines selected for an increase in CT_{max} , and the blue lines and symbols (below the dash line) represent the response of the lines to decrease CT_{max} (as compared with the control lines). The highly asymmetrical response to selection is possibly generated by biochemical limits at high temperature (Morgan et al. 2020). (d) Possible effect of environmental variation, represented by acclimation, on evolvability in the presence of negative directional epistasis. With acclimation, CT_{max} generally increases but shows less phenotypic and genetic variation. Source: Panels (c) and (d) were drawn from Morgan and colleagues (2020).

Causes of variation in evolvability

Understanding the causes of variation in evolvability requires identifying characteristics of underlying GP maps and their sources of variation. It then requires understanding how and how fast GP maps evolve and to what extent this evolution reflects structural properties of developmental systems versus the effects of natural selection on those systems.

Characteristics of the GP maps affecting evolvability and its variation: Differences in evolvability may stem from differences in the number of loci affecting the trait, in the effect sizes of alternative alleles, in the prevalence of pleiotropic effects that structure the GP map into developmental modules, and in the patterns

of epistasis that confer specific properties to the GP map such as robustness (figure 2). The following questions address these issues.

Question 1: How are evolvability and molecular genetic variation related? Techniques such as next-generation sequencing, genome-wide association studies, gene knockdown and gene editing have improved our understanding of how sequence variation generates phenotypic variation. For phenotypes influenced by genes with large effect (e.g., age at maturation in salmon; Barson et al. 2015), identifying such genes may help explain evolvability and its variation within and among populations (e.g., Jensen et al. 2022). However, the amount of additive genetic variation

that remains unexplained in many genome-wide association studies (e.g., Génin 2020, Wainschtein et al. 2022) indicates that genetic variants with large phenotypic effects are too rare to explain the amount of heritable variation responsible for microevolutionary changes of most quantitative traits. The difficulty in linking molecular genetic diversity to heritable phenotypic variation is further highlighted by studies of developmental processes that show how nonlinear GP maps obscure the relationships between DNA sequence variation and phenotypic variation (Hallgrímsson et al. 2009, 2023) and by increasing evidence that most traits are influenced by many loci (e.g., Yengo et al. 2022). Accordingly, measures of molecular genetic diversity and measures of heritable variation such as mean-scaled evolvability tend to be uncorrelated (Mittell et al. 2015).

Still, the absence of statistical correlation between these two types of variation does not mean that they are independent, and understanding the relationship between them remains necessary to comprehend how heritable phenotypic variation on which selection acts is generated. Furthermore, knowing when both types of variation could or should be correlated is of particular interest in conservation biology. Preserving evolutionary potential is crucial for facilitating evolutionary rescue—that is, when adaptive evolution restores positive growth of a population that declines because of environmental change and resulting maladaptation (Gomulkiewicz and Holt 1995). However, efforts to preserve the evolutionary potential in conservation programs are primarily focused on preserving molecular genetic diversity (e.g., Robuchon et al. 2023), possibly because molecular genetic data are now often cheaper and easier to obtain than quantitative genetic parameters, which are difficult to estimate in threatened species. Although such efforts can reduce extinction risks by reducing inbreeding depression (i.e., allowing genetic rescue; Frankham et al. 2017), the resulting presence of genetic diversity does not guarantee the evolvability of fitness and evolutionary rescue. Consequently, knowing when molecular data can or cannot predict evolvability may help conservation programs where evolutionary rescue is required.

Experimental evolution or artificial selection experiments combined with whole-genome sequencing (evolve and resequence; e.g., Wein and Dagan 2019 on bacteria) should contribute to our understanding of the mechanisms affecting the relationship between molecular genetic diversity and evolvability, but applying such knowledge in conservation biology will require general principles that could be broadly applied to a range of different traits and species. In the absence of such knowledge, overarching efforts to preserve genetic diversity remains the best options for preserving evolvability (Kardos et al. 2021, Willi et al. 2022).

Question 2: How are mutational variation and evolvability related? In principle, evolvability should increase with the magnitude of variation generated by new mutations (the *mutational variance*). Accordingly, positive correlations between mutational variance and mean-scaled evolvability have been observed in morphological and life-history traits of *Drosophila* (Houle 1998, Houle et al. 2017). However, a fundamental question is whether differences in mutational variation, and the resulting evolvability, stem from differences in locus-specific mutation rates, mutational effect sizes, or mutational target size (the number of loci with potential effects on the trait). Each mechanism has received empirical support (e.g., mutation rate, Chan et al. 2010; effect size, Estes et al. 2004; mutational target size, Besnard et al. 2020), and they all may nonexclusively contribute to variation in evolvability.

However, evaluating their relative importance is fundamental for understanding evolvability and its evolution because mutation rates, effect sizes, and target sizes involve molecular processes occurring at different organizational levels (e.g., DNA repair fidelity, gene duplication, cooption of motives in gene-regulatory network) that evolve at different rates. Understanding the contribution of these mechanisms to evolvability is challenging, however, because the different processes are investigated via different methods involving different timescales and where selection is rarely considered (e.g., mutation accumulation, experiments, comparative genomics). However, recent experiments on *Arabidopsis thaliana*, revealing the covariation of within-genome mutation rates (locus and nucleotide-specific mutation rates) with within-species polymorphism and between-species substitution rates further suggest how mutations contribute to evolvability (Weng et al. 2019). Specific epigenetic features associated with these site-specific mutation rates in *A. thaliana* (Monroe et al. 2022) also provides a mechanism by which evolvability itself may evolve, such as through the evolution of protein–DNA epigenetic interactions.

Question 3: What is the relationship between pleiotropy, modularity and evolvability? It has become clear that evolvability of any trait depends on relationships with other traits and, more particularly, on the degree of underlying pleiotropy that leads to genetic correlations among traits alongside the alignment of selection (Etterson and Shaw 2001, Hansen 2003, Houle and Rossoni 2022). This issue is well known by breeders when they fail to increase multiple traits of interest that are negatively genetically correlated, such as the protein and oil content of soybean seeds (Bandillo et al. 2015). Consequently, univariate measures of additive genetic variation in single traits may often overestimate evolvability when the focal trait is genetically correlated with traits under opposing forms of selection.

Empirical estimates of conditional evolvability accounting for genetic correlations have confirmed the negative effect of pleiotropy on evolvability (e.g., Sztepanacz and Houle 2019, Opedal et al. 2023). However, such estimates generally include only few measured traits and neglect other unmeasured traits that are plausibly entangled with the focal trait. Because conditional evolvability may further decrease when additional traits are considered, conditional evolvability may still be overestimated for most traits. As automated and high-throughput phenotyping methods improve (Houle et al. 2010), we should aim to estimate *G* matrices (i.e., genetic variance–covariance matrices, *G*) that encompass a large majority of potentially associated traits. Such studies would reveal the degree to which conditional evolvability decreases when the number of traits considered increases.

However, the constraining effects of genetic correlations among traits can be counteracted by the higher genetic variation that can emerge if many loci affect many traits (Hansen 2003). Furthermore, variation in pleiotropic effects across loci also reduces genetic correlations and opens the possibility of compensatory changes that can alleviate detrimental effects of correlated responses to selection. Some degree of modularity could ultimately emerge, whereby sets of traits become substantially genetically independent. Understanding the effects of modularity on evolvability therefore requires a population-genetics perspective incorporating the dynamics of the underlying genetic and developmental mechanisms. It also requires further efforts to relate the magnitudes and signs of genetic variances and covariances to the patterns of selection acting on different traits (e.g., Blows et al. 2004) or with the patterns of divergence among

lineages, assuming that the divergence results from adaptation (e.g., Hohenlohe and Arnold 2008, Houle et al. 2017).

Question 4: What is the relationship between robustness and evolvability? By reducing the phenotypic expression of genetic changes, robustness is expected to decrease short-term evolvability (Ancel and Fontana 2000). However, robust phenotypes could ultimately be more evolvable if they accumulate conditionally neutral mutations that could create paths to novel phenotypes following either environmental or genetic changes (Gerhart and Kirschner 1997, Wagner 2008). This latter hypothesis, however, relies on specific assumptions about the structure of the GP map and the fitness landscape, and alternatives reasonable assumptions can equally generate a negative relationship between evolvability and robustness (Mayer and Hansen 2017). Ultimately, initial robustness may decrease evolvability over short timescales and under limited environmental variation, but it may increase evolvability over long timescales or when populations encounter rarer, more extreme and stressful environments. In an experiment on *Escherichia coli*, Zheng and colleagues (2019) observed that the rate of evolution of a fluorescent protein toward a new fitness optimum was faster in populations harboring cryptic genetic variation compared to populations without such variation. This study illustrates the type of approach that should help us assess the link between cryptic genetic variation and evolvability. However, a positive effect of robustness on the ability to respond to selection requires that the genetic variation accumulated under benign conditions is adaptive once expressed, a condition that is still debated, as is illustrated in the next question.

Question 5: Is cryptic genetic variation adaptive? Cryptic genetic variation can be adaptive if it is released by environmental changes to produce phenotypes that are beneficial in the new environmental conditions (Hoffmann and Parsons 1997, Rutherford and Linquist 1998, Hermisson and Wagner 2004). Waddington (1953, 1956) showed that selection on such environmentally induced phenotypes could lead to a process of genetic assimilation where trait expression shifts from being environmentally induced to being mostly genetically controlled. The hypothesis of adaptive cryptic genetic variation has been supported by theoretical (e.g., Hermisson and Wagner 2004, Masel 2005) and empirical studies (e.g., McGuigan et al. 2011, Zheng et al. 2019). However, novel traits or new genetic variation expressed under stressful conditions do not necessarily facilitate adaptation to such conditions. For example, despite observing an increase in additive genetic variance in desiccation resistance in *Drosophila* after a bottleneck event, van Heerwaarden and colleagues (2008) failed to observe the increased response to selection predicted by the new level of additive variance. Furthermore, the beneficial effect of cryptic genetic variation on fitness can be limited to specific novel environments possibly determined by the population's selection history. Accordingly, Grainger and colleagues (2021) found that the evolution of fecundity and developmental rates in *Drosophila melanogaster* populations exposed to cooling climate in the fall was contingent on the type of selection experienced by previous generations during the summer. Echoing the review by McGuigan and Sgro (2009), we emphasize that more studies are required like those discussed above and on an array of systems to understand the conditions under which potentially adaptive cryptic genetic variation can accumulate and be favourably expressed in nature. Such work would help clarifying the relationships between cryptic genetic variation and robustness and how these properties of the GP map affect evolvability.

Question 6: Do more plastic traits evolve faster? Phenotypic plasticity, when different phenotypes are produced by a genotype exposed to different environments, is often suggested to facilitate genetic adaptation, because it may produce variants that can increase fitness and allow persistence for long enough for adaptive evolution to occur via genetic accommodation. However, such a plasticity-first hypothesis (West-Eberhard 2003) is controversial and difficult to test, not the least because the effects of plasticity on evolution remain debated. If plasticity and evolvability are controlled by different mechanisms, adaptive phenotypic plasticity that shifts the phenotype toward the new fitness optimum should decrease the rate of evolution, because it decreases the strength of selection (e.g., Borenstein et al. 2006, but see Paenke et al. 2007). In contrast, phenotypic plasticity could accelerate evolution by increasing evolvability if selection for plastic genotypes that are more sensitive to environmental variation also selects genotypes that are more sensitive to mutations (Fierst 2011, Draghi and Whitlock 2012). This congruence hypothesis was initially suggested by Ancel and Fontana (2000).

Attempts to elucidate the effects of plasticity on evolution have considered the relative role of plasticity and genetic adaptation in populations differentiation. In a review of reciprocal transplant experiments, Stamp and Hadfield (2020) showed that genetic and plasticity-induced phenotypic differences among populations are generally positively correlated and that plasticity is the major contributor to population divergence for two thirds of the traits considered (see also Noble et al. 2019, Rohner et al. 2022, Rohner and Berger 2023). Together with the positive correlation observed between genetic and residual variation among traits (Hansen et al. 2011), these results give supports to the congruence hypothesis, although within-trait variation in heritability also implies some degree of independence between the different components of phenotypic variation (e.g., Ketola and Kronholm 2023). Correlation is not causation, however, and the correlation observed between plastic and genetic changes in population differentiation does not necessarily imply that plastic changes preceded genetic changes or even that both type of changes result from similar alteration of developmental pathways.

More studies are therefore needed to elucidate the relationship between phenotypic plasticity and evolvability. This can be achieved by comparing the phenotypic outcome of genetic or environmental perturbations on developmental pathways or by analyzing the consequences of artificial selection on variance (e.g., stabilizing, fluctuating, or disruptive selection) on the different components of phenotypic variation. For example, Pélabon and colleagues (2010) found that phenotypic variation among versus within individuals (i.e., developmental instability) responded differently to stabilizing and fluctuating selection, suggesting different genetic control of these sources of variation. Modular and clonal organisms are particularly useful in this context because they allow a better partitioning of the different components of the phenotypic variation. Similarly, animal-pollinated plants with different levels of phenotypic variation expected in vegetative versus reproductive organs may prove particularly useful to understand the evolution of the different components of phenotypic variation (Fenster and Galloway 1997).

Question 7: What is the strength and type of directional epistasis affecting phenotypic traits? Recent studies have highlighted that directional epistasis can be substantial and therefore must be considered in efforts to predict evolutionary responses to selection. For example, artificial selection on maximal heat tolerance (CT_{max}) in zebrafish showed that it is easier to generate an evolutionary

decrease in CT_{max} than an evolutionary increase (figure 3c; Morgan et al. 2020). This may be because biochemical or physiological constraints prevent genetic changes that increase CT_{max} from being phenotypically expressed above certain temperatures. This implies that genetic or environmental changes that shift CT_{max} closer to these thermal limits decrease the effects size of mutations that should affect CT_{max} . This generates negative directional epistasis where genetic changes that increase CT_{max} interact with other such genetic changes and decrease their effect sizes. Consequently, additive genetic variation and therefore evolvability should decrease when CT_{max} evolves toward higher values (figure 3; Hansen 2013) or when acclimation to higher temperature increases CT_{max} (figure 3d). This example illustrates the expected effect of directional epistasis on the evolvability of quantitative traits and suggests that estimating additive genetic variance of such traits may not be sufficient to fully predict their evolvability in different (opposite) directions and over many generations of selection. One should keep in mind that it is not the presence of epistasis, as is classically estimated by the epistatic variance (e.g., Lynch and Walsh 1998), that is important for the evolutionary dynamics of quantitative traits but the directionality of epistasis, which affects evolvability in a nontransient manner (Hansen 2013).

Although changes in additive genetic variation alongside changes in the trait mean have long been acknowledged for *threshold traits*—that is, traits with discrete phenotypic states such as color polymorphisms, sex determination in reptiles, or partial migratory behavior (Dempster and Lerner 1950, Reid and Acker 2022)—empirical estimates of directional epistasis in continuously distributed phenotypic traits have been scarce (but see Pavlicev et al. 2010, Bourg et al. 2024, Le Rouzic et al. 2024). Consequently, it remains unclear how much directional epistasis could affect evolutionary predictions and at which timescale. However, a recent meta-analysis of line-cross data of size-related traits in animals shows that negative epistasis toward larger size is predominant for such traits and the median estimate predicts that evolvability would on average be halved when going from the small to the large parental lines used in the line crosses (Bourg et al. 2024). This could substantially constrain the evolution toward larger size in many organisms.

Estimating the type and strength of directional epistasis in different categories of traits could therefore improve our evolutionary predictions and help explain the differences in evolvability seen among different types of traits. This should be facilitated by newly developed statistical tools that allow estimating patterns of directional epistasis, either from artificial selection time series where patterns of directional epistasis can be inferred from their potential effects on the response to selection (Le Rouzic et al. 2011) or from line-cross data that allow building GP maps from which patterns of epistasis can be inferred (Bourg et al. 2024). Acquiring such broader knowledge about the patterns and strength of directional epistasis in different categories of trait may also help incorporate results of evolvability research into conservation biology. Indeed, estimating directional epistasis of specific key ecological traits in a threatened population for which evolutionary rescue is desirable may rarely be possible because of the difficulties of obtaining such genetic details. Using broad principles may remain the best way to improve our predictive models to determine whether this population is able to adapt to more extreme environment.

Evolution of evolvability: Beyond understanding the mechanistic causes of variation in evolvability, the next ambition is to un-

derstand how such mechanisms can themselves evolve. Many scenarios of adaptive and nonadaptive evolution of evolvability have been suggested. Lynch (2007, 2024) argued that the evolution of genetic architecture may be dominated by genetic drift and mutations, because the impacts of selection on genomic changes will typically be weak in multicellular organisms with long-term effective population sizes below 10,000 (the drift-barrier hypothesis, but see Whitney and Garland 2010, Monroe et al. 2022). Evolution of evolvability may also result from a tendency for developmental systems to become more robust against mutations when their complexity increases (Kauffmann 1993, Siegal and Leu 2014). In addition, with directional epistasis, additive genetic variance is affected by changes in the trait mean. Consequently, evolvability may evolve as an indirect consequence of directional selection on the trait mean (Carter et al. 2005, Hansen et al. 2006). Furthermore, if relaxed selection increases evolvability by allowing accumulation of genetic variation (Payne and Wagner 2019), evolution of traits that decrease the strength of selection, such as adaptive plasticity or key innovations allowing for example individuals to escape competition, may also increase evolvability (Galis 2001).

Alternatively, genetic architecture may have been selected to facilitate adaptive levels of evolvability, because more evolvable genotypes, populations, or species are better able to adapt and persist in changing environments (Lloyd and Gould 1993, Le Rouzic et al. 2013, Tufto 2015). Similar arguments have been developed to explain the evolution of genes promoting variation (Wagner 1981), sex, and recombination (Maynard Smith 1978, Bell 1982). Other adaptive hypotheses for the evolution of evolvability concern the generation of more localized or coordinated effects of mutations via developmental modularity and integration to reduce pleiotropic costs of phenotypic evolution (Raff 1996, Gerhart and Kirschner 1997) and the evolution of robustness and canalization when, as in conservative bet-hedging, short-term evolution causes a long-term decrease in fitness (Haaland et al. 2019). In addition, observations of lower mutation rates of genes subjected to purifying selection (e.g., Monroe et al. 2022) suggest that some feature of the genetic architecture affecting evolvability may respond to selection. The two following questions address this debate.

Question 8: Do changes in mutation rate under stressful conditions result from selection for evolvability? Some researchers have argued that the increase in mutation rate sometimes observed under stressful conditions is adaptive because it enhances evolvability (e.g., Metzgar and Wills 2000, Wei et al. 2022). However, no studies have yet convincingly demonstrated direct selection on evolvability instead of indirect selection of evolvability, and alternative hypotheses such as the drift-barrier hypothesis provide nonadaptive explanations for such changes (Lynch et al. 2023). Testing adaptive hypotheses about the evolution of mutation rate and evolvability is difficult, however, because natural selection can only favor traits that immediately benefit individuals, as opposed to potentially benefiting descendants. Consequently, adaptive hypotheses for the evolution of evolvability generally involve some components of group or lineage selection (Dawkins 1988, Lloyd and Gould 1993, Lynch 2007), as for any bet-hedging traits.

Empirical evidence of selection at these levels has been reported (e.g., Goldberg et al. 2010, Moreira et al. 2021), but rigorous tests of adaptive hypotheses concerning the evolution of evolvability at different levels (genome, population, or species) are still needed. Such tests should consider adaptive and nonadaptive scenarios at the different levels, as was recently done by Barnett and colleagues (2025). There, the rapid shift in natural selection on mat formation in the bacterium *Pseudomonas fluorescens*

favoring cellulose-producing or non-cellulose-producing cells resulted in lineages evolving higher mutation rates and higher evolvability at the loci responsible for the shift in cellulose production.

Question 9: Does modularity result from constraints embedded in developmental systems or from the effect of natural selection on those systems? We cannot assume that modularity necessarily results from natural selection. Accordingly, several models of neutral evolution of modularity, where natural selection is either absent or plays a limited role, have been suggested (see Wagner et al. 2007 for a review). However, there are cases in which modularity and integration appear to have evolved as a result of natural selection. In the plant *Dalechampia scandens*, variation of involucral bracts that subtend the cluster of male and female flowers to form the blossom is decoupled from variation of the leaves from which they are developmentally derived (Armbruster 2023). Such variational modularity of the bracts seems adaptive given their role in the pollination process, as well as in the protection of the developing fruits (Pélabon et al. 2011). Inversely, fine-tuning of the pollination process is often achieved by *connation*, the fusion (integration) of floral parts, which may be a response to the stabilizing selection resulting from pollination accuracy (Armbruster et al. 2009). Eventually, whether modularity represents a constraint imposed by developmental systems or the effect of natural selection on those systems will depend on the rates at which pleiotropy and modularity evolve and whether such an evolution is a response to selection; that is, it coincides with changes in the patterns of multivariate selection. These considerations underline the relevance of studying the evolution of *G* matrices.

Theoretical studies that integrate some aspects of development suggest that *G* matrices can evolve rapidly under selection (e.g., Milocco and Salazar-Ciudad 2020, 2022). In addition, Pavlicev and colleagues (2008) reported genetic variation for pleiotropy that affects covariation among traits in mice. But on the other hand, artificial selection on correlated traits and experimental evolution studies modifying the orientation of the fitness landscape have provided inconclusive evidence about the adaptive evolution of genetic correlation and modularity (e.g., Bolstad et al. 2015, Sztepanacz and Blows 2017, Gomez et al. 2019). These experimental studies may suffer a lack of power to efficiently modify pleiotropic effects in a limited number of generations. Still, their results tend to be confirmed by comparative analyses of *G* matrices that often report a relative stability of *G* across populations and species (e.g., Delahaie et al. 2017, McGlothlin et al. 2022) and therefore suggest some possible constraints on the evolution of modularity. This interpretation remains contentious because relatively constant fitness landscapes could generate such consistency in *G* and information concerning the dynamics of fitness landscapes remains scarce. More studies are therefore necessary to understand whether and how selection can modify modularity, a conclusion shared by other properties of the GP map such as plasticity and robustness. Combining results on short-term evolution provided by comparative studies across populations and species with those of long-term evolution provided by historical data such as the fossil record, as in Uyeda and colleagues (2011), could provide valuable insight into the timeframe necessary to witness the evolution of these properties of the GP maps.

Consequences of variation in evolvability

Given our limited knowledge concerning the rate and causes of evolution of GP maps, it remains unclear over what timescales

evolvability influences evolution. It is also often difficult to predict how such influences are manifested. The last three questions address these issues.

Question 10: Is the lack of evolvability responsible for macroevolutionary stasis? Many traits exhibit phenotypic stasis for millions of years (figure 1d; Gingerich 1993, Hansen and Houle 2004, Uyeda et al. 2011, Hunt et al. 2015), but rapid trait evolution is often observed on short timescales. In addition, contemporary estimates of population evolvability are too high to constrain evolution over timescales relevant for population and species divergence (Hansen and Houle 2004, Holstad et al. 2024). Several authors have therefore suggested that genetic correlations may substantially reduce evolvability and constrain evolution at macroevolutionary timescales (e.g., Hansen and Houle 2004, Galis and Metz 2007, 2008). This hypothesis makes two testable predictions. First, estimates of conditional evolvability should be small enough to constrain population or species divergence. Second, the constraining effect of evolvability should decrease over longer periods of time if pleiotropy and modularity evolve.

Testing the first prediction requires more complete *G*-matrices to fully account for the constraining effect of pleiotropy (see question 3). Concerning the second prediction, Holstad and colleagues (2024) observed a weaker relationship between measures of evolvability and species divergence compared with population divergence, which supports the idea of decreasing constraints at longer timescales. However, the relationship they observed between rates of evolution and the time interval over which those rates were measured was not compatible with an evolutionary scenario where evolvability gradually increased over time (see also Uyeda et al. 2011). To reconcile these observations, Holstad and colleagues (2024) suggested that trait stasis for up to one million years could be generated by fluctuating selection around stationary adaptive optima (i.e., no change of the long-term average optimum). Under this scenario, the divergence among populations tracking asynchronized fluctuating optima can be constrained by low evolvability. Testing this hypothesis requires further information concerning the long-term dynamics of fitness landscapes. This is the topic of our next question.

Question 11: Are contemporary measures of natural selection relevant to understand macroevolutionary consequences of evolvability? Natural selection has been shown to fluctuate on short timeframes (year to year; e.g., Siepielski et al. 2009, Reynolds et al. 2010, Acker et al. 2021), although some of these estimates are likely to be substantially inflated by sampling variance (Morrissey and Hadfield 2012). Increasing the number of rigorous estimates of temporal and spatial dynamics of fitness landscapes measured over longer time periods and using appropriate statistical methodologies will help resolve this issue. The increasing number of longitudinal studies of wild populations (e.g., Soay sheep on St Kilda, great tit populations in England and Netherlands) should make this possible, but such studies are still taxonomically restricted. We therefore require more long-term selection studies, while accepting that contemporary selection dynamics may not necessarily reflect patterns of selection that have shaped animal and plant species during the past thousands or millions of years. This is not least because of the increasing effects of global changes on contemporary populations, as pollution, habitat changes, and global warming may generate patterns of selection different from those experienced by plant and animal species in the last 10,000 years.

Question 12: Is evolvability relevant for macroevolutionary processes such as speciation and extinction? It has been repeatedly suggested that evolvability could influence speciation and extinction rates (Jablonski 2008). For example, Goldberg and colleagues (2010) suggested that reduced evolvability generated by self-incompatibility in species with bisexual flowers causes the increased extinction rates in Solanaceae. Similarly, the scattered phylogenetic distribution of asexual taxa has long been interpreted to reflect elevated extinction rates because of low evolvability (e.g., Bell 1982, Moreira et al. 2021). In addition, Kolbe and colleagues (2011) found that survival of bivalve species through the Pliocene–Pleistocene extinction interval was correlated with greater morphometric variation, which can be broadly interpreted as evolvability (see also Liow 2007, Jablonski 2023). Yang (2001) also proposed that increased modularity between young and adult stages associated with holometabolous insect development, as compared with hemimetabolous development, increased evolvability and consequently species diversity among holometabolous insects.

These examples are suggestive but assessing whether evolvability causally affects species persistence and speciation rate is highly challenging. First, observed relationships between species persistence and morphological variation may be mediated by geographic and ecological breadth which influence patterns of selection (Love et al. 2022). Furthermore, the relationships between evolvability, speciation and species diversification are often complex. For example, in Goldberg and colleagues (2010), self-incompatibility was associated with higher species diversification. However, this pattern did not result from higher speciation rates of self-incompatible taxa (speciation rate was higher in self-compatible taxa) but from a lower extinction rate of those taxa, which may be better able to maintain adaptive potential (Clo et al. 2019 but see Sztepanacz et al. 2023). Similarly, if diversification and speciation are favored by the evolution of key innovations (Galis 2001, Maia et al. 2013), it remains unclear whether key innovations increase evolvability by generating new variation that results in further changes (e.g., Stroud and Losos 2016, Jablonski 2023) or whether increased evolvability generates novelties and key innovations—for example, because of the accumulation of new genetic variation under relaxed selection (Galis and Metz 2007). These possibilities may be resolved by comparing the evolutionary response of clades exposed to similar opportunities. A possible example concerns the comparison of the diversity in bill characteristics in the speciose Hawaiian honeycreepers versus their relatives, the Hawaiian thrushes, that colonized the archipelago at the same time. Lovette and colleagues (2002) concluded that the bill diversity associated with the higher number of honeycreeper species resulted from a clade-specific ability to evolve novel morphologies. Unfortunately, the number of clades where such comparisons can be made is limited, and the influence of contingency generated for example by different selection histories remains difficult to consider.

The effects of modularity on evolvability over macroevolutionary timescales have also been inferred from correlations between measures of modularity and measures of population or species divergence (e.g., Du et al. 2019). These studies remain mostly suggestive, however, because they lack well-defined models of evolution underpinning the quantitative relationships between measures of modularity and measures of evolutionary diversification such as the number of species (Houle and Pélabon 2023).

Jablonski (2023) argued that new methods for integrating fossil and present-day data should offer promising avenues to evaluate the role of evolvability in macroevolution (see also

Love et al. 2022, Voje et al. 2023). Such approaches will need to overcome core difficulties, such as the fact that studies of evolvability of extant species typically concern specific traits, whereas macroevolutionary studies concern taxon- or clade-specific evolvability (Love et al. 2022). Similarly, the weak correlation often observed between speciation and morphological evolution (e.g., Hunt and Rabosky 2014) complicates the establishment of a link between contemporary measures of evolvability quantified as morphological divergence and macroevolutionary measures of evolvability quantified as number of species.

Conclusions

In recent years, evolvability has matured into a cohesive and cross-cutting research topic in evolutionary biology. Although the concept was originally developed independently in different disciplines, recent progress has established important bridges between research programs and agendas. This progress is evidenced by our highlighted questions, which often combine concepts, theories, and empirical approaches from multiple disciplines. Accordingly, efforts to answer these questions will necessarily further stimulate interdisciplinary and multifaceted research. This will, among other advances, require more fully integrating traditional quantitative genetics with modern genomic and phenomic capabilities among others.

Furthermore, the door is now open to more explicitly apply cutting edge insights stemming from the evolvability concept across more applied fields, including conservation biology and evolutionary medicine (e.g., Gatenby and Brown 2020, Mansur and Greaves 2023). The resulting insights regarding the causes, constraints, and implications of rapid evolution and the timeframes over which such processes act and evolve should reveal the potential for rapid desired evolution (e.g., in evolutionary rescue) and suggest new approaches to prevent rapid undesired evolution (e.g., in antibiotic resistance). The formal introduction of evolvability into such active fields should also generate reciprocal stimulation, because progress in such directions will prompt further developments to operationalize the evolvability concept.

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Author contributions

C. Pélabon (Conceptualization, Funding acquisition, Project administration, Visualization, Writing - original draft, Writing - review & editing), G.A. Agudelo-Cantero (Writing - review & editing), Y.G. Araya Ajoy (Writing - review & editing), G.H. Bolstad (Writing - review & editing), C. Cheng (Writing - review & editing), F. Galis (Writing - original draft, Writing - review & editing), F. Guillaume (Writing - review & editing), T.R. Haaland (Visualization, Writing - review & editing), B. Hallgrímsson (Writing - review & editing), T.F. Hansen (Funding acquisition, Writing - review & editing), A. Holstad (Writing - review & editing), D. Houle (Writing - review & editing), G. Hunt (Writing - review & editing), A. Isaksen (Writing - review & editing), L. Milocco (Writing - review & editing), N. Mubalegh (Writing - review & editing), L. Nuño de la Rosa

(Writing - review & editing), S.H. Orzack (Writing - original draft, Writing - review & editing), A. Porto (Writing - review & editing), J.M. Reid (Writing - original draft, Writing - review & editing), J.L. Sztepanacz (Writing - review & editing), E.A.B. Undheim (Writing - review & editing), C. Villegas (Writing - review & editing), K.L. Voje (Writing - review & editing), C. Walling (Writing - review & editing), and J. Wright (Writing - review & editing)

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RESEARCH ARTICLE

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Network properties of human disease genes with pleiotropic effects

Sreenivas Chavali^{*†}, Fredrik Barrenas[†], Kartiek Kanduri and Mikael Benson

Abstract

Background: The ability of a gene to cause a disease is known to be associated with the topological position of its protein product in the molecular interaction network. Pleiotropy, in human genetic diseases, refers to the ability of different mutations within the same gene to cause different pathological effects. Here, we hypothesized that the ability of human disease genes to cause pleiotropic effects would be associated with their network properties.

Results: Shared genes, with pleiotropic effects, were more central than specific genes that were associated with one disease, in the protein interaction network. Furthermore, shared genes associated with phenotypically divergent diseases (phenodiv genes) were more central than those associated with phenotypically similar diseases. Shared genes had a higher number of disease gene interactors compared to specific genes, implying higher likelihood of finding a novel disease gene in their network neighborhood. Shared genes had a relatively restricted tissue co-expression with interactors, contrary to specific genes. This could be a function of shared genes leading to pleiotropy. Essential and phenodiv genes had comparable connectivities and hence we investigated for differences in network attributes conferring lethality and pleiotropy, respectively. Essential and phenodiv genes were found to be intra-modular and inter-modular hubs with the former being highly co-expressed with their interactors contrary to the latter. Essential genes were predominantly nuclear proteins with transcriptional regulation activities while phenodiv genes were cytoplasmic proteins involved in signal transduction.

Conclusion: The properties of a disease gene in molecular interaction network determine its role in manifesting different and divergent diseases.

Background

Decades-long research efforts have resulted in the identification of a large number of human disease genes [1-3]. Most of this research has been based on experimental and clinical studies of individual diseases and genes. A conceptually different approach was recently described, namely to study the network properties of human disease genes on a disease-wide scale. The studies were based on analyzing disease genes databases, such as the Online Mendelian Inheritance in Man (OMIM) [4]. The disease genes were classified as monogenic, polygenic or complex and their properties in molecular interaction networks were elucidated [5,6]. Further, it was shown that phenotypically similar diseases are often caused by functionally

related genes [7-9]. This has led to the exploitation of molecular interaction networks to find novel candidate genes exploring neighbors of a disease-causing gene in a network as they are more likely to cause either the same or a similar disease [7,8].

Pleiotropy, in the context of human genetic diseases, implies that different pathological effects of different mutations within the same gene predispose an individual to different disorders [10]. While the previous studies have studied the properties of disease genes classified based on the number of genes involved in a phenotype, it is paramount to study the genes classified based on the number of phenotypes they are involved in. This would aid in identifying disease genes that are specific to diseases (specific genes), which can be exploited for therapeutic intervention. This would also help to find pleiotropic genes that are shared between different diseases (shared genes) to understand shared pathogenesis and hence mechanisms underlying co-morbidity [10-12].

* Correspondence: sreenivas.chavali@gu.se

[†] The Unit for Clinical Systems Biology, University of Gothenburg, Medicinaregatan 5A, Gothenburg SE405 30, Sweden

[†] Contributed equally

Full list of author information is available at the end of the article

Network properties of shared genes associated with phenotypically similar diseases have been examined so far, whereas those of pleiotropic genes with effects on divergent phenotypes and genes associated with specific diseases have not been examined. We hypothesized that the network properties of a gene in molecular interaction network and its tissue co-expression with its interactors determines the number of disease phenotypes it is associated with.

Here, we retrieved human disease genes and the associated diseases from Morbid Map (OMIM). We classified the shared disease genes into genes associated with phenotypically similar diseases (phenosim genes) and those that are associated with phenotypically divergent diseases (phenodiv genes) based on CIPHER score [13]. For instance, *AKT1* which is associated with ovarian cancer, breast cancer, colorectal cancer and schizophrenia was classified as a phenodiv gene while *TYRPI* which is associated with brown albinism and rufous albinism was classified as a phenosim gene. We demonstrated that shared genes were more central than specific genes while phenodiv genes were more central than phenosim genes. Shared genes had a higher number of disease gene interactors compared to specific genes. However, shared genes had a relatively restricted tissue co-expression with its interactors compared to specific genes. Essential genes, mutations in which lead to lethality, are known to be high degree nodes (hubs), thus occupying a central position in protein interaction network. When compared with specific, shared and phenosim genes essential genes had higher measures of centrality, as expected. However, essential genes and phenodiv genes had comparable connectivities (degrees) instigating us to explore for other network attributes of lethality and pleiotropy. We found that essential and phenodiv genes were intra-modular and inter-modular hubs, with the former being highly co-expressed with their interactors contrary to the phenodiv genes. Gene Ontology analysis identified the essential genes to be predominantly transcription factors residing in nucleus while phenodiv genes were cytoplasmic proteins involved in signal transduction. This study demonstrated that the effect of a disease gene on the number of different and phenotypically divergent diseases is associated with its properties in a molecular interaction network.

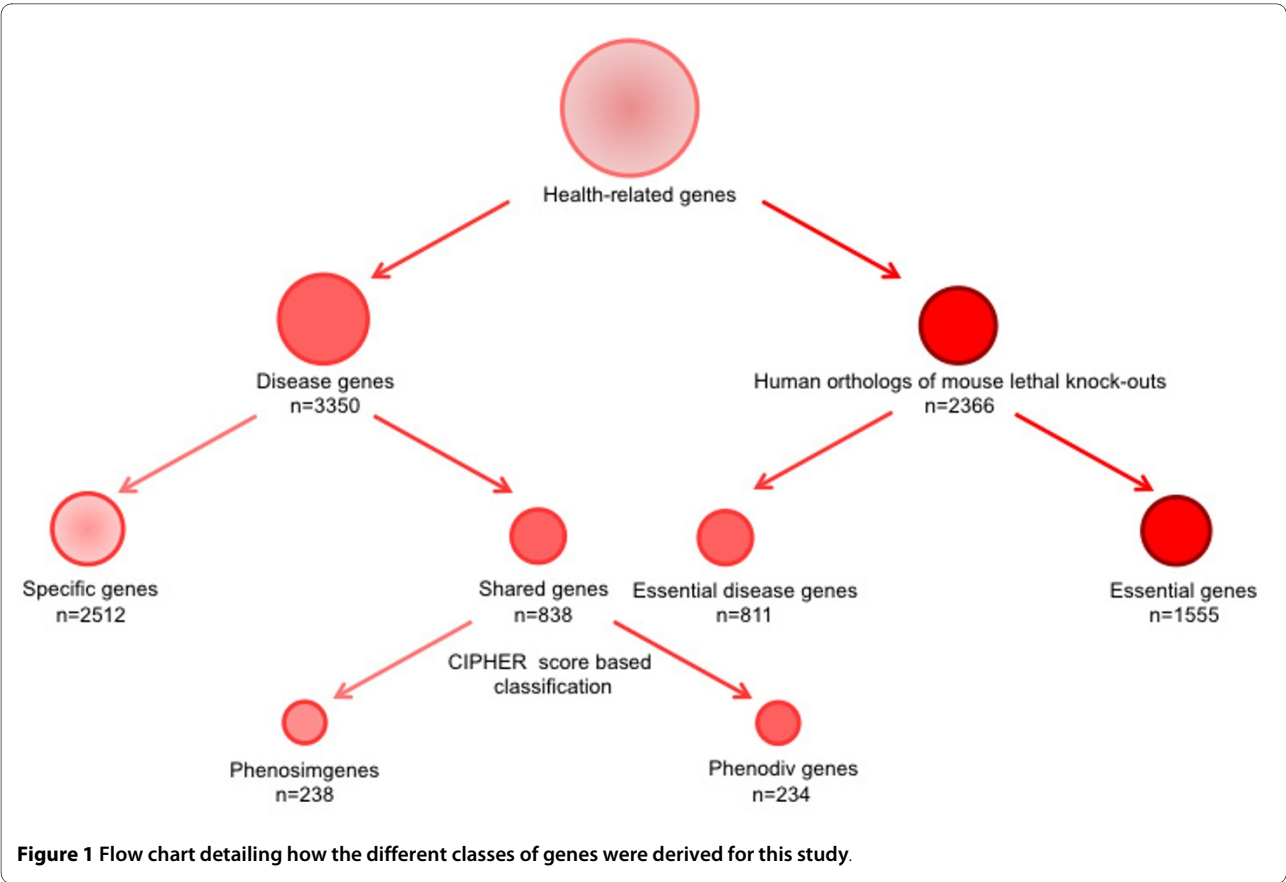
Results

Centrality of human disease genes in protein interaction network

We retrieved a list of 3350 human disease genes from OMIM Morbid Map. If a gene is associated with only one disease it was classified as specific disease gene and if it is associated with more than one disease it belonged to the shared disease genes category (Figure 1). The importance

of a node in a molecular network is often correlated to its centrality [14]. There are different measures that capture the centrality of a node in a network. We constructed a human protein interaction network using a modified version of CRG interactome [15] and investigated for differences in four different centrality measures namely degree, closeness, betweenness and eccentricity between specific and shared genes. The degree of a node provides the information about how many links (edges) that node has to other nodes in the network. Closeness is defined as the reciprocal average distance (number of links in the shortest path) to every other node- a node with high closeness is thus, on average, close in graph distance to the other nodes. Betweenness is a global centrality measure, which determines the centrality of a node in a network based on the total number of shortest paths going through the given node. Thus, nodes that occur on many shortest paths between other nodes have higher betweenness. The eccentricity of a node is the distance to the farthest reachable other node in a network, thus focusing on a maximal property where closeness focuses on an average.

Shared genes were more central than the specific genes in the protein interaction network as indicated by all the four measures of centrality (Table 1; Figure 2 panels A through D). These topological differences could be affected by shared genes in phenotypically similar diseases; such genes would be expected to be topologically similar to specific genes. To account for this we used CIPHER to distinguish between shared genes associated with phenotypically divergent and similar diseases (Phenodiv and phenosim genes respectively). Our analysis established that phenodiv genes were significantly more central than phenosim genes (Table 1; Figure 3 panels A through D). The observed differences in all the measures of centrality among the genes belonging to the four categories were well demonstrated by the distribution profiles (Figures 2 and 3). Further, phenotypic similarity of diseases (demonstrated by an increasing CIPHER score) showed a significant correlation with the centrality measures in the protein interaction network; we observed significant negative correlations of degree, closeness and betweenness with CIPHER score (Spearman's $\rho = -0.24$, -0.23 and -0.26 ; $P < 0.001$ for all comparisons; Figure 4 panels A through C) while eccentricity was positively correlated (Spearman's $\rho = 0.19$; $P < 0.001$; Figure 4D). In a protein interaction network human disease genes, relative to non-disease genes, are known to have a higher tendency to interact with protein products of other disease genes [4,5]. Since the shared genes are involved in many diseases they would be expected to interact with more disease genes compared to the specific ones. Confirming this, we observed that disease genes were overrepresented among the interactors of shared genes compared to those of specific ones (Figure 5; $P < 0.001$). This obser-



vation prompts us to speculate that there is a higher likelihood of finding a disease gene in the network neighborhood of shared genes. Phenodiv and phenosim genes had a comparable proportion of disease-gene interactors ($P = 0.5$). The enrichment of disease genes among the interactors of shared genes highlighted their role in shared pathogenesis.

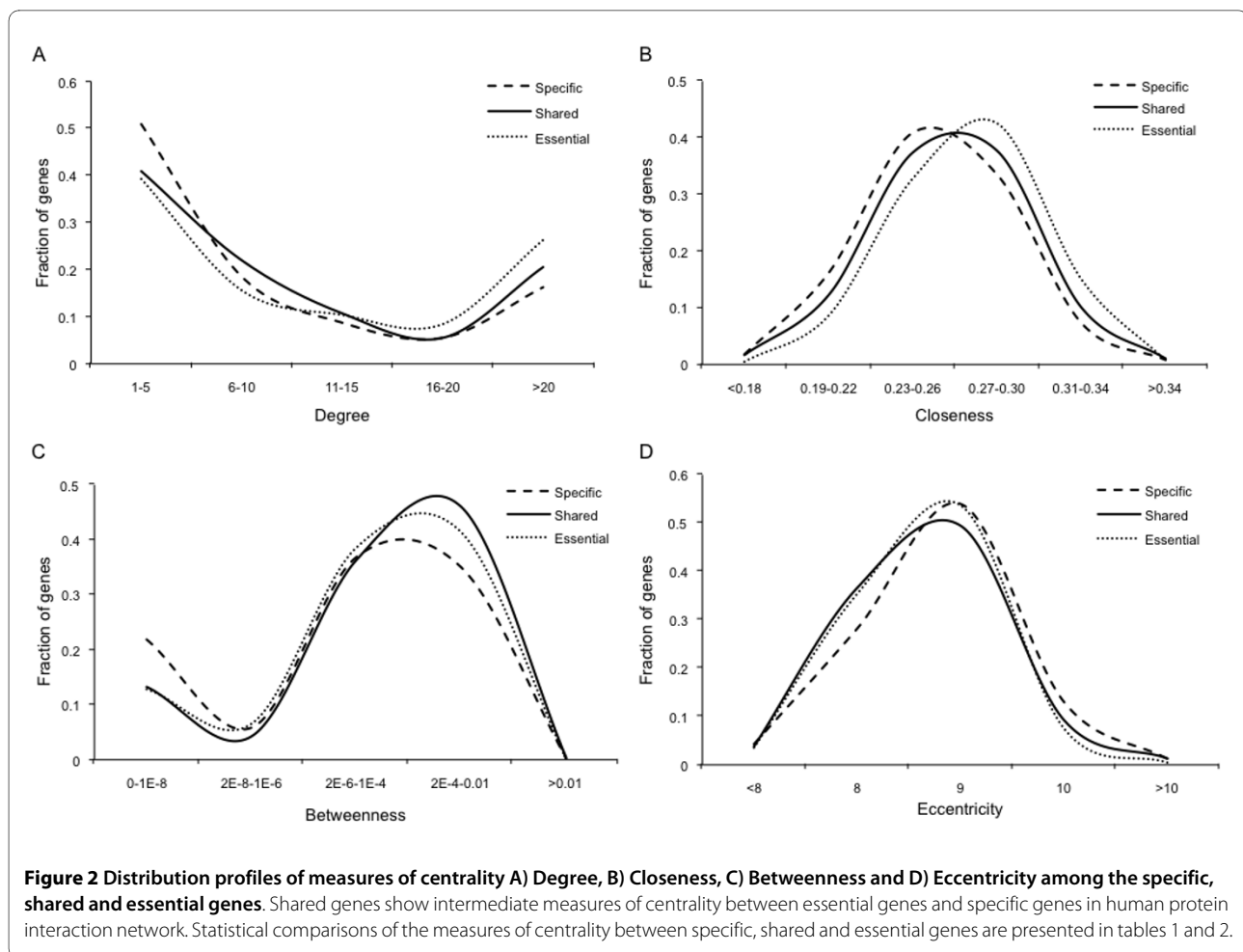
Genes that are essential for early development, functional changes in which might lead to abortions, are termed as essential genes. Essential genes are known to show a tendency to be associated with hubs [4,14]. Some of the human orthologs of mouse lethal (essential) genes are known to be associated with human genetic diseases

[4]. Essential genes associated with human diseases were classified as essential disease genes ($n = 811$) while others belonged to the essential genes set ($n = 1555$; Figure 1). As shared genes show higher centrality we speculated that these might be enriched with essential disease genes compared to specific genes. Shared genes showed an enrichment of essential disease genes compared to specific (44% to 17%; Fisher's Exact $P < 0.001$; Figure 6). Phenodiv and phenosim categories had comparable proportions of essential disease genes. These essential genes can be presumed to be vital for organism survival, sequence variants in which may lead to lethality. We observed that essential genes were significantly more

Table 1: Comparison of measures of centrality of specific with shared genes and phenosim with phenodiv genes in human protein interaction network

Disease gene classes	Degree		Closeness		Betweenness		Eccentricity	
	Mean $\pm$ S.D.	P-value *	Mean $\pm$ S.D.	P-value *	Mean $\pm$ S.D.	P-value *	Mean $\pm$ S.D.	P-value *
Specific genes	13.29 $\pm$ 24.21	<0.001	0.26 $\pm$ 0.04	<0.01	3.3 $\times 10^{-4}$ $\pm$ 0.001	<0.001	8.65 $\pm$ 1.37	<0.001
Shared genes	16.28 $\pm$ 27.73		0.26 $\pm$ 0.03		5.9 $\times 10^{-4}$ $\pm$ 0.002		8.61 $\pm$ 1.06	
Phenosim genes	12.72 $\pm$ 20.72	<0.001	0.27 $\pm$ 0.04	<0.001	3.1 $\times 10^{-4}$ $\pm$ 0.0007	<0.001	8.75 $\pm$ 0.88	<0.01
Phenodiv genes	20.44 $\pm$ 34.85		0.27 $\pm$ 0.04		8.7 $\times 10^{-4}$ $\pm$ 0.003		8.47 $\pm$ 1.17	

*P-values were determined using Mann-Whitney U test



central than specific and shared genes (Table 2; Figure 2). Phenosim genes differed significantly only in degree and closeness while phenodiv genes had all measures of centrality comparable to essential non-disease genes except betweenness. Phenodiv genes had significantly higher betweenness than essential genes (Table 2; Figure 3).

Tissue-specificity of disease genes and their interactors

To discern whether centrality in a protein interaction network relates to a broader tissue distribution we checked for differences in the gene expression of the four categories of disease genes in 79 different human tissues [16]. We could not find any significant difference in the distribution of shared and phenodiv genes compared to specific and phenosim genes respectively ($P = 0.05$ and 0.23 respectively). A previous study suggests significant over-expression of disease genes and their complexes in normal tissues where defects cause pathology [17]. This led us to check for differences in the 'local' interactomes for the four classes of disease genes in the different tissues. For this, we integrated the protein interaction data with that of the gene expression information. When we

checked for co-expression of interactors in the protein interaction network with those of the disease genes we identified that specific genes are more often co-expressed with their interactors than the shared genes (Figure 7A; Mann-Whitney $P < 0.001$; Mean $\pm$ S.D. = 45.9 ± 24.6 and 42.4 ± 24.1 respectively). We could not find any differences among phenodiv and phenosim genes ($P = 0.45$). Further, we identified that specific genes have a higher tendency of tissue co-expression with their disease-gene interactors compared to shared genes (Figure 7B; Mann-Whitney $P < 0.001$; Mean $\pm$ S.D. = 42.6 ± 24.6 and 40.2 ± 24.1 respectively). This, though somewhat unexpected, is in line with the earlier observation of increased co-expression of specific genes with their interactors compared to shared genes. Similarly, we could not find differences in co-expression of disease-gene interactors when shared genes were classified into phenodiv and phenosim genes ($P = 0.05$). Essential genes had higher tissue co-expression with their interactors compared to the different classes of disease genes except specific genes which had comparable proportions (Table 3). Notably, phenodiv genes had significantly lesser tissue co-expression.

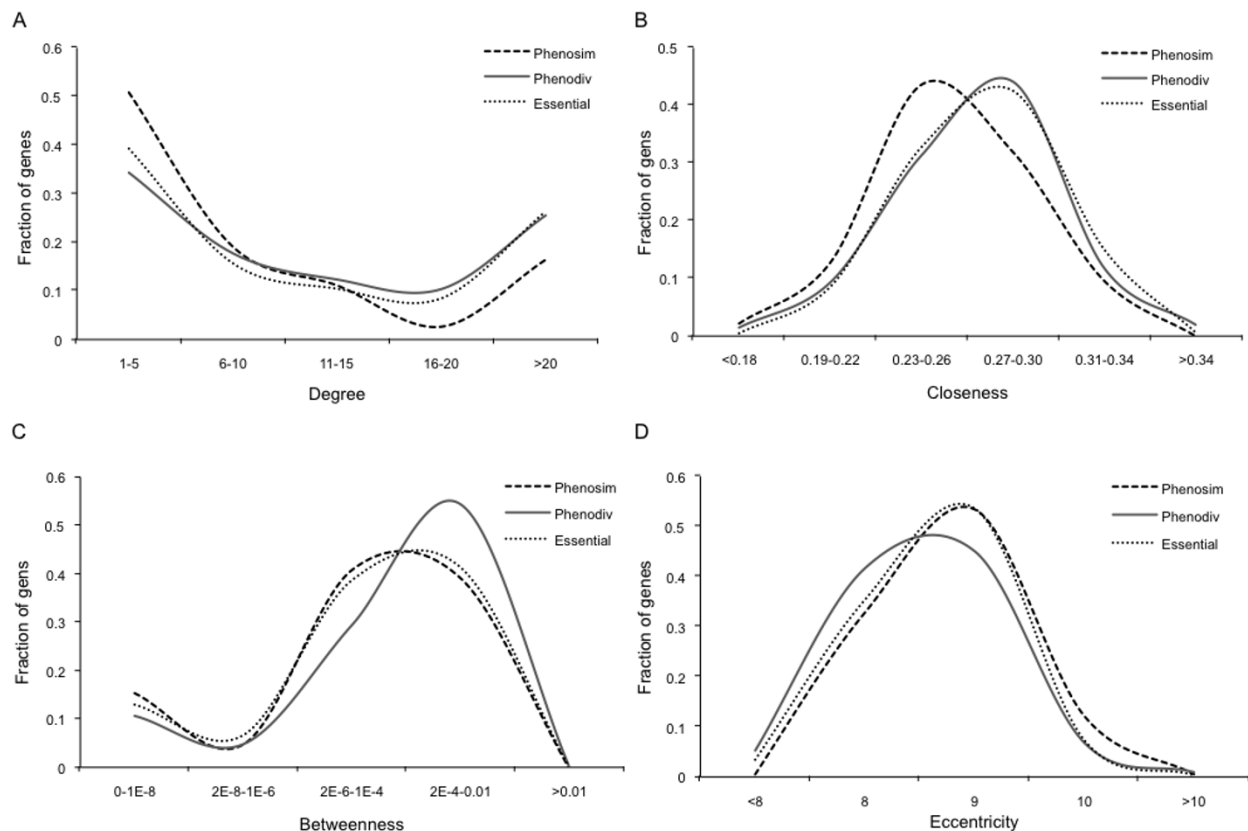


Figure 3 Distribution profiles of measures of centrality A) Degree, B) Closeness, C) Betweenness and D) Eccentricity among the phenosim, phenodiv and essential genes. Phenodiv genes have higher measures of centrality compared to Phenosim genes. Phenodiv genes have comparable measures of centrality to essential genes except higher betweenness. Statistical comparisons of the measures of centrality between phenosim, phenodiv and essential genes are presented in tables 1 and 2.

Pleiotropy and network modularity

In the human protein interaction network, phenodiv genes and essential genes had comparable degrees, closeness and eccentricity. However, phenodiv genes had greater global centrality as indicated by higher betweenness than essential genes (Mann-Whitney $P < 0.001$; Mean $\pm$ S.D. = $8.7 \times 10^{-4} \pm 0.003$ and $5.5 \times 10^{-4} \pm 0.002$ respectively). This prompted us to check for differences in clustering coefficient among essential and phenodiv genes. Clustering coefficient quantifies the cohesiveness of the neighborhood of a node and is defined as the ratio between the number of edges linking nodes adjacent to a node and the total possible number of edges among them. Thus, clustering coefficient characterizes the overall tendency of nodes to form clusters or groups. We observed that phenodiv genes had significantly lesser clustering coefficient than the essential genes (Mann-Whitney $P = 0.018$; Mean $\pm$ S.D. = 0.12 ± 0.19 and 0.17 ± 0.24 respectively). In addition, phenodiv genes showed relatively restricted tissue co-expression with their interactors in contrary to essential genes. Taken together, these results

imply that these phenodiv and essential genes are classes of high degree genes (hubs) which are inter-modular and intra-modular respectively (Figure 8). Such hubs have been extensively studied recently [18] and biochemical differences have been reported between these two types of hubs. We hypothesized that the varying topological properties along with their different tissue co-expression profiles with their interactors could be explained by different biological functions. Hence, we used Gene Ontology (GO) to functionally characterize the essential and phenodiv genes for their cellular component, molecular function and biological process (Additional files 1 and 2). GO analysis of essential genes identified organelle, intracellular membrane bound organelle (1022 and 946 genes in contrast to a random expectation of 701.5 and 607.1 genes respectively; $P < 0.001$) to be the most significant cellular components with most of the genes in nucleus (737 genes as against a random expectation of 392.3 genes; $P < 0.001$). Contrarily, cytoplasm and membrane (123 and 121 genes in contrast with the random expectation of 82.9 and 88.1 respectively; $P < 0.001$) were over-

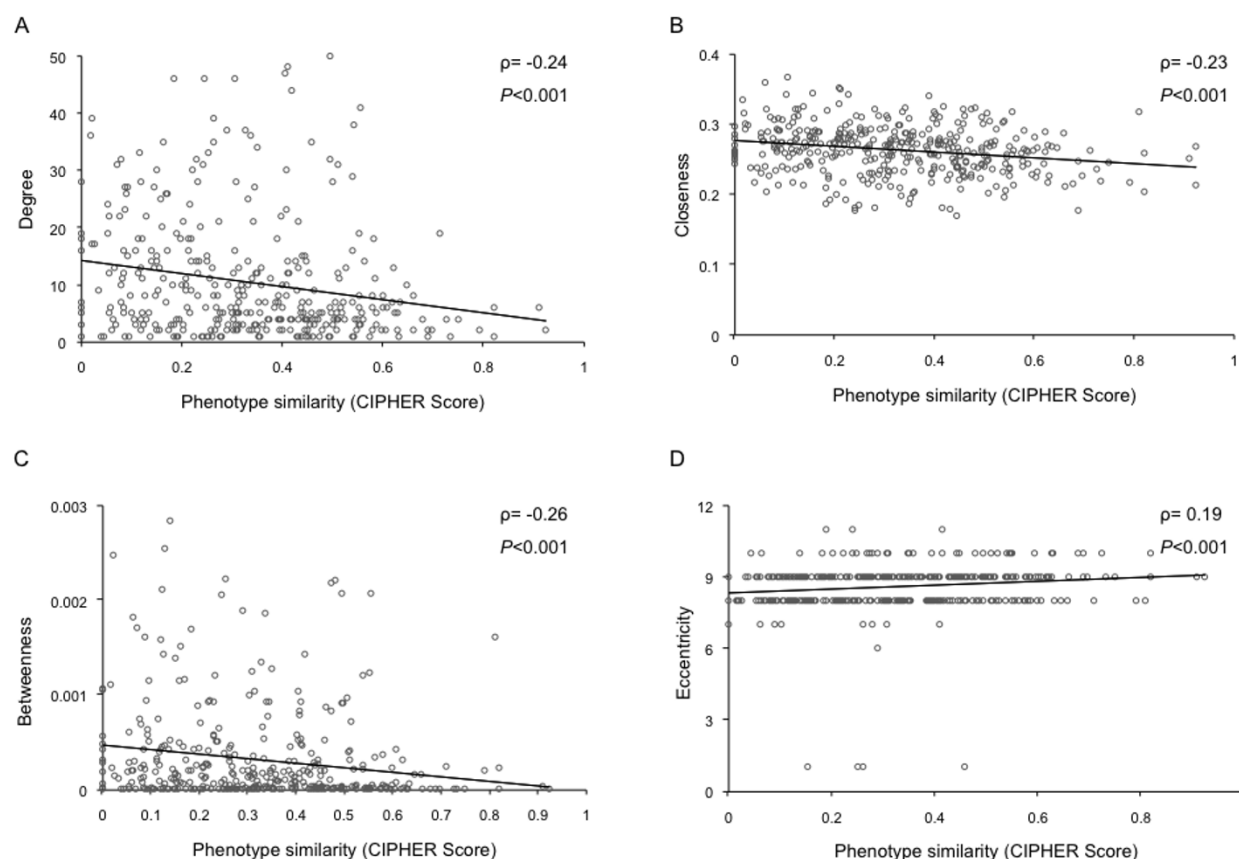


Figure 4 Correlation of measures of centrality. A) Degree, B) Closeness, C) Betweenness, D) Eccentricity with phenotypic similarity estimated by CIPHER Score. ρ represents Spearman's rho. The trend lines with negative slope in panels A, B and C indicate negative correlation of degree, closeness and betweenness with phenotypic similarity with the corresponding correlation coefficient and significance. The positive slope of the trend line in panel D demonstrates a positive correlation between eccentricity and phenotypic similarity. Thus, with the increasing phenotypic similarity, the respective disease associated genes show decreasing centrality. Very high degree values in the panel A have been removed in order to aid better visualization; however they have been considered for estimating correlation.

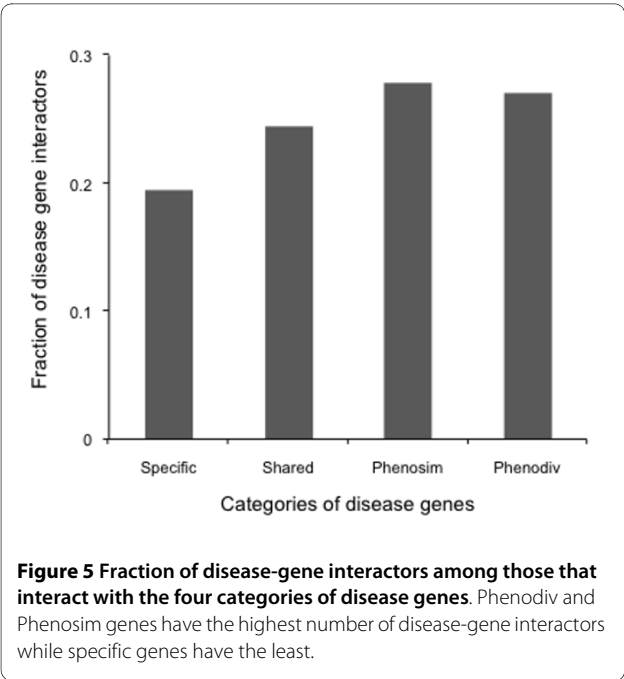
represented in phenodiv genes. The overrepresented molecular function categories for the essential genes include protein binding and nucleic acid binding (945 and 505 genes against random expectation of 588.1 and 280.8 respectively; $P < 0.001$) while apart from protein binding overrepresented molecular function classes for phenodiv genes were catalytic activity and signal transducer activity (84 and 62 genes to a random expectation of 65.2 and 28.2 respectively; $P < 0.01$ and < 0.001 respectively). Intriguingly, both these classes of genes are involved in similar biological processes.

Discussion

The phenotypic consequence of a variation in a gene is known to be affected to a large extent by the topological position of its protein product in the molecular interaction network. Thus, the functional importance of a gene is signified by its centrality in a protein interaction network. Previously, we and others have shown that the con-

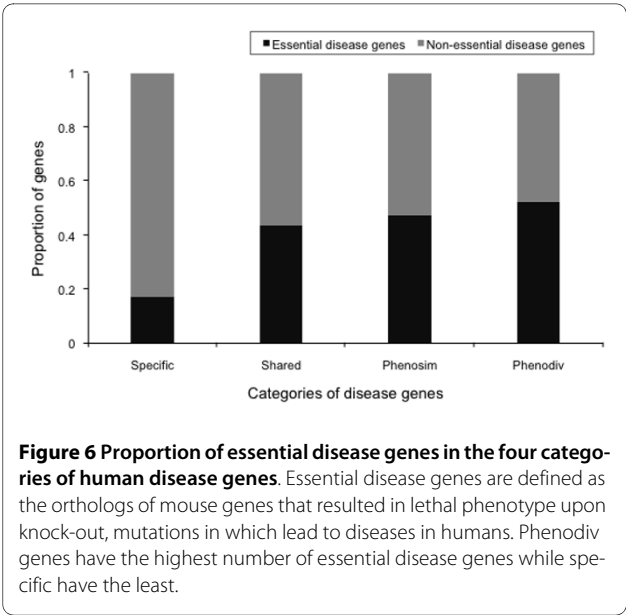
tribution of variations in a single gene to bring about an associated phenotype is a function of its centrality [4-6]. Accordingly, based on centrality different gene classes leading to a phenotype are ordered as essential genes (being the most central), monogenic disease genes, complex disease genes and non-disease genes (being the most peripheral). However, the network properties of a gene, mutations in which lead to various phenotypes have not been explored.

Based on the current understanding of the human protein interaction network and the results presented here, we demonstrated that the pleiotropic genes (shared genes) had an intermediate centrality compared with essential genes and genes associated with only one disease (specific genes). However, classification of the shared genes based on the similarity of the associated phenotypes demonstrated that phenodiv genes leading to divergent phenotypes were more central than phenosim genes. Thus based on increasing order of centrality these



different disease genes could be arranged as Specific, Phenosim and Phenodiv genes. We note that the observed correlation of measures of centrality with phenotypic similarity provides support that the interpretations might not have been affected by considering median CIPHER value as a cut-off to classify phenosim and phenodiv genes.

Co-expression with interactors is a prerequisite to bring about the function of a gene. Thus, specific genes with a very small network neighborhood would always be co-expressed with their interactors. Conversely, essential genes are hubs with high co-expression with their interactors. This attribute explains as to why mutations in these genes lead to lethality. Contrary to both specific and essential genes, shared genes showed decreased co-expression with their interactors. In addition to an intermediate centrality in the protein interaction network, this could be considered as an important functional property



of genes with pleiotropic effects. For instance, the phenodiv gene *AKT1* is associated with divergent phenotypes including schizophrenia, colorectal cancer, ovarian cancer and breast cancer. The network of its interactors varies among these diseases (Figure 9A) with some expressed in all disease tissues to very few not expressed in any. On the contrary, the disease-specific genes are co-expressed with all their interactors in the respective disease tissue. This is explicitly demonstrated by the co-expression of *CLINT1* associated with schizophrenia with all its interactors in the brain region associated with the pathophysiology of the disease (Brodmann area 10: anterior prefrontal cortex; Figure 9B). Similarly, *RRAS2*, *PMS1* and *PHB* which are associated only with ovarian cancer, colorectal cancer and breast cancer are co-expressed with their interactors in the respective disease tissues (Figure 9 panels C through E).

The similar measures of centrality between essential and phenodiv genes, except betweenness, led us to investigate the properties that determine essentiality (lethality

Table 2: Comparison of measures of centrality of Essential genes with that of specific, shared, phenosim and phenodiv genes in human protein interaction network

Measure	Essential Mean $\pm$ S.D.	Specific P-value*	Shared P-value*	Phenosim P-value*	Phenodiv P-value*
Degree	21.57 $\pm$ 35.89	<0.001	0.012	<0.001	0.44
Closeness	0.27 $\pm$ 0.03	<0.001	<0.001	<0.001	0.83
Betweenness	5.5 $\times 10^{-4}$ $\pm$ 0.002	<0.001	0.23	0.12	<0.001
Eccentricity	8.63 $\pm$ 0.87	<0.001	0.99	0.039	0.039

Mean $\pm$ S.D. values for the specific, shared, phenosim and phenodiv genes are provided in Table 1. *P-values were determined using Mann-Whitney U test

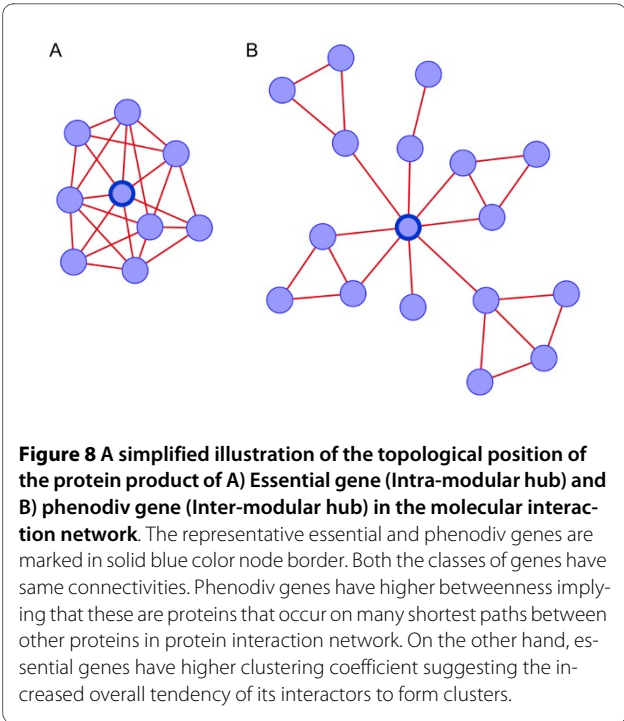
Table 3: Comparison of tissue co-expression of interactors of Essential genes with that of specific, shared, phenosim and phenodiv genes

Gene Class	Tissue co-expression of interactors	
	Mean $\pm$ S.D.	P-value*
Essential	46.4 $\pm$ 23.2	
Specific	45.9 $\pm$ 24.6	0.42
Shared	42.4 $\pm$ 24.1	<0.001
Phenosim	42.5 $\pm$ 24.0	<0.001
Phenodiv	42.0 $\pm$ 22.9	<0.001

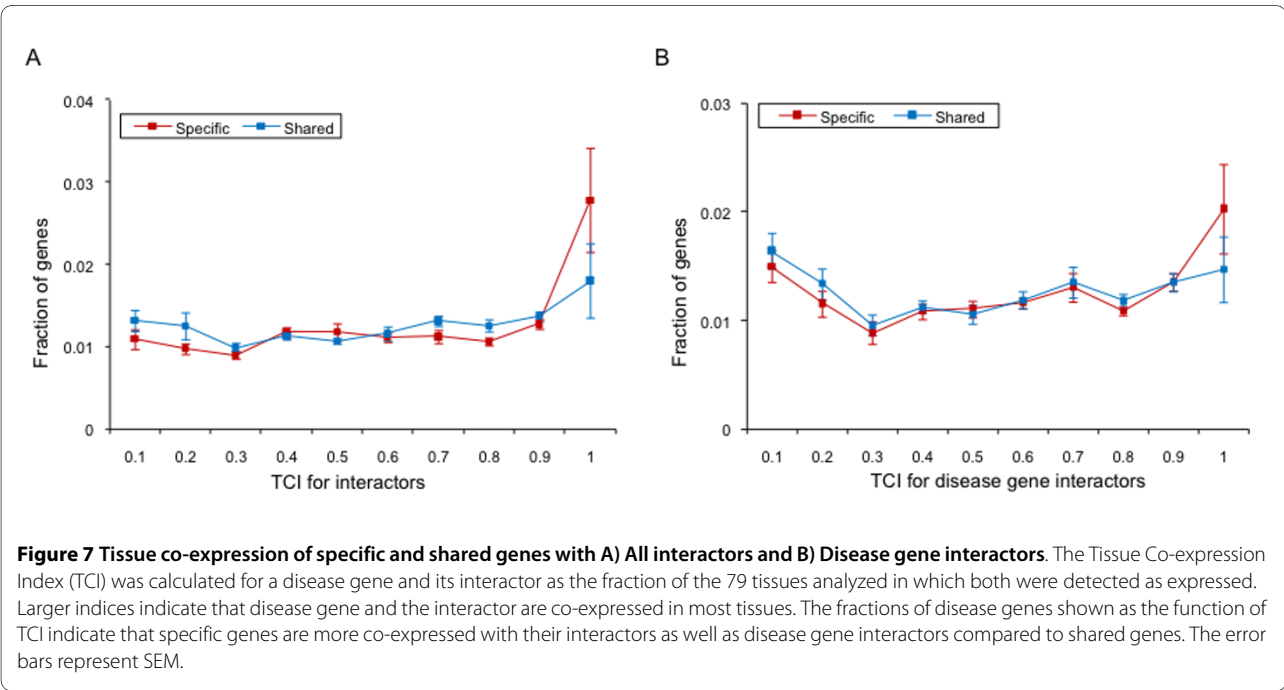
*P-values were determined using Mann-Whitney U test

of mutants) and pleiotropy. One of the most striking observations made here was that the essential genes and phenodiv genes were intra-modular and inter-modular hubs, with the former being highly co-expressed with its interactors contrary to the latter. Furthermore, essential genes were predominantly involved in transcription regulation while phenodiv genes in signal transduction.

This study could be affected by knowledge bias pertaining to the disease genes and their associated phenotypes as presented in OMIM, the human protein interaction network and the information on tissue expression. With an increasing number of genetic studies it is likely that some of the specific genes will be identified as shared and some of the phenosim as phenodiv genes. Based on the



trend we observed here, it is tempting to speculate that essential disease genes in the specific and phenosim genes categories may have a higher likelihood for this transition. An expansion of knowledge of the diseases and disease genes, protein interactions and tissue expression would aid in better comprehension of the properties associated with genes causing pleiotropic effects. Further,



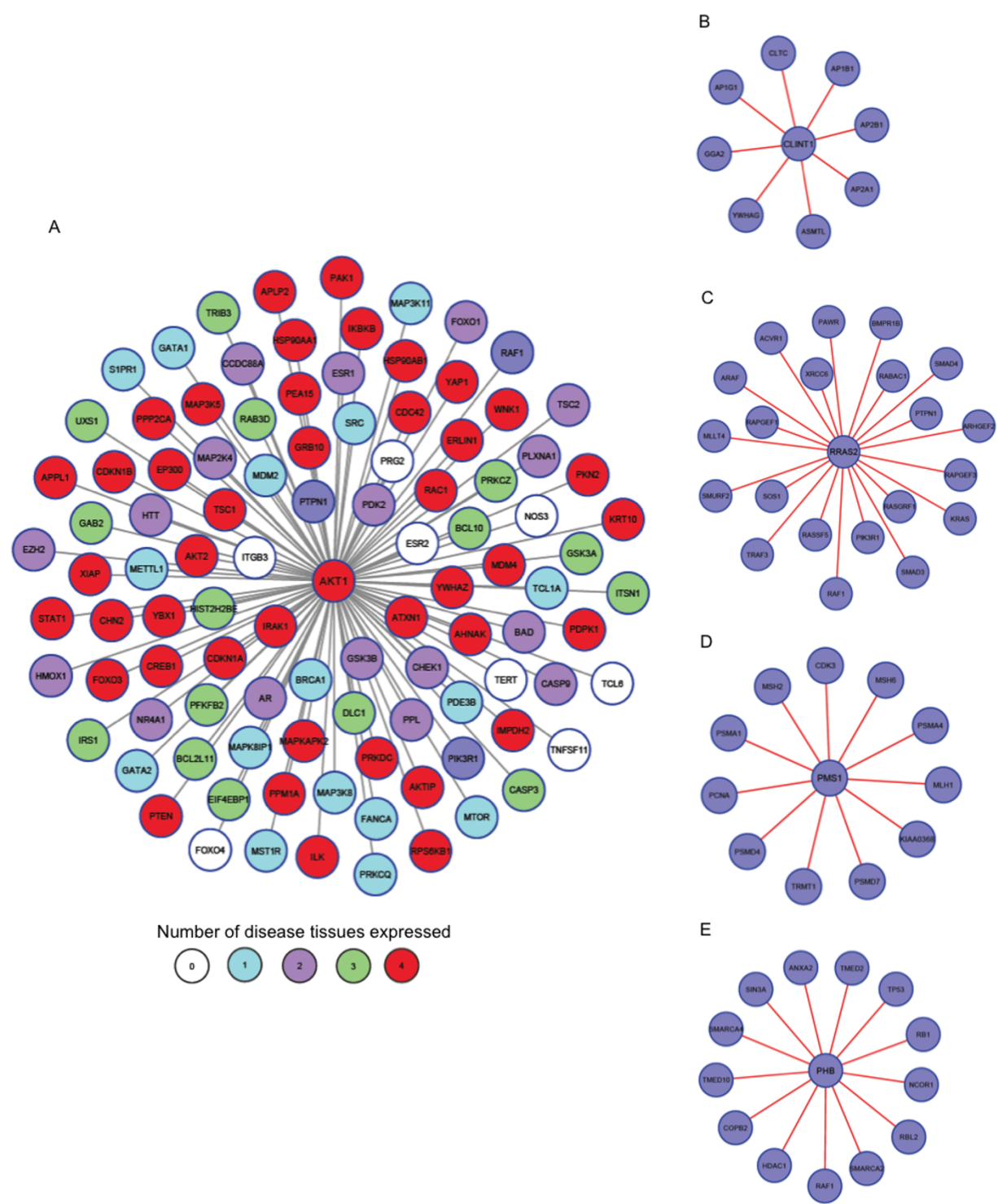


Figure 9 Networks of interactors of disease genes in disease tissues. A) Network of interactors of the phenodiv gene *AKT1* which is associated with schizophrenia, ovarian cancer, colorectal cancer and breast cancer. The color of the nodes indicates the number of disease tissues in which the interactors are expressed in. As indicated by the co-expression of interactors, *AKT1* interacts with diverse interactors under different pathological conditions. *AKT1* as a specific example of phenodiv genes demonstrates that phenodiv genes have more interactors (higher connectivity) and show relatively restricted co-expression with their interactors across different tissues. Network of interactors of specific genes- B) *CLINT1* C) *RRS2* D) *PMS1* and E) *PHB* associated with schizophrenia, ovarian cancer, colorectal cancer and breast cancer respectively. As has been observed for the class of specific genes, *CLINT1*, *RRS2*, *PMS1* and *PHB* have lesser number of interactors and are co-expressed with all their interactors in the respective disease tissue.

it will be interesting to study the temporal co-expression of these genes with their interactors in various tissues.

Conclusions

Here we demonstrated that the ability of a disease gene to influence the cellular network, signified by its centrality and tissue co-expression with its interactors, determines its pleiotropic effects.

Methods

Dataset

We obtained the list of human genetic diseases and associated 5024 diseases from the OMIM MorbidMap (downloaded in February 2009) [19]. Based on the number of disease associations the genes were then classified as specific- genes associated with one disease ($n = 2512$) and shared- those associated with more than one disease ($n = 838$). To determine the phenotypic divergence among diseases associated with the shared genes we used CIPHER (Correlating protein Interaction network and PHENotype network to pRedict disease genes) [13]. CIPHER provides a similarity score of phenotypes for diseases in MorbidMap, based on their OMIM descriptions using Medical Subject Headings (MeSH) terms. Thus, a lower CIPHER score represents a higher phenotypic divergence. We retrieved CIPHER score for all diseases associated with the same gene and recorded the lowest score for each shared gene. This resulted in identification of 472 shared genes with phenotypic similarity/divergence information. Using the median CIPHER score (0.33) of the entire dataset as cut-off, we then categorized these genes into genes associated with phenotypically similar diseases (Phenosim genes with CIPHER scores ≥ 0.33 ; $n = 238$) and genes associated with phenotypically divergent diseases (Phenodiv genes with CIPHER scores < 0.33 ; $n = 234$). We defined essential genes ($n = 2366$) as previously described [4], by retrieving a list of human orthologs of mouse genes that resulted in lethal phenotype in embryonic and postnatal stages upon knockout as catalogued in Mouse Genome database [20]. Of the essential genes 811 were associated with human diseases and are classified as essential disease genes while 1555 essential non-disease genes were categorized as essential genes (Figure 1).

Human protein interaction network

We constructed a human protein interaction network using a modified version of CRG interactome [15]. CRG interactome is by far the largest protein interaction network including protein-protein interactions supported by at least one direct experimental evidence demonstrating physical association between two human proteins. To validate proper annotation of the proteins, we retrieved Entrez gene identifiers for all the proteins that were listed in the CRG interactome with Ensembl gene identifiers.

After removing entries that lacked Entrez gene identifiers, the modified CRG interactome consisted of 10,092 proteins with 79,211 interactions.

Tissue expression

In order to examine differences in tissue distribution among the shared and specific genes, and the Phenosim and Phenodiv genes and co-expression with their interactors (from the human interaction network) we used previously described dataset of complete set of interactions with details of cells and tissues in which each interaction can occur [15]. This dataset was derived based on GNF Atlas expression data [16] which details tissue expression patterns of genes across 79 different non-disease human tissues. As previously described [15] we determined the expression of interactors for AKT1 in disease tissues- schizophrenia (GSE17612), colorectal cancer (GSE14333), breast cancer (GSE19615) and ovarian cancer (GSE18520). The datasets of disease tissues from patients were retrieved from NCBI Gene Expression Omnibus [21]. Expression of interactors of CLINT1, PMS1, PHB and RRAS2 were determined in schizophrenia, colorectal cancer, breast cancer and ovarian cancer disease tissues respectively. The interactors are considered co-expressed if they are expressed together in tissues in the datasets considered.

Gene Ontology

Overrepresentation of Gene Ontology (GO) categories [22] was determined using the GOSim package from Bioconductor. Statistical significance estimation for overrepresented GO categories in real datasets (phenodiv and essential genes) was done by considering all GO categories without defining any levels in the GO hierarchy in order to avoid loss of information. We considered only categories with at least 5 total genes to prevent categories appearing to be significantly over-represented due to chance.

Statistical Analysis

We compared the measures of centrality- degree, closeness, betweenness and eccentricity, between different classes of genes by Mann-Whitney U tests. Correlation between phenotypic similarity, as determined by CIPHER score and the measures of centrality was determined using Spearman's rank correlation. Different classes of disease genes were compared for disease essential genes using Fisher's Exact test. For GO analysis the number of genes in real dataset (N) with GO classifications was determined. In each GO category the number of genes in real dataset was counted. From the total number of genes represented in GOSim a set of N genes were randomly sampled and the number of genes present in each GO category was counted. This was repeated 10,000 to generate "randomly expected gene lists" for each GO category.

Statistical significance was estimated as the ratio of number of times a GO category had more number of expected genes than observed for the real dataset genes to the number of random gene lists considered. Statistical analyses were performed using R.

List of Abbreviations

OMIM: Online Mendelian Inheritance in Man; CIPHER: Correlating protein Interaction network and PHENotype network to pRedict disease genes; GO: Gene Ontology; Phenosim genes: Genes associated with phenotypically similar diseases; Phenodiv genes: Genes associated with phenotypically divergent diseases;

Additional material

Additional file 1 Over-represented Gene Ontology categories in Essential genes.

Additional file 2 Over-represented Gene Ontology categories in phenodiv genes.

Authors' contributions

SC and FB designed, processed, interpreted the data and wrote the manuscript. KK has contributed in acquisition and interpretation of the data. MB has conceived and supervised the study. All the authors read and approved the final manuscript.

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Author Details

The Unit for Clinical Systems Biology, University of Gothenburg, Medicinaregatan 5A, Gothenburg SE405 30, Sweden

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